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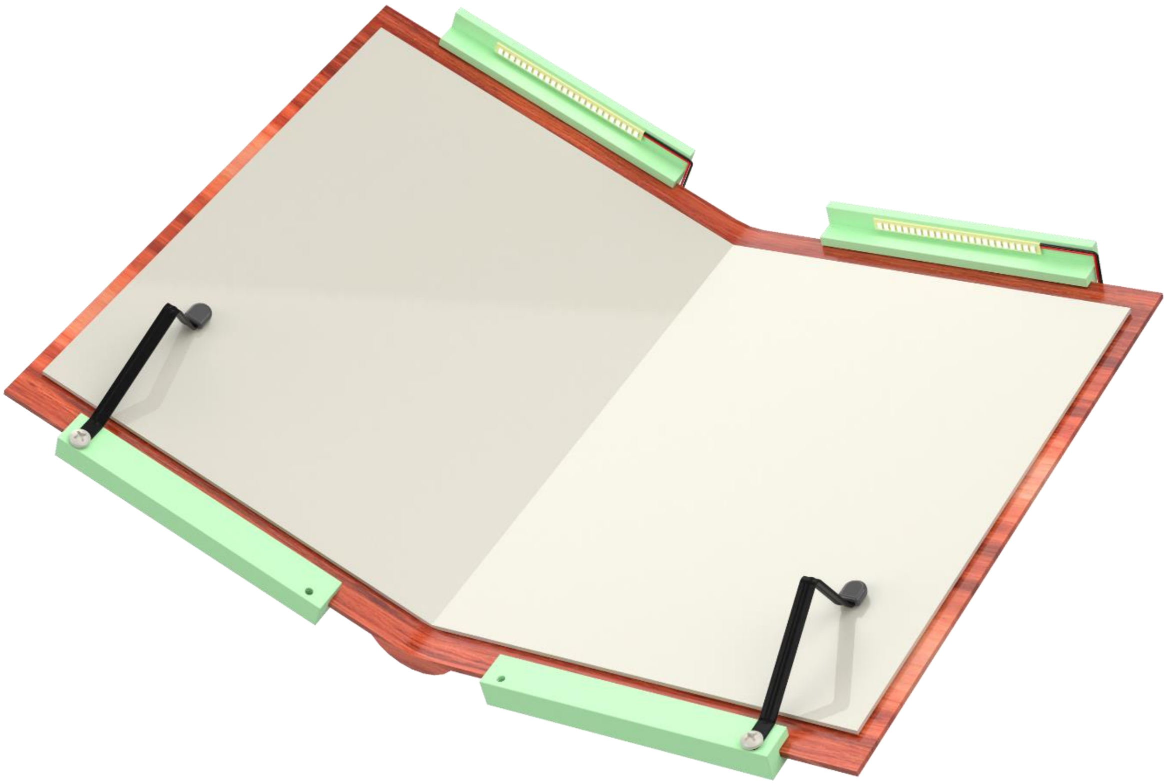
Book Nook Report



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MCEN 5045: Design For Manufacturability

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Table of Contents

Table of Figures

Table of Tables

Executive Summary

This report details the design and manufacturing proposal for BookNook, a hardcover book attachment that enhances the reading experience through integrated page management and illumination. The product combines an adjustable lever arm system to enable smooth page turning along with an associated book light to supplement ambient lighting and provide focused book illumination. The BookNook attaches to hardcover books via a clamp system that accommodates a variety of book sizes while maintaining consistent tension through a wave form spring mechanism.

The product development process follows a methodical approach: component modeling, assembly analysis, and redesign implementations. Initial efforts focused on detailed 3D modeling of all components with corresponding 2D technical drawings to establish manufacturing specifications and tolerances.

The design comprises three distinct subsystems: the clamp system, page holder system, and book-light circuit. To better understand the system's operation, we developed black-box and glass-box diagrams, along with Ishikawa diagrams to analyze both overall system functions and individual component relationships. Additionally, patent research provided insight into existing solutions and opportunities for differentiation.

Once determining the initial design, material and manufacturing analysis was done on each part, followed by some FEA modeling to prove the product will not break for the multiple types of hardcovers it can be used on. A DFA and DFM analysis were completed with the final calculated values seen below, along with the compared values after our design improvements. As you can see, there is a large jump from our initial to our final design.

Design attempt	Practical Minimum Part %	DFA Complexity Metric	Unit Cost
Initial Design	50%	31.01	$19.79
Final Design	92.9%	19.1	$14.19
Improvement Spec.	85% increase	38.7% improvement	28.3% increase

Overall, the BookNook has a whopping 92.9% practical minimum part count, but it’s cost has been optimized to save 28.3% of its original cost with its DFA metric improving by 38.7%. The BookNook would be sold for $27.88, right in the same price point of current booklights on the market with a break-even point at 50,000 units.

Gantt Chart

The chart below represents the Gantt Chart for the timeline of our project.

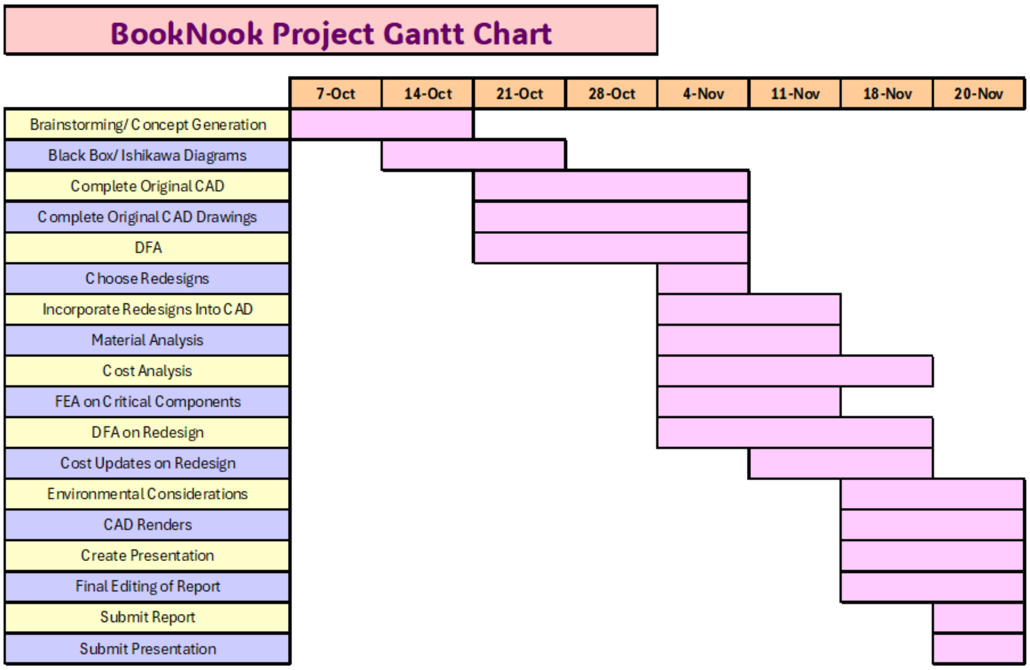


Figure 1: Gantt Chart

Black Box and Glass Box Diagrams

The first step in transforming the design concept into a tangible product involved analyzing the system's inputs and outputs through a black box diagram. This diagram was developed during the product's conceptualization phase to identify the desired outputs and necessary inputs. By mapping these elements early on, it became possible to streamline the design by eliminating unnecessary features and clarifying the level of user interaction required.

Figure 2: Black Box Diagram

The next step in the design process involved creating a glass box diagram to reveal the internal components and their relationships within the system. This diagram expanded upon the black box analysis by identifying the specific subsystems needed to achieve the desired outputs. By mapping out the lighting subsystem, clamping mechanism, and power subsystem along with their interconnections, it became possible to understand how each component contributed to the overall functionality. This transparent view of the system's architecture helped guide decisions about component selection, spatial arrangement, and the integration of mechanical and electrical elements.

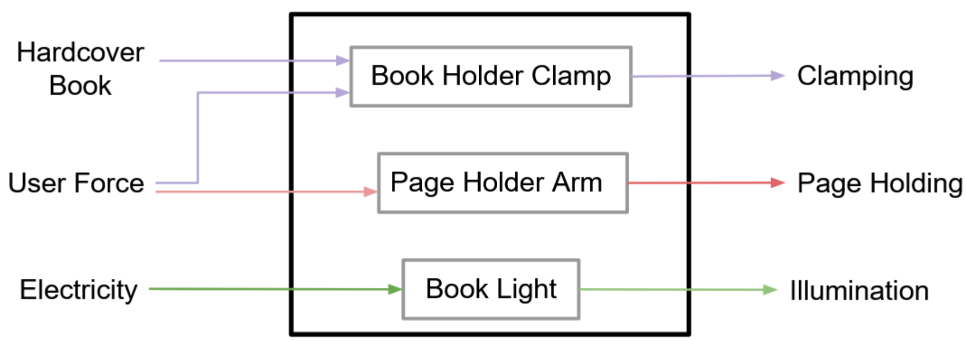


Figure 3: Glass Box Diagram

Ishikawa Diagram

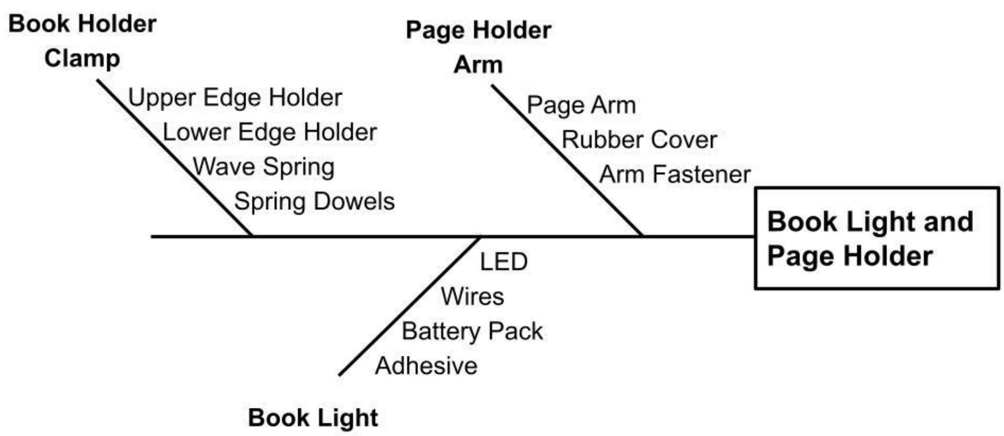


Figure 4: Ishikawa Diagram

The Ishikawa diagram above, Figure 4, was developed from initial BookNook designs. The interfaces and subsystem complexity disparities identified indicate potential areas for redesign in future iterations.

User Experience/operation

The BookNook has a design that allows it to fit to many different sizes of hardcover books. The user will need to separate the cover of a book and fit the booker on. The most common cover size of a hardcover fiction book is 6" x 9" and the spring has been designed to perfectly hold onto a 9" tall book. The spring will need to be stretched to allow the two cover holders to fluctuate in size to grip onto the book.

Once in place, the user will turn the light on with a switch located on the battery pack. When the user wants to hold open the page, they will need to input around 1-2 lbs. of force to lift the metal arm and pivot it to hold open a page. The rubber tip on the arm will hold to page down and won't slide off your page.

If you find the need to mount the BookNook to a different size cover, you will want the arm that holds the page open to move to the opposite side. To accommodate for this the arm and its associated screw can be unscrewed and screwed into the opposite side. A representation of this can be seen here:

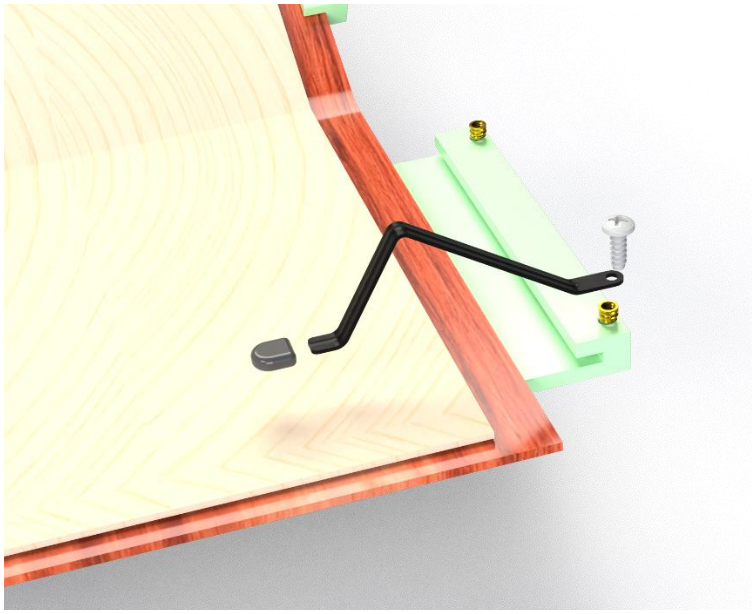


Figure 5: Tension Repositioning of the Arm Placement

If the user needs to replace the battery for the LED, an off-the-shelf battery pack is included that has a removable cover. NOTE: For purposes of this project, we are treating the battery pack and removable cover as one part for simplicity.

A single unit of the BookNook can be consolidated into a 4.25”x3.25”x1.25” box for convenient and lightweight packaging. The whole package, assuming a cardboard box outer shell, would weight ~0.32 lbs. Not only will this add ease for the consumer buying the product, but it will also save room during transportation.

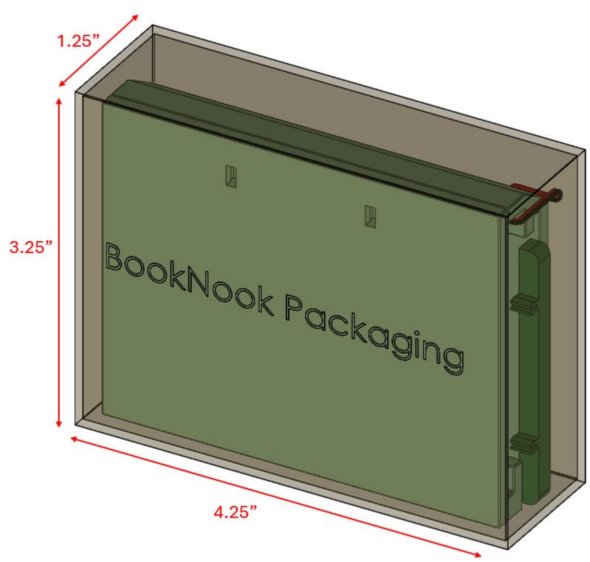


Figure 6: BookNook Packaging Dimensions

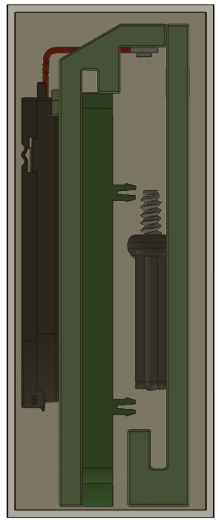


Figure 7: Side View of BookNook Packaging

In this configuration, the consumer will only need to snap fit the wave spring into place and attach the tension arm to their desired side. Instructions need to be provided to guide the user's set-up experience.

Patent Search

A patent search was conducted to identify related innovations and gather insights from existing solutions. By examining how other designers addressed similar challenges, valuable ideas emerged that informed the development process and helped refine the product's unique features.

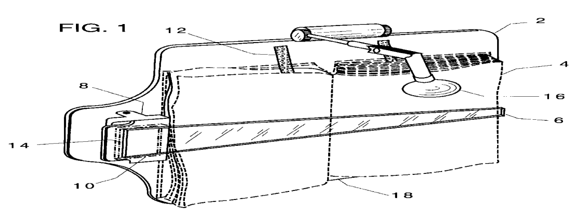


Figure 8: US Patent 5829787A

US Patent 5829787A, granted in 1998, possesses the combination of a booklight, page holder and hold onto the book, seen above in Figure 7. The booklight features a hinge for adjustable hinge to direct light. The page holder is one arm, composed of clear plastic stretches across an open book acting as a clip on one side to enable page turning. The book is secured to a holder via elastic straps inside its cover.

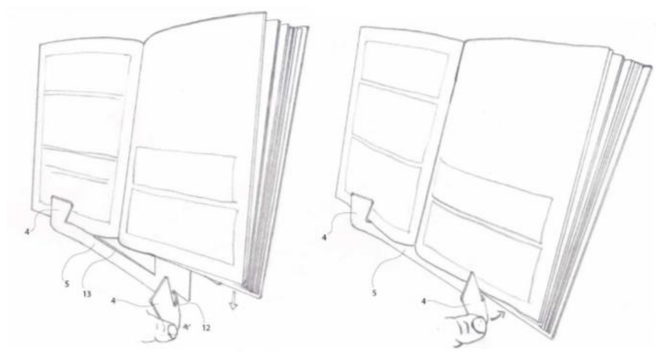


Figure 9: Patent 20170004116U

Korean patent 20170004116U, awarded in 2016, describes a reading stand with an integrated page holder, as seen in Figure 8. The page holder functions by retracting a thin plastic tab, cut out from the body, and positioning it over the pages to secure them in place.

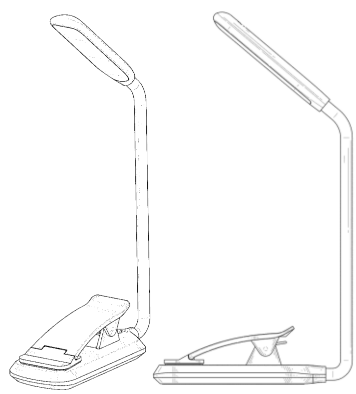


Figure 10: Patent D87092S1

US patent D87092S1, awarded in 2019, presents a book light design that secures a book by clamping over both the cover and pages for stable operation. The battery is housed in the lower portion of the clamping mechanism, while the light remains fixed in position through an integrated holder.

Bill of Materials: Original Design

The complete bill of materials includes all components of the assembly, both manufactured and those pulled off the shelf. The item number, part number, quantity, description, and material can be found below.

Item #	Part #	Qty	Name	Material
1	1	1	Upper Edge Holder	PC Plastic
2	2	1	Lower Ede Holder	PC Plastic
3	3	1	Wave Spring	PET
4	4	5	Spring Dowels/Hinge Rod	Off the Shelf
5	5	1	LED	Off the Shelf
6	6	1	Battery Pack	Off the Shelf
7	7	1	Battery	Off the Shelf
8	8	2	Wires	Off the Shelf
9	9	2	Double Sided Adhesive	Off the Shelf
10	10	1	Hinge Mount	PC Plastic
11	11	1	Hinge Swivel	PC Plastic
12	12	2	Hinge Fasteners	Off the Shelf
13	13	1	Page Arm	20G 316 Stainless Steel
14	14	1	Rubber Cover	TPE
15	15	1	Arm Fastner	Off the Shelf
16	16	3	Heat Set Inserts	Off the Shelf

Table 1: Initial BOM

Material Analysis and Process Selection

While designing the BookNook, several types and functions of parts were analyzed that would only work with specific materials. After examining each subsystem and separating the part list into off-the-shelf components and specially manufactured components, the key parts requiring custom manufacturing or justification of material choice were identified. The specific components and the processes used to determine the appropriate manufacturing materials and methods for the BookNook are explained below.

Part Number + Name	Material	Manufacturing Method
1 - Upper Edge Holder	PC Plastic	Injection Molding
2 - Lower Edge Holder	PC Plastic	Injection Molding

3 - Wave Spring	PET Plastic	Injection Molding
4 - Spring Dowels	316 Stainless Steel	Off The Shelf
9 - Double Sided Adhesive	3M 9482PC	Off The Shelf
10 - Hinge Mount	PC Plastic	Injection Molding
11 - Hinge Swivel	PC Plastic	Injection Molding
14 - Arm	316 Stainless Steel	Sheet Metal Stamping
15 - Rubber Cover	TPE	Injection Molding

Table 2: Material and Manufacturing Process of Each Part

1 - Upper Edge Holder, 2 - Lower Edge Holder, 10 - Hinge Mount and 11 - Hinge Swivel

These four components make up the general body, size and weight of the BookNook. They each need to be lightweight rigid bodies that have multiple components attached to them through different means and need to be chosen to accommodate several design constraints while being the same material for simplicity.

A suitable material for these components must possess adequate rigidity to maintain book covers in tension while accommodating heat-set inserts during assembly. The material should also provide sufficient thread engagement strength to withstand overtightening of fasteners without stripping or deformation, while ideally maintaining a low density to minimize overall product weight.
Based on the requirement that the component function as a rigid body, especially one that incorporates a threaded screw hole, a material with a high Young's modulus is preferred. Because this material constitutes the majority of the BookNook, a lower density is also desirable to maintain a lightweight product. After identifying materials that satisfy both attributes, high temperature resistance will serve as the deciding factor in selecting the optimal option. A Young's modulus greater than 1 GPa is required to ensure that the material cannot be deformed by hand. The target overall weight of the assembly is approximately ¼ lb (0.1134 kg). With a total volume of 6 in^3 (98.32 cm^3), the desired material density is approximately 1.15 Mg/m^3.

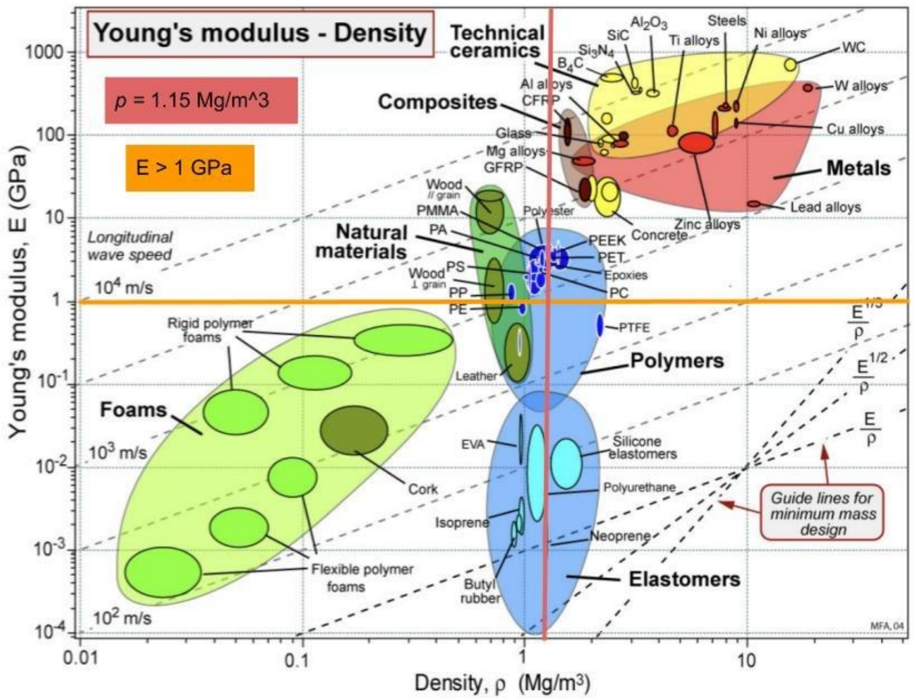


Figure 11: BookNook Body Ashby Chart

After analyzing an Ashby Chart, many polymers were identified that meet the exact needs. The noticeable materials are PC, PET and Polyester. The following materials and their respective temperature resistances (Heat deflection temperatures) are:

PC TempResistance = ~130 C

PET TempResistance= ~70-80 C

Polyester TempResistance= 60-150 C

Taking these values into account, a generally high heat resistance is desired for a heat set material, eliminating PET and Polyester to choose a reliable material.

To summarize, the injection molded structure of the product will be made of PC plastics to have a general light target weight, a strong Young's modulus to avoid deformation and a high temperature resistance to accommodate the heat set inserts. The molds will be slightly complex for the lower and upper edge holder as they require additional pins to be inserted before material flows.

3 - Wave Spring

The wave spring needs to hold some spring tension to keep the book holders on the cover of the book. This component also requires some elasticity as you also need the wave spring to extend and contract when removing and installing the book holders on the covers of books. This also will have the heat set inserts put through to fasten it to the book holder covers.

For a material to be suited for the wave spring, it must be able to handle cyclical loading and possess a high yield strength to prevent over-drawing by users, while maintaining enough deflection to allow for easy removal from book covers. The selected material must not expand at elevated temperatures, ensuring the effectiveness of heat-set inserts during assembly. Additionally, the material needs sufficient strength to withstand the cyclic loading experienced by each interacting component while maintaining rigidity to preserve the structural integrity of the BookNook. The material must exhibit immediate elastic response when subjected to tensile forces without experiencing failure, and it should have a relatively high yield stress to prevent product damage from excessive user force during operation.

To choose a model from these design constraints, a lower Young's Modulus ($E < 5GPa$) was prioritized to account for the need to flex, but also not fail when overpulled. This also requires slightly higher fracture toughness. The Ashby chart of fracture toughness vs Young’s modulus is perfect for this component.

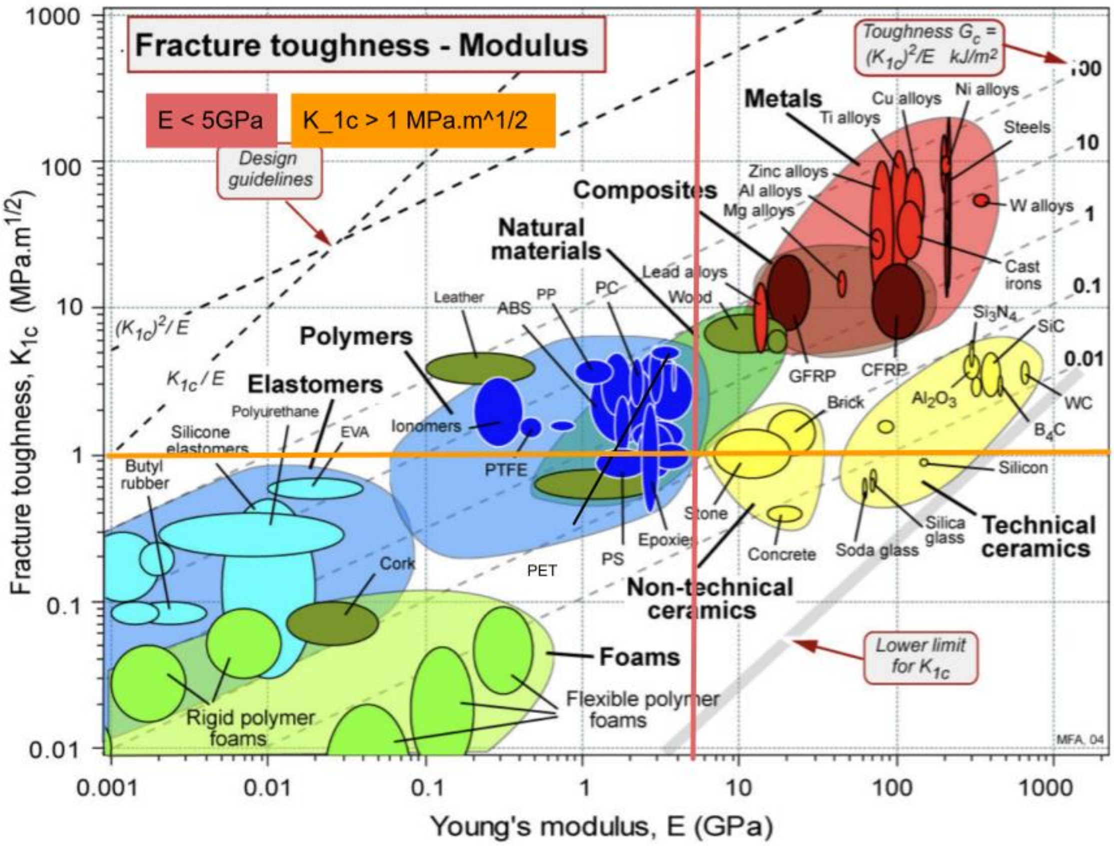


Figure 12: Ashby Chart for the Wave Spring

Both Young's modulus and Fracture toughness were important to prioritize, but as for which one was more important, that would be decided between one that has a low Young's Modulus (PTFE) and one that has a high Fracture Toughness (PET K_1c = 6.5MPa.m^½). After narrowing down to 2 materials; PTFE and PET, PTFE or PET were analyzed for the final material choice. After further analysis, PTFE was determined to be soft and slippery, traits that are directly opposite of the desired function. With a need of a material that is capable of being structurally sound PET was chosen for the material for the spring.

Injection molds are necessary for this part to get the necessary geometry. The molds will be slightly complex like the other molds as they require additional pins to be inserted before material flows. Material will also likely flow in at two different locations to make sure each corner of the spring is filled evenly. Injection molding PET will require a vendor to be careful with material temperature and state, but it is possible. C-PET (crystalline PET) and A-PET (amorphous PET) are both possible subtypes of PET that the spring can be made out of. C-PET will likely be used since there is no need to prioritize opaqueness or stiffness over strength. A comparison of the two materials can be found below with the relevant factors to highlight the choice of C-PET instead of A-PET.

Property / Aspect	Crystalline PET (C-PET)	Amorphous PET (A-PET)
Appearance	Opaque / milky white	Clear and transparent
Melt temperature	265 – 280 °C	250 – 270 °C
Heat resistance (HD T)	~150 °C	~70 – 80 °C
Toughness / Impact strength	Moderate	Higher (less brittle)
Stiffness / Strength	High	Moderate
Surface finish	Matte to semi-gloss	Glossy / high clarity
Common applications	Hot-fill containers, food trays, mechanical parts, housings	Clear packaging, clamshells, display parts
Cost of processing	Higher (energy + cycle time)	Lower

Table 3: Crystalline vs Amorphous PET Comparison

4 - Spring Dowels

The Spring Dowels will be used as both fasteners to the edge holders and wave spring and used in the hinge assembly of the light to eliminate the need for multiple fasteners.

Design Constraints:

- Needs not expand at a higher heat to make heatset inserts effective
- Strong enough to handle the cyclic loading put on each component they interact with.
- Needs to be rigid and not deform when pulled on.

To choose material for these, a material with a very high heat resistance (thermal conductivity and a high Young's Modulus are needed. The high heat resistance aimed for was greater than 10 W/mK, enough to not have to worry about the heatset inserts falling out. The Ashby chart with thermal conductivity and thermal diffusivity will be used, though thermal diffusivity itself is not a factor to consider.

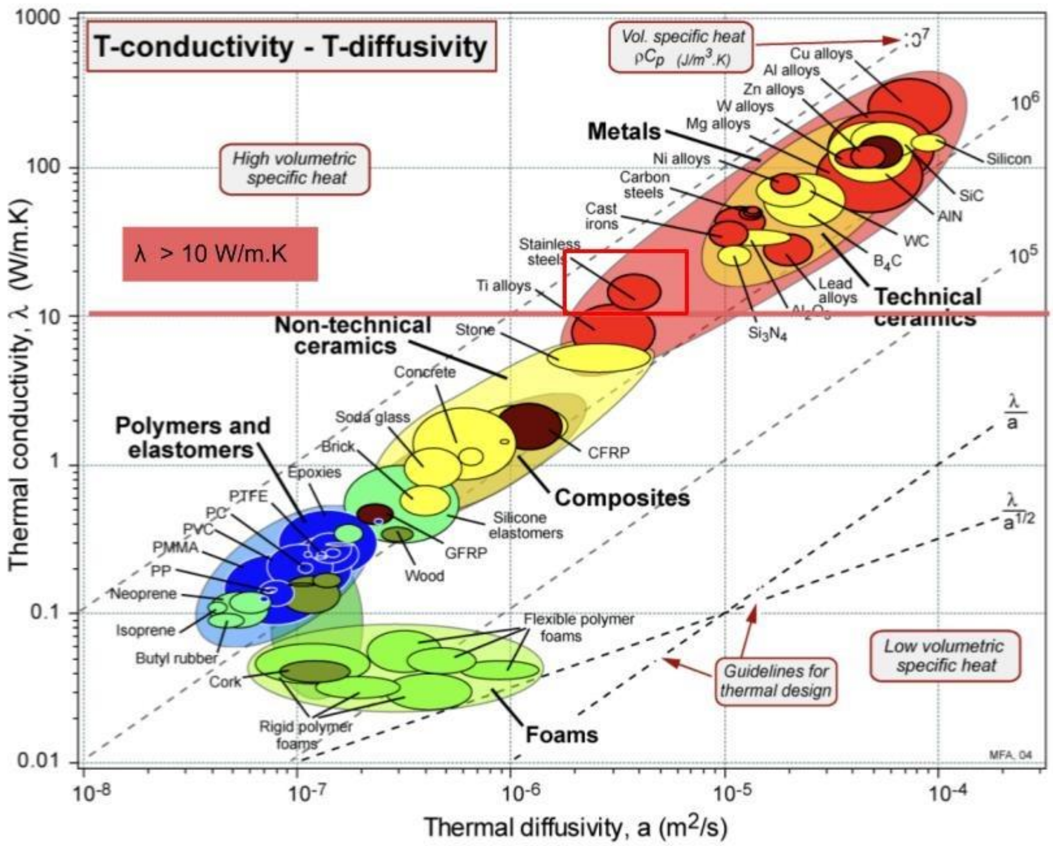


Figure 13 : Spring Dowels Ashby Chart

Cost is more impactful than thermal diffusivity, so the most cost-effective metal was chosen for the Spring Dowel. Steel, Cast Iron and Carbon Steel were the three materials chosen to compare, with Stainless Steel having the most reasonable cost per pound of around $.50/pound.

These specific dowels can be bought off the shelf in the exact dimensions required and should have the desired results.

9 - Double Sided Adhesive

The double-sided adhesive required material that could cheaply affix two plastics together. There was no requirement for the adhesive to hold a certain amount of force or resist heat or electricity. The most important consideration was a minimal profile and the adhesive to stick to plastics specifically. Currently, the only interface that requires the adhesive is between the PC of the upper edge holder and the battery holder, an off-the-shelf part. These two parts will need to be attached together some way, with access to the top of the battery holder readily accessible to remove and replace the battery. A weighted decision matrix was conducted to compare adhesive transfer tape [1], Loctite Plastic Bonding [3], and Gorilla Grip Clear [2].

Criteria	Weight	Transfer Tape	Plastics Bond	Gorilla Clear Grip
Bond Strength	0.2	5	7	5
Cure Time	0.3	9	4	5
Flexibility	0.2	5	8	8
Temperature Rating	0.15	8	5	3
Ease of Application	0.15	8	5	5
Score		7.1	5.7	5.3

Table 4: Weighted Decision Matrix for Adhesives

3M 9482PC adhesive was chosen as it can adhere PC Plastic to PC Plastic and rated the highest given the desired criteria. This can be bought off the shelf from McMaster-Carr cheaply and is ideal for the function.

14 - Arm

For the arm, it needed to swivel around a fastener point at the bottom of the lower edge holder. The arm also needed to sustain repeated elastic deformation to the point where the resulting tension could hold the pages of a book open.

Design Constraints:

- Needs to be a generally thin metal and have a low elastic modulus to allow for cyclic loading and bending.
- Needs to have enough strength that it does not plastically deform (high yield stress)
- Needs to be able to have a screw tightened into it without cracking.
- Needs to be able to slide past and around smooth plastic to not have additional forces needed to move it

With the need for a low elastic modulus (for metals, $E \approx 100$ GPa) and a high yield stress or strength ($\sigma \approx 100$ MPa), the Ashby chart of Modulus vs. Strength was chosen to determine an adequate material.

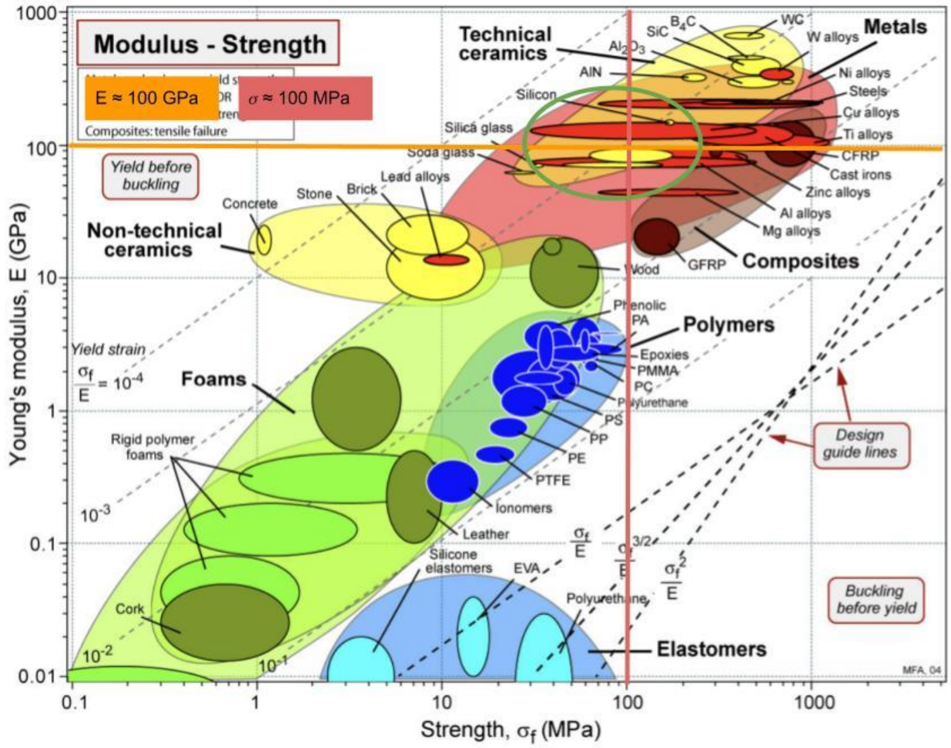


Figure 14: Ashby Chart for the Arm

The main materials that line up with the design requirements are copper alloys, steel, aluminum alloys and zinc alloys. Once general material was selected new key requirements were implemented including price, over molding capability, and manufacturing methods. Sheet metal is a natural choice for the arm since it can be stamped and bent. Additionally stainless steel is regularly processed into sheet metal and retains the low cost, high manufacturability desired. The manufacturing plan for the arm is to have stainless steel stamped, cut, and bent into shape. Its final finish is not as important as it will be covered by the rubber cover at the end of the manufacturing process.

15 - Rubber Cover

For the rubber cover that would be attached to the arm, a material that interacts and holds pages open without damaging the paper is necessary. The rubber cover will mainly interface with book paper but also interfaces with the arm and needs to securely fasten to the end.

<u>Design Constraints:</u>

- Needs to be able to slip fit on to the arm

- Needs to have enough texture/friction to hold onto a page
- Needs to be malleable enough to not damage a piece of paper
- Needs to not be a material that does not stick—generally a matte finish.

Given the design constraints, the desired material requirements are the ability to stretch and slip onto an arm and have high friction and texture. With this, a very low Young's Modulus (E < 10^-2 GPa) is necessary to allow stretching. Additionally, the material must be dense enough to support a constant half-pound force. The rubber cover cannot deform under the force applied to the page (ρ ≈ 1 Mg/m^3). The Ashby Chart of density vs Young's Modulus is what was used to decide the material.

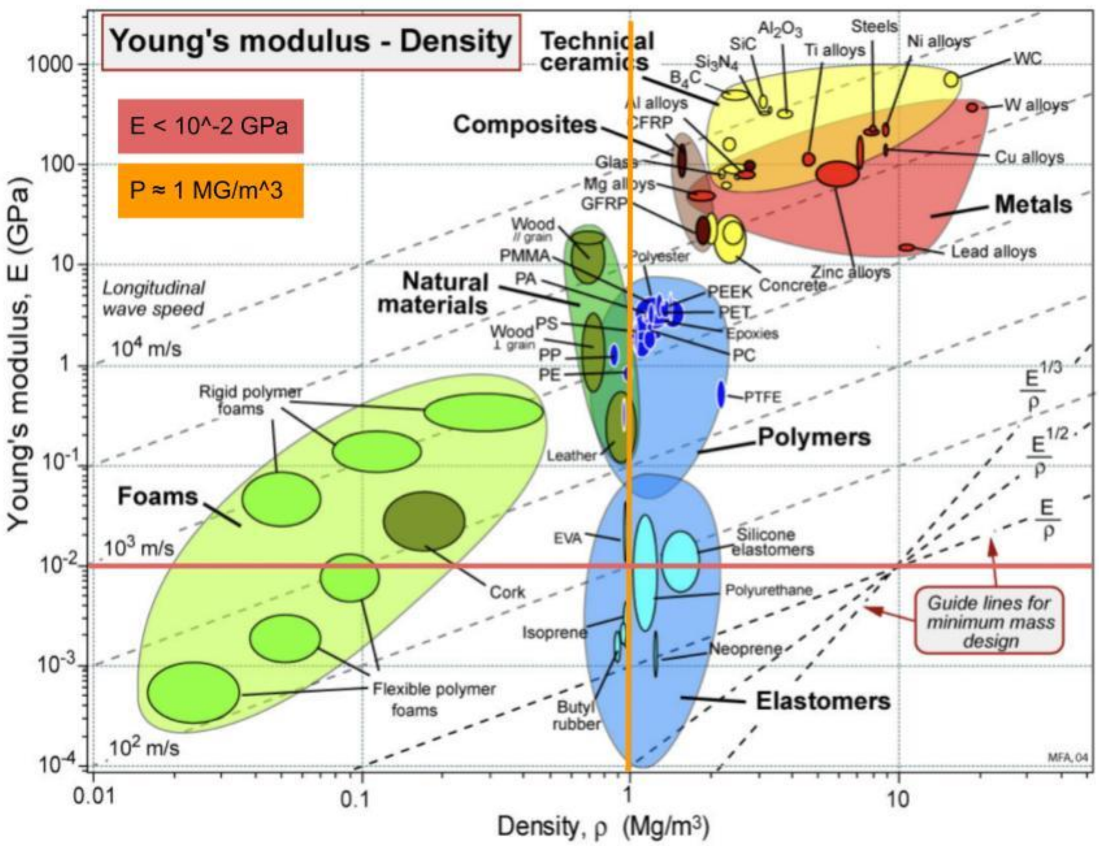


Figure 15: Ashby Chart for the Rubber Cover

Given the design constraints, an elastomer is the general family of material that best fits the requirements. Butyl rubber and Isoprene are the two closest materials to these requirements. Isoprene is a rather slippery material and would have the chance of slipping off of the arm, so butyl rubber is the ideal choice of material. However, the fabrication requires vulcanization which does not allow the material to be remolded after curing. A similar material, TPE, was found that has similar mechanical properties, but is much easier to process and over mold for ideal manufacturing methods. Table 5 below shows the similarities and differences of why TPE was chosen.

Category	**Butyl Rubber (IIR)**	**TPE (Thermoplastic Elastomer)**
Material Type	Thermoset elastomer	Thermoplastic elastomer
Density (Mg/m^3)	0.92–0.94	0.88–1.30
Young's Modulus (GPa)	0.002–0.007	0.001–0.6
Elasticity	High	High
Chemical Resistance	Excellent	Good–excellent (type-dependent)
Recyclability	No (cannot be remelted)	Yes (reprocessable)
Processing Method	Thermoset; requires vulcanization	Thermoplastic; melt-processable
Ability to Weld or Heat-Bond	Not possible	Yes; can be hot-welded and injection molded.

Table 5: Butyl Rubber vs TPE Comparison

TPE can and will be injection molded and pressed onto the bar. Its manufacturing process is ideal for the rubber cover, and it has desirable mechanical properties as well.

Finite Element Analysis

Wave Spring:

The purpose of the wave string is to provide a rigid structure to the BookNook, while allowing the user to adjust it to the proper width for the hard cover of their book. This means, some analysis will be needed to ensure the spring is able to stretch about an inch without seeing failure.

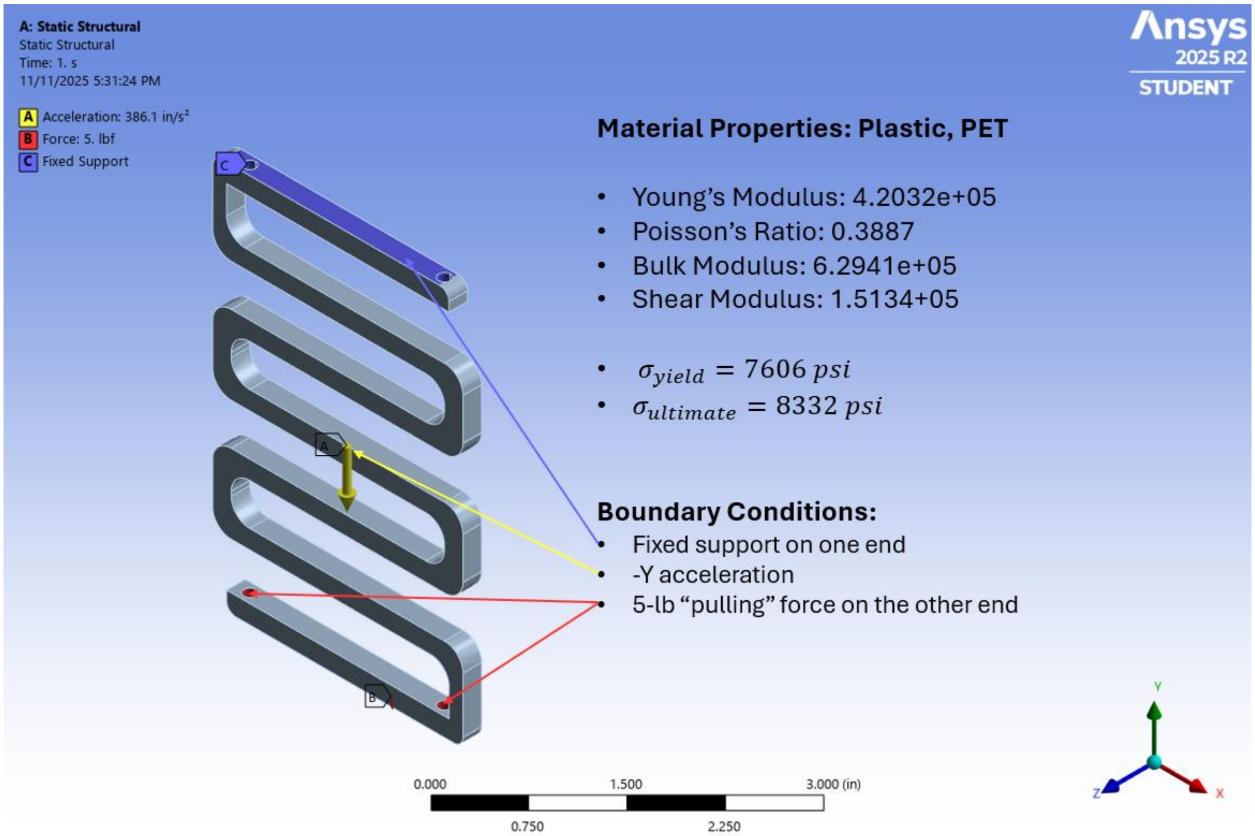


Figure 16: Model Set-up

As discussed in the material analysis, the wave spring will be made from PET Plastic, due to its ability to flex without immediate failure. For the Ansys model, the appropriate material properties of the spring, 7.6ksi yield stress and 8.3ksi ultimate stress were added.

For the boundary conditions shown in Figure 15, one end of the wave spring was fixed, and then a 5-lb force was applied to the other end to simulate the load path going through the wave spring when being stretched.

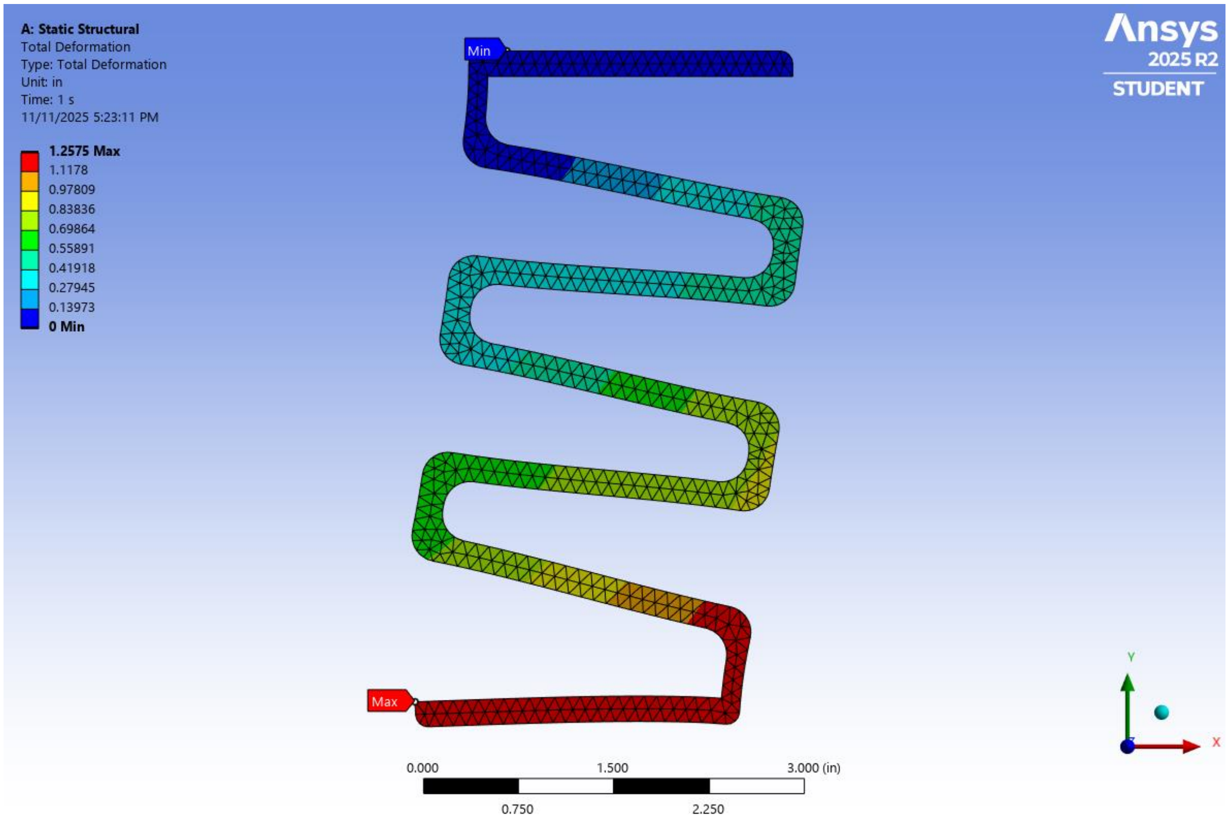


Figure 17: Wave Spring Deformation

In Figure 16 when 5 lbs. of force is applied to the wave spring it will deflect around 1.2 inches. The tetrahedral mesh may make the model appear more rigid than reality, so the deflection may even be ever so slightly larger than pictured.

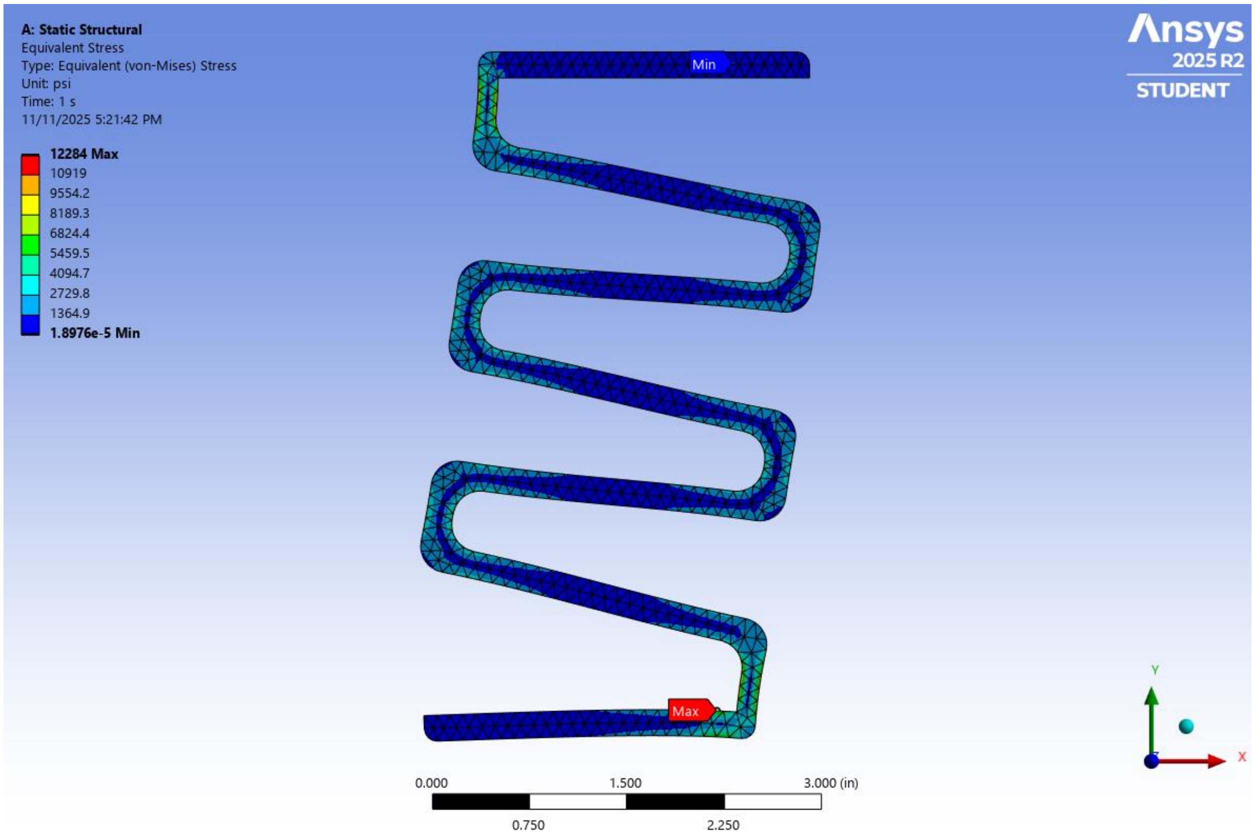


Figure 18: Wave Spring Stress

Looking at a von-mises stress model, the maximum stress predicted is well above the yield stress of the material, however much of the wave spring remains significantly lower than the yield stress as shown in Figure 17.

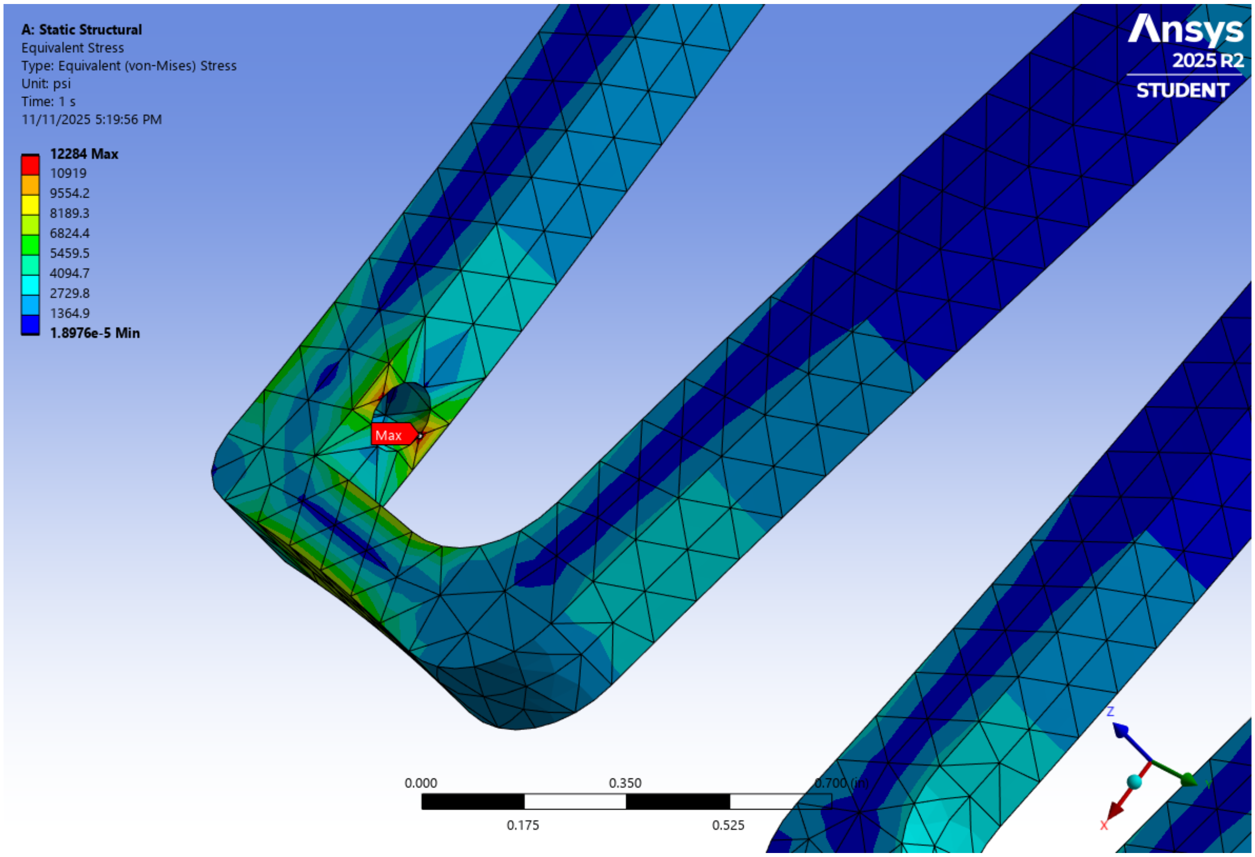


Figure 19: Stress Singularity in Model

Looking closer at the location where the maximum stress occurs (Figure 18), we see that it is where the fixed boundary condition was set in the model. For this reason, we believe that this localized stress concentration is caused by singularities in the model and does not simulate reality. Fixing the boundary gives the model conservatism, and in reality, the wave spring would be able to deflect a small amount, which would reduce overall stress at this location.

Tension Arm:

In the BookNook, the tension arm serves the purpose of holding down the pages through clamping force. It is meant to deform enough to be lifted over the pages, while remaining under the yield stress of the material (Stainless Steel 316 Sheet Metal). We created a simple analysis to determine the integrity of the tension arm.

Figure 20: Boundary Conditions of Tension Arm Model

The model has 2 very simple boundary conditions to simulate the stresses in the tension arm. A fixed support is added to the end where a fastener would normally sit and hold the wave spring to the lower edge holder. Then, a half-pound of force is applied to the other end, which mimics someone pulling the end up and setting it on a page.

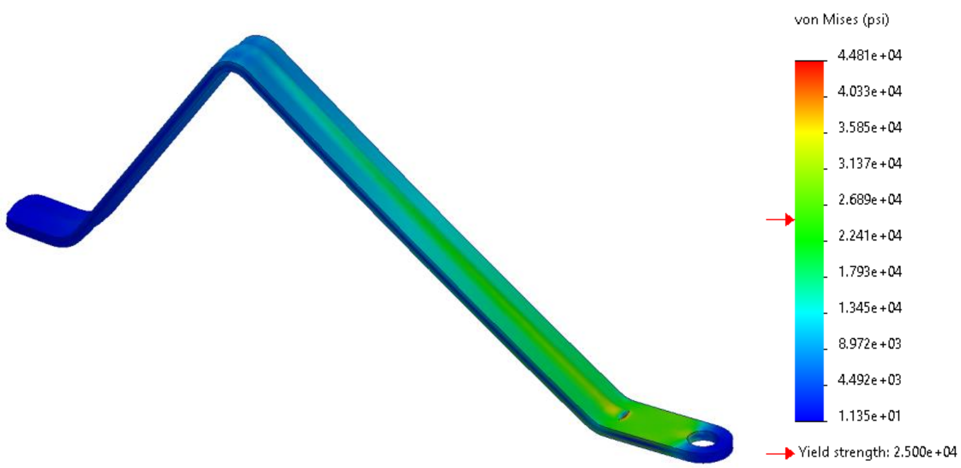


Figure 21: Tension Arm Stress

The results of the stress analysis shows that the part is below the yield stress, except for the singularity near the fixed support. This does not raise concerns as it is a properties of the boundary condition.

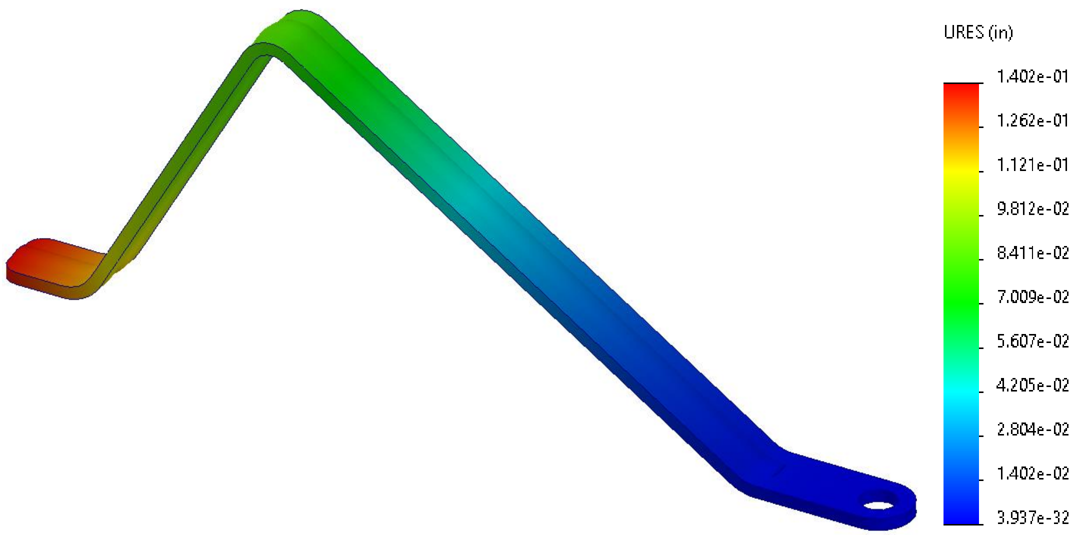


Figure 22: Tension Arm Deformation Results

Deformation in tension arm shows that the tension arm will deflect a little over a 1/8th of an inch when 0.5-lbf is applied to one end. This is a reasonable amount of deflection, as the point of the tension spring is to be able to lift onto the page to hold it down.

Design of Assembly Analysis

Design of assembly (DFA) analysis is a process used to illuminate shortcomings and struggles that an assembly line worker would encounter. The analysis focuses on issues that result from design decisions, or lack thereof. Using DFA, part redesigns can be made on the foundation of avoiding handling, orientation, and insertion issues while also reducing the number of secondary operations such as screwing and glueing. By applying this process, production output is increased while the cost efficiency of each assembly is significantly improved.

An analysis spreadsheet is used to categorize each component individually and determine what failure criteria they each meet. The original spreadsheet is shown below to create initial metrics that can be used to support the data influenced design changes that were made to the BookNook.

DFA Analysis Worksheet

Assembly Name: BookNook **Team:** 3 **Date:** 11/8/2025

If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.

Part		DFA Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations				
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (N_I)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
1	Upper edge holder	1	5	1	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	Lower edge holder	1	3	1	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	Wave spring	1	4	1	0	H	1	0	0	1	0	0	0	1	0	0	1	0	0	0	0
4	Spring dowels	4	1	1	0	L	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
5	Light	1	1	1	0	M	1	0	1	0	1	0	0	1	0	0	1	0	1	0	1
6	Battery Pack	1	1	0	0	M	1	0	1	0	0	0	0	1	0	0	0	0	1	0	0
7	Battery	2	1	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	Wires	2	2	1	0	L	2	0	0	1	1	0	0	0	0	0	0	0	0	0	0
9	Double Sided Adhesive	1	2	0	0	H	0	0	0	0	0	0	0	1	0	0	0	0	1	0	0
10	Hinge mount	1	4	1	0	M	1	0	1	0	0	0	1	1	0	0	0	0	0	0	0
11	Hinge swivel	1	2	0	0	M	0	0	1	0	0	0	1	1	0	1	0	0	0	0	1
12	Hinge rod	1	2	0	0	L	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
13	Hinge fasteners	2	2	0	1	L	0	0	0	0	0	0	0	1	0	0	1	1	0	0	0
14	Arm	1	2	1	0	L	1	0	0	0	0	0	0	1	0	0	0	0	0	0	1
15	Rubber cover	1	1	0	0	L	1	0	0	0	1	0	0	1	1	0	0	0	0	0	0
16	Fastener	1	2	0	1	L	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
17	Heat set inserts	4	2	0	0	L	2	0	0	0	0	0	1	1	1	0	0	0	0	1	0
	Totals	26	37	9	2	0	13	0	4	2	3	0	5	13	4	3	3	2	3	1	3
	Design for Assembly Metrics	31.01612484		34.6%	←Theor. Effy.	Pract. Effy.→	50.0%	0.44		0.56			2.78				1.33				
	Targets																				

Figure 23: DFA Initial Design

The key metrics for each category can be seen in the "Design for Assembly Metrics" row. Each number corresponds to the associated columns it encompasses. The metrics are based on a ratio between the number of issues compared to the theoretical minimum number of parts (TMP). The theoretical minimum number of parts is every part that is necessary to the function of the product. Any components that improve user function or could be changed to omit the part are not included. By comparing the TMP to the issues an objective measure of inefficiencies is created.

Practical Part Count

The initial DFA analysis included determining the TMP and practical minimum parts (PMP). To determine the parts necessary to the BookNook's function, functional requirements of holding a book, providing a reading light, and a way to hold the pages in place were kept in mind. To perform these three functions, any aesthetic components, user experience related parts, and any excessive components were eliminated to obtain a TMP of 9. Parts of the hinge assembly were removed as a light was the only necessary component for function. Parts like adhesives, heat sets, and any fasteners were removed as snap fits could make these parts obsolete. Finally, the battery pack was also excluded in the TMP since two wires could connect the batteries to light.

In the next analysis, parts were brought back if deemed practical to the design. This included the battery pack, a single piece of the hinge assembly to mount the light, and the rubber cover to increase friction for page holding. After reintroducing these components PMP was compared to the original total part number to evaluate a practical part efficiency of 50%. Using a guideline practical part efficiency of 60% highlighted the noticeable number of inefficiencies in the initial design. By reducing the number of unnecessary parts, the total cost and logistics can be minimized for a more efficient design.

Quality (Error Proofing)

Error proofing analysis looked to identify any components that can be assembled in the wrong orientation and any places where the wrong part could be placed. The baseline metric for the initial design was 0.44. The battery pack, light, and the hinge assembly were the only parts that could be assembled incorrectly. Without a discrete location for the battery pack, it could be adhered anywhere on the upper edge holder. Without a marker for the light, it suffers from the same issue. The hinge swivel and mount have no indicators for which side of each are supposed to be in line. Either of the components could be pinned upside down from the initial design intent.

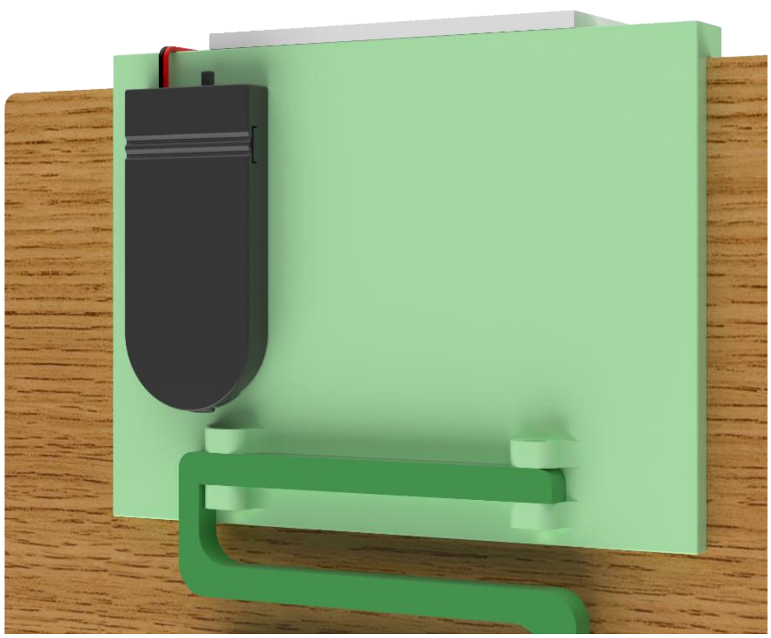


Figure 24: Battery pack without location marker

Figure 25: Hinge swivel without indication marker

The initial Quality metric is relatively low compared to metrics analyzed later. However, the analysis illuminated large potential issues in having a variable battery pack location. The necessity of each hinge assembly component was also brought into question. In summary, these findings emphasize the importance of adding orientation cues and location features to reduce ambiguity and strengthen the robustness of the assembly process.

Handling

Handling analysis considered the issues that an assembler might have dealing with the components. The criteria analyzed were identifying any parts that were sharp or fragile, any parts that may become nested or tangled when mass amounts of them are near each other, and any necessity for additional tools like tweezers or magnifying glasses that would be necessary to install the component.

An initial design metric of 0.56 was evaluated. This number is a good indicator that the initial design does not create many issues related to handling. Additionally, the nature of the components, like the size of the rubber cover, the fragility of the light, and the potential for tangled wave springs are mostly unavoidable. Alternate substances such as resin could be applied to the arms for increased friction but that would add an additional step to the assembly line ultimately creating a more difficult process. Removing the light or the wave spring is also not available as functionality would be forfeited. Due to the nature of the components causing handling issues no design changes were made to account for easier assembly.

Insertion (Locate and Secure)

The insertion analysis highlighted concerns that assemblers encounter placing and installing components. Examples of these issues include precise placements from small hole tolerances, parts being strenuous to insert, lack of visibility during insertion, or the need to hold down a component to install.

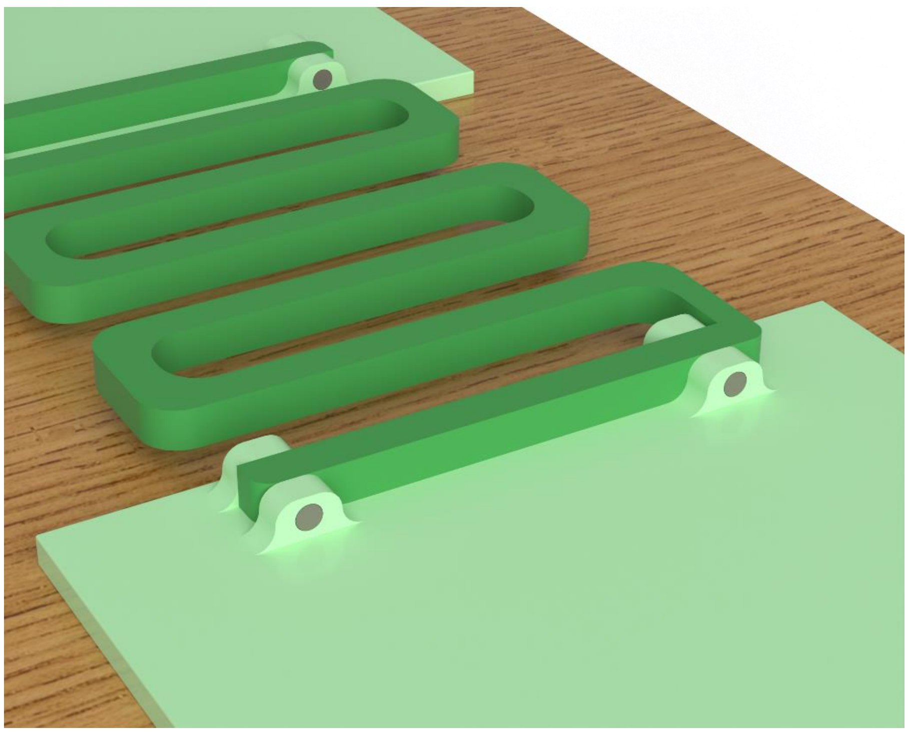


Figure 26: Dowel Pin View

The insertion metric was evaluated at 2.78, the highest metric on the DFA spreadsheet and had the most severe problems that would occur during assembly. Some key components that were largely contributing to this issue were the dowel pins. The dowel pins required press fitting into an 1/8" hole that made them have issues in each criterion evaluated in this section. Additionally, there are 5 of them that each require the same assembly procedure. The heat set inserts suffered similar issues, however visibility was maintained during their installation. While causing issues in assembly, they provide a reliable solution to allow the readjustment of the arms. Without reasonable changes to the heat set inserts new ways to fasten the wave spring and hinge assembly were identified as key areas for design changes.

Additional contributions to the high 2.78 metric stemmed from the necessity to hold down most of the parts during assembly. The lack of mass in the parts requires a securing force (holding down) to install many of the components. Potential modifications could include the creation of an assembly stand to hold the BookNook in place, but no direct design changes were made to accommodate the numerous places where the product must be held down. In summary, addressing these high-burden insertion steps through alternative fastening solutions and reduced-tolerance requirements will be critical for improving overall assembly efficiency.

Secondary Operations

Secondary operations analysis looks for any areas where a step past placing the part is needed. This can include screwing, glueing, soldering, welding, re-orienting the assembly, and testing. This analysis is extremely important to keep in mind for assembler's health. Repetitive motion from secondary operations can lead to potential health risks and should be avoided. One important note was that while the dowel pins did not require secondary operations the assemblers were determined to be at risk for repetitive motion from inserting the dowel pins. This consideration was considered when determining practical design changes.

The initial metric of 1.33 indicates a fair number of secondary operations being necessary in the assembly process. Each column in this section was comparable with no standout operation showing up repeatedly. Multiple parts require the adhesive to be applied to fasten them in place, heat must be applied to the heat inserts, three reorientations are required assuming an optimal assembly process, and the light fixture and arm must be tested to ensure functionality. Finding simpler ways to speed up the hinge assembly would cause the most significant decrease in secondary operations. Alternative fastening methods would also reduce the number of glueing instances necessary, however glueing is the simplest way to fasten the battery and light. Their flat rectangular surfaces work well with the thin strips of double-sided adhesive that are used in the product. Finding ways to reduce the number of parts could potentially reduce the number of reorientations and overall optimize the design. Testing is necessary for quality control and cannot be avoided unless outsourced.

The analysis determined that design changes to reduce testing and glueing would not be beneficial without causing other problems related to outsourcing and necessitating alternate secondary operations. Trying to eliminate components to reduce related to the hinge assembly would help to avoid unnecessary reorientations and screwing. This was targeted in design changes to alleviate unnecessary stress and excess movements during the assembly process.

DFA Conclusion

DFA analysis showed strong baselines regarding the error proofing and handling of the design, with low values of 0.44 and 0.56 respectively. However, the practical part efficiency of 50%, insertion of components, and secondary operations highlighted areas of improvement that could improve the assembly process. Design changes regarding these issues can be found in the "Design Revisions" in the next section.

Design Revisions

The design changes that were made reflect the inefficiencies discovered in the hinge assembly, the lack of assembler accommodations when installing parts, and the practicality of components based on prospective use. Each design change was made with the intent of reducing assembly or manufacturing complexity, while maintaining the functionality and user experience of the product. These can be summarized in the following four revisions:

Redesign Change 1: Hinge Assembly

The hinge assembly was originally intended to provide a pivoting light that could be adjusted to illuminate the page in various orientations. There was also potential for future designs to utilize this hinge for a fastening mechanism if the user had a BookNook on both sides of their book. While the hinge assembly showed great potential for user experience it consisted of parts that were expensive and ultimately deemed unnecessary. Additionally, the LED strip was the only component that was necessary to the function of the product. The additional plastic hinge mount, swivel, and fastener pin were all extra components that were adding to the material costs and assembling time without contributing towards the function.

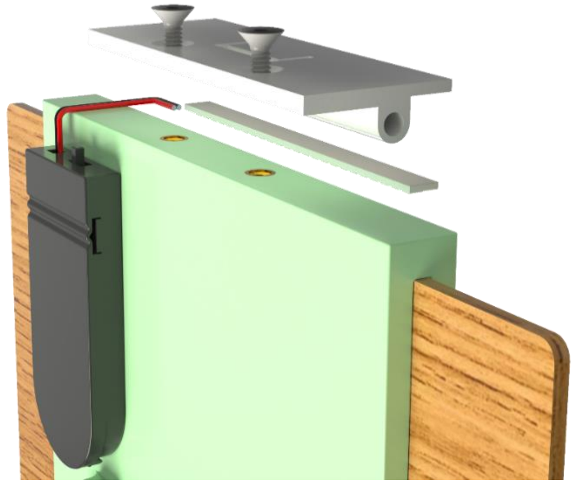


Figure 27: Hinge assembly to be omitted

The entire hinge assembly was omitted in favor of an additional extrusion that was added to the upper edge holder. This extrusion became the new mounting platform for the LED strip.

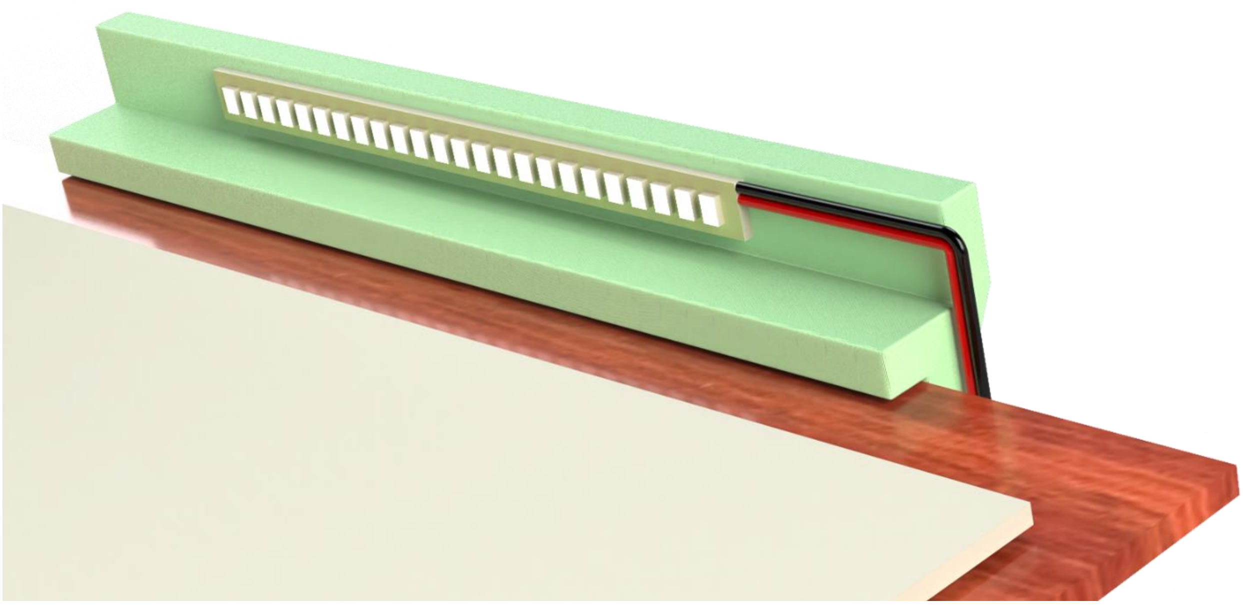


Figure 28: Hinge assembly after it is omitted

This simple addition was an effective way to omit four parts without greatly increasing the complexity of the upper edge holder mold. Additionally, the added extrusion did not require complex internal geometries since the wires connecting the battery pack to the LED strip are exposed. This design change helped to simplify DFA issues that arose from the dowel pin insertion and helped improve the practical part efficiency of the final design by eliminating non-essential parts.

Redesign Change 2: Snap Fit Wave Spring

The original wave spring was fastened with dowel pins on four locations. The upper and lower edge holder each featured two mounting holes that were part of the injection mold.

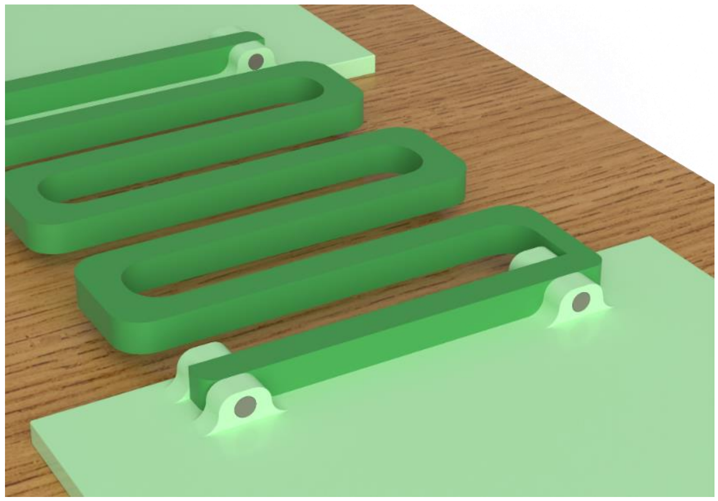


Figure 29: Insertion issues for dowel pins

During DFA analysis, the dowel pins were seen to cause four instances of insertion issues. Since the dowels were the most common part of the initial design, with 5 necessary for completion, they became a key component to target to reduce time to assemble. Targeting the dowel pins would also reduce repetitive motion that an assembler would need to endure to complete the product.

The redesign omitted the dowel pins fastening the wave spring by replacing the mounting holes with through holes. The wave spring itself was given four snap fits that are pressed into the upper and lower edge holders in the same location as they were previously fastened.

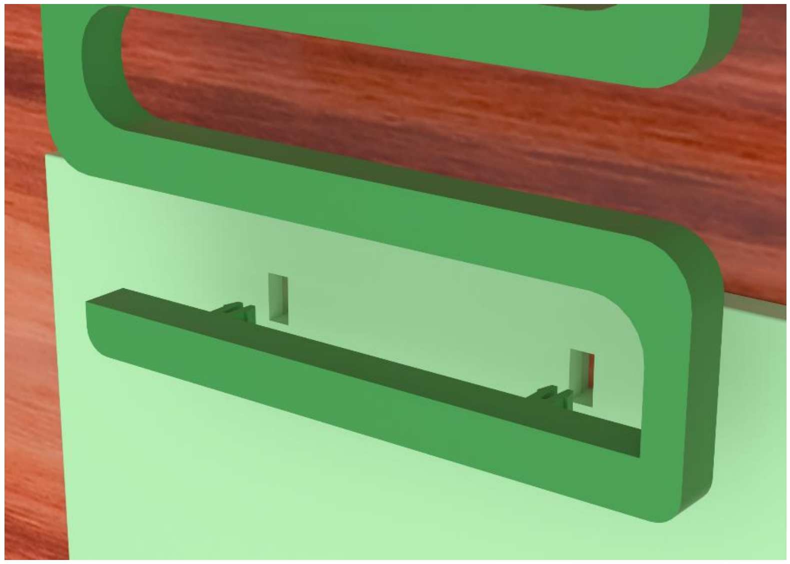


Figure 30: Snap fit replacement of dowels

The redesign accomplished a couple unseen benefits as well. By simplifying the fastening method, the assembly could be outsourced to the user. The simple snap fit completely removed the need for an assembler to connect the wave spring and simplified the packaging constraints now that the wave spring would be placed in packaging unattached. This redesign resulted in decreased DFA metrics for insertion and secondary operations as well as improving the practical part efficiency. Exact metrics are discussed in the "Adjusted DFA" section of the report.

Redesign Change 3: Battery Pack Alignment Markers

The initial design did not include any location markets for the battery pack's location. With the double sided adhesive able to be placed anywhere, the location of the battery pack was completely ambiguous. This shortcoming not only makes it impossible to consistently install the battery pack but also creates potential issues with wire lengths that must be installed as well. Without a consistent distance for the battery pack to the LED strip, there is no way to standardize the length of wire required to complete the assembly.

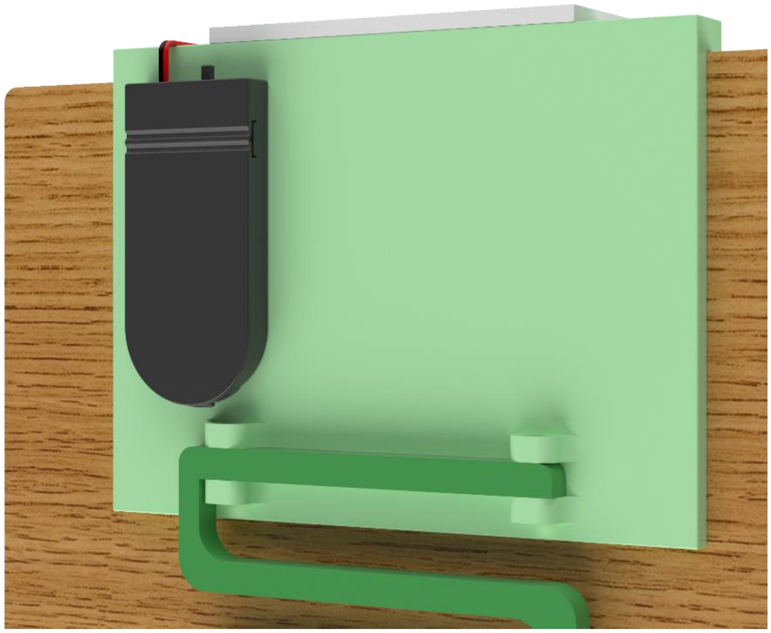


Figure 31: No markers for battery pack

To fix these issues, two small extrusions were added onto the upper edge holder. These corner markers provide guiderails for the corresponding corners of the battery pack.

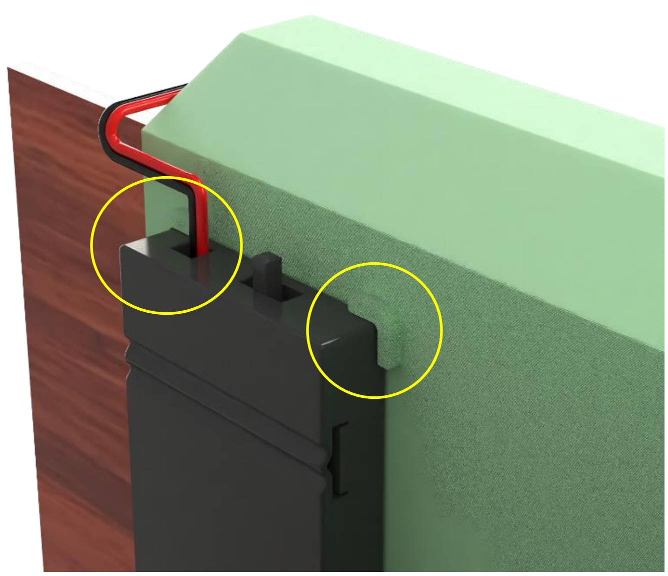


Figure 32: Markers added for battery pack to improve assembly

This design could still allow for the battery pack to be installed on its back, but an assumption was made that retaining access to insert batteries was enough to inform the assemblers of the correct side to adhere. The redesign adds slight complexity to the upper edge holder mold and increases the required amount of material as well. While these do not help support manufacturing processes the aid it provides assemblers greatly outweighs these downsides. By standardizing the location of each battery pack, the consistency of the product is maintained. Before this change, it was impossible to guarantee consistent placements and created the potential for snowballing complications related to the length of wire needed to connect the batteries to the light. This standardization was essential to ensure consistent quality in the eyes of the consumer, and to aid the assemblers in creating a repeatable process.

Adjusted Manufacturing Processes

A single manufacturing process was changed after the rigorous analysis. The initial plan to over mold the rubber cover onto the arm turned out to be a bottleneck in the production rate of the arm. By separating the rubber over mold from the arm's stamping process, the arm's throughput doubles. The

Because of this insight, the rubber cover's manufacturing process was changed to injection molding and assembled onto the arm after. This will require additional assembling time but greatly increase the number of parts produced per hour, reducing the machine wear, energy consumption, and number of machine operations.

Adjusted Bill of Materials

Adjusted Bill of materials to reflect components present in the final design.

Item #	Part #	Qty	Name	Material
1	1	1	Upper Edge Holder	PC Plastic
2	2	1	Lower Edge Holder	PC Plastic
3	3	1	Wave Spring	PET
4	5	1	LED	Off the Shelf
5	6	1	Battery Pack	Off the Shelf
6	7	1	Battery	Off the Shelf
7	8	2	Wires	Off the Shelf
8	9	2	Double Sided Adhesive	Off the Shelf
9	13	1	Page Arm	20G Stainless Steel
10	14	1	Rubber Cover	TPE
11	15	1	Arm Fastener	Off the Shelf
12	16	2	Heat Set Inserts	Off the Shelf

Table 6: Bill of Materials for Final Design

Adjusted DFA

DFA Analysis Worksheet																					
Assembly Name: BookNook										Team: 3									Date: 11/16/2025		
If the answer is Yes to any of the metrics or questions enter a 1. If the answer is No then enter 0. Each cell must have a number.																					
Part		DFA Complexity		Functional Analysis / Redesign Opportunity				Error Proofing		Handling			Insertion				Secondary Operations				
Part Number	Part Name	Number of Parts (N_p)	Number of Interfaces (N_I)	Theoretical Minimum Part	Part Can Be Standardized (if not already standard)	Cost (Low/Medium/High)	Practical Minimum Part	Assemble Wrong Part/ Omit Part	Assemble Part Wrong Way Around	Tangle, Nest, or Stick Together	Flexible, Fragile, Sharp or Slippery	Pliers, Tweezers, or Magnifying Glass Needed	Difficult to Align/ Locate	Holding Down Required	Resistance to Insertion	Obstructed Access/ Visibility	Re-orient Workpiece	Screw, Drill, Twist, Rivet, Bend, or Crimp	Weld, Solder, or Glue	Paint, Lube, Heat, Apply Liquid or Gas	Test, Measure or Adjust
1	Upper edge holder	1	5	1	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
2	Lower edge holder	1	3	1	0	H	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	Wave spring	1	4	1	0	H	1	0	0	1	0	0	0	1	0	0	0	0	0	0	0
5	Light	1	1	1	0	M	1	0	1	0	1	0	0	1	0	0	1	0	1	0	1
6	Battery Pack	1	1	0	0	M	1	0	0	0	0	0	0	1	0	0	1	0	1	0	0
7	Battery	1	1	1	0	M	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	Wires	2	2	2	0	L	2	0	0	1	1	0	0	0	0	0	0	0	0	0	0
9	Double Sided Adhesive	1	2	0	0	H	1	0	0	0	0	0	0	1	0	0	0	0	1	0	0
14	Arm	1	2	1	0	L	1	0	1	0	0	0	0	1	0	0	0	0	0	0	1
15	Rubber cover	1	1	0	0	L	1	0	0	0	1	0	0	1	1	0	0	0	0	0	0
16	Fastener	1	2	0	1	L	0	0	0	0	0	0	0	1	0	0	0	1	0	0	0
17	Heat set inserts	2	2	0	0	L	2	0	0	0	0	0	1	1	1	0	0	0	0	1	0
	Totals	14	26	8	1	0	13	0	2	2	3	0	1	8	2	0	2	1	3	1	2
	Design for Assembly Metrics	19.07878403		57.1%	Effy. Pract. Effy.→		92.9%	0.25		0.63			1.38				1.13				
	Targets																				

Figure 33: DFA Final Design

The initial DFA metrics had some noticeable issues that were addressed in the redesigns. After implementing the redesigns there was a significant change in the practical part efficiency, improving from 50% to an astounding 93%. This indicates that the changes optimized the parts used to ensure almost every part was necessary to the function or user experience. The error proofing metric was decreased by 43%. Error proofing was not targeted in design changes but naturally improved from standardizing the battery pack's location and omitting the hinge assembly. The handling metric increased by 13% due to the removal of parts that were not contributing to the metric. Since the components that were causing handling instances remained but overall, the number of parts decreased the metric increased. This is a reasonable tradeoff for the overall improvement in assembly. The insertion metric decreased by 50%, a large improvement from the initial value of 2.78. Removing the dowel pins in both the hinge assembly and wave spring assembly were pivotal to this decrease. While the final 1.38 value indicates there is still room for improvement, large contributors like the heat set insert are necessary to ensure the longevity of the BookNook. Finally, the secondary operations metric decreased by 15%. Since the need to test subsystems and adhesives still needed to be applied after redesigns, there were no significant improvements in this category.

The BookNook saw significant DFA improvements after implementing the design changes. The reduction of parts and key insights from the initial DFA analysis helped identify negative contributors, such as the dowel pins, to the assembly process. The changes made not only helped to reduce the assembly process but also greatly reduced the number of parts that contributed towards optimizing the logistics of creating the BookNook. Altogether, these improvements demonstrate a streamlined, more intentional design that not only enhances assembly efficiency but also strengthens the overall manufacturability and long-term viability of the BookNook.

Economic Analysis

This analysis assesses manufacturing costs of the initial BookNook design to determine unit cost, pricing considerations, and impact of design changes. Cost calculations for the BookNook assume production in a 25,000-square-foot manufacturing facility located in the United States, with all materials procured at current bulk purchase market rates. The analysis accounts for single-operator machine staffing throughout the production process. These assumptions are taken into account for cost using the unit cost eqaution shown in appendix B.

Table 7 summarizes the adjusted costs for the BookNook between the initial and final designs. The final design eliminated parts, adds snap fit and alignment features, impacting tooling cost.

Part #	Part Name	Initial Qty	Final Qty	Initial Unit Cost ($)	Final Unit Cost ($)
1	Upper Edge Holder	1	1	$1.27	$1.42
2	Lower Edge Holder	1	1	$1.29	$1.35
3	Wave Spring	1	1	$2.83	$2.99
4	Spring Dowels	5	-	$4.00	-
5	LED	1	1	$0.96	$0.96
6	Battery Pack	1	1	$3.29	$3.29
7	Battery	1	1	$2.01	$2.01
8	Wires	2	2	$0.10	$0.10
9	Double Sided Adhesive	2	2	$0.10	$0.10
10	Hinge Mount	1	-	$0.77	-
Part #	**Part Name**	**Initial Qty**	**Final Qty**	**Initial Unit Cost ($)**	**Final Unit Cost ($)**

11	Hinge Swivel	1	-	$0.82	-
12	Hinge Fasteners	1	-	$0.04	-
13	Page Arm	1	1	$0.75	$0.75
14	Rubber cover	1	1	$0.81	$0.81
15	Arm Fastner	1	1	$0.07	$0.07
16	Heat Set Inserts	4	2	$0.68	$0.34

Table 7: Unit cost comparison by part

Table 7 summarizes unit cost calculations for all manufactured and off-the-shelf parts in the assembly. Manufactured parts total $8.82 (33.8% of unit cost) while off-the-shelf components total $17.24 (66.2%). The wave spring carries the highest unit cost among manufactured parts at $2.99 for final design. This elevated cost stems from its complex geometry, which requires a collapsing core mold, precise ejection systems, and controlled cooling to prevent warpage. Despite its cost, the wave spring provides essential functionality to the BookNook and justifies the tooling investment.

Order of Magnitude Estimation

Table 8 compares calculated unit costs against order of magnitude estimation (OME) calculations for all manufactured parts. OME estimates manufacturing cost primarily on the mass of the part, approximately three times material cost.

Part #	**Part Name**	**Unit Cost ($)**	**OME Part Manufacturing Cost ($)**
1	Upper Edge Holder	$1.42	$0.44
2	Lower Edge Holder	$1.35	$0.48
3	Wave Spring	$2.99	$0.08
13	Page Arm	$0.75	$0.04
14	Rubber cover	$0.81	$0.00

Table 8: Unit cost compared to OME

Table 8 reveals a significant discrepancy between unit cost and OME values. This applied particularly to the wave spring at $2.99 vs $0.08, respectively. The order of magnitude estimates method does not capture cost related to complex tooling needed for the relatively light wave spring, recognizing of course that OME is a magnitude estimate. Unit cost calculations provide more accurate estimates for complex parts and therefore are used for pricing decisions.

Price Analysis	Initial Design	Final Design
Unit Cost	$19.79	$14.19
Break Even Price	$27.88	$20.10

Table 9: Price analysis for initial design verses final

The break-even price targets a 25% profit margin and 50,000-unit production run. Quality booklights currently on the market range from $12.99 to $34.99. At $20.10 minimum price, the BookNook sits in the middle of this range and offers additional page-holding functionality that justifies a mild premium. Targeting a profit margin of 50% lifts price to a still competitive mid-tier $23.65 break-even price.

Table 10 summarizes the impact of cost between design changes of the original and adjusted designs.

Metric	Original Design	Adjusted Design	Change
Total Parts	17	12	-5
Manufactured Parts Cost	$8.54	$7.32	-$1.22 (14.3%)
Off-the-Shelf Cost	$11.25	$6.87	-$4.38 (38.9%)
Unit Cost	$19.79	$14.19	-$5.60 (28.3%)
Break-Even Price	$27.88	$20.10	-$7.78 (27.9%)

Table 10: Cost impact of design change

The adjusted design achieves a 28.3% reduction in unit cost primarily through part elimination and enables targeting higher profit margins while maintaining a competitive price point against booklights with added functionality.

DFE (Design For Environment)

Before mass producing the product, it is extremely important to consider the environmental impact the BookNook will have.

The highlighted values in the carbon footprint per unit weight graph (Figure 33) represent the materials that the BookNook uses for its manufactured components.

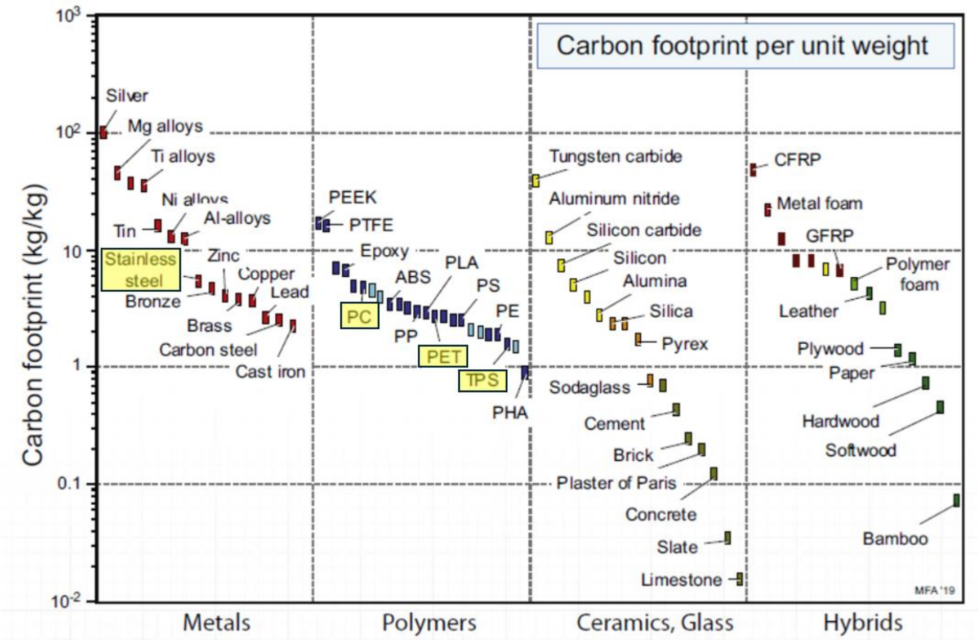


Figure 34: Carbon Footprint Per Unit Weight

By taking each material and finding the total amount of weight they make up, the total carbon footprint of the product can be calculated.

Material	Material Mass in Product (kg)	Carbon Footprint Per Unit Weight (kg/kg)	Carbon Footprint (kg)
PC Plastic	0.0771	~ 5	0.3855
PET	0.0318	~ 3	0.0954
TPS	0.000181	~ 1.5	0.0002715
Stainless Steel	0.00322	~ 8	0.02576
1 Unit Total			0.5069

Table 11: Carbon Footprint of BookNook

As shown in 11, the total carbon footprint of the manufactured components is roughly 0.5 kg. This means that if 50,000 units are produced, the carbon footprint will be about 25,000 kg.

Considering the lifecycle energy consumption (Figure 34), it can be noted that much of the energy consumed will be during the manufacturing process. The production of the part is significantly lower due to the design's simplicity, and the product requires very little energy to function when in use. Finally, the product is very lightweight (0.3 lbs. total mass) and compact, meaning it will not require significant energy once it is disposed of.

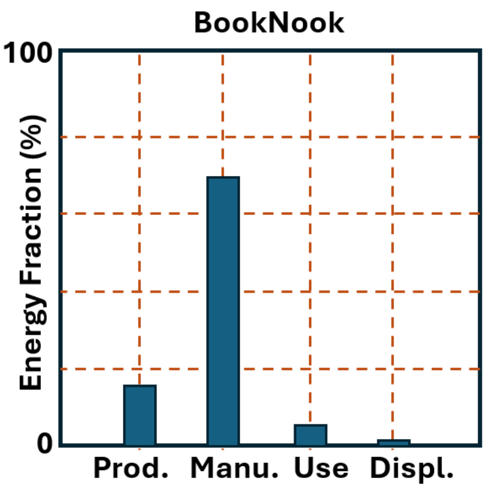


Figure 35: BookNook Lifecycle Energy Consumption

In summary, the materials selected in the BookNook fall into the lower range of carbon footprint. The lightweight design will be beneficial not only from a user standpoint, but also a transportation standpoint. Finally, apart from the stainless-steel tension arm, the material production energy for the plastics used are also on the lower end of the spectrum.

Deltas

As seen in the executive summary, the main aspects of our initial and final design changes can be seen below.

Design attempt	Practical Minimum Part %	DFA Complexity Metric	Unit Cost
Initial Design	50%	31.01	$19.79
Final Design	92.9%	19.1	$14.19
Improvement Spec.	85% increase	38.7% improvement	28.3% improvement

There is at least a 25% improvement for each major specification in the BookNook after the redesign, making effective use of materials, costs and assembly time by users and manufacturers.

Test Plan

Prior to packaging the BookNook for consumers to purchase, it is important to test that it functions as intended. When considering the components that may be most likely to fail, we can conclude that two tests need to be completed to verify the product.

1. Book Light

The lighting fixture shall be tested prior to adhering the LED strip to the top edge holder of the book. This test will ensure that the battery pack and wiring to the LED is appropriate, and the LED strip itself is not faulty. This test will be as simple as placing two verified CR2032 batteries into the battery pack, and flipping the switch to make sure the light turns on. In cases where the light turns on, but is dim or flickering, the LED strip will also be deemed faulty. Once the LED strips functionality is verified, it may be adhered to the top edge holder, and the assembly may resume. It is also important to note that the battery will be removed, and a plastic shim will be inserted into the battery cover to prolong battery life.

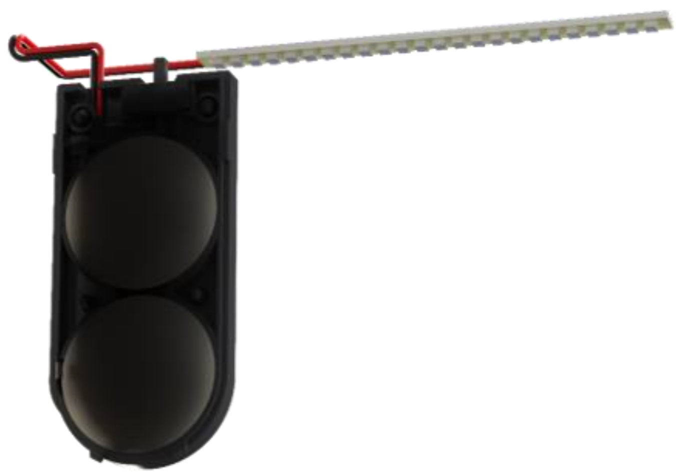


Figure 36: Battery Testing Setup

2. Wave Spring

The wave spring adds structure to the entire assembly, so it becomes another critical component that will need to be tested. For this test, the wave spring will be placed in a fixture that holds both ends. Then it will be pulled along its longitudinal direction under a load of 3 lbs. The test conductor shall visually inspect that the wave spring deflects at least 0.5 inches and shall also verify that it does not yield or fracture under the load. Once the wave spring is removed from the fixture, the test conductor shall visually inspect the wave spring for deformities. If there are permanent bends or disfigurations in the wave spring it shall be discarded as a faulty component.

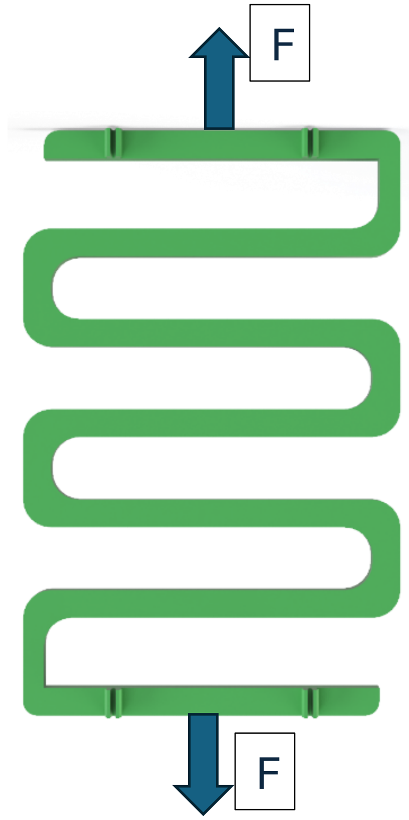


Figure 37: Wave Spring Test FBD

Conclusion

Through systematic DFA analysis and targeted redesigns, the product achieved substantial improvements across all key metrics: practical minimum part efficiency increased from 50% to 92.9%, the overall DFA metric improved by 38.7%, and unit cost decreased by 28.3% from $19.79 to $14.19.

The most impactful design changes centered on eliminating the hinge assembly and replacing dowel pin fasteners with snap fits. These modifications reduced the total part count from 17 to 12 components and simplified the assembly process and enabling user assembly. The addition of battery pack alignment markers addressed critical error-proofing concerns without significantly increasing mold complexity. Material selection ensured each component met its functional requirements while maintaining cost efficiency—PC plastic for structural rigidity and heat-set compatibility, PET for the wave spring's fatigue resistance, and TPE for the rubber cover.

At a break-even price of $20.10, the BookNook positions competitively within the $12.99 to $34.99 booklight market while offering differentiated functionality through its integrated page-holding arm. The 50,000-unit production run achieves economies of scale necessary for tooling amortization, with manufactured components comprising about 51.6% of unit cost. Environmental analysis confirms a modest carbon footprint of approximately 0.5 kg per unit, with most energy consumption occurring during the manufacturing phase rather than product use or disposal.

Appendices

Sources

[1] "7628A69 – McMaster-Carr," McMaster-Carr, accessed Nov. 19, 2025. [Online]. Available: https://www.mcmaster.com/7628A69/.

[2] "Glues & Epoxy – Ace Hardware," Ace Hardware, accessed Nov. 19, 2025. [Online]. Available: https://www.acehardware.com/departments/paint-and-supplies/tape-glues-and-adhesives/glues-and-epoxy/1663764?store=16046&gclsrc=aw.ds&gad_source=1&gad_campaignid=20158973401&gbraid=0AAAADtqLJFOlknP14hG36f5Lifl6FFAQ&gclid=CjwKCAiA8vXIBhAtEiwAf3B-g5jg6WttdnTnBt2JqEvdCG1u8M16v1i8PRFCDv5hm78_MlybsCP1RoCEOIQAvD_BwE.

[3] "Loctite Plastic Bonding System High Strength Cyanoacrylate Clear Plastic Bonder 4 gm – Ace Hardware," Ace Hardware, accessed Nov. 19, 2025. [Online]. Available: https://www.acehardware.com/departments/paint-and-supplies/tape-glues-and-adhesives/glues-and-epoxy/1008713?store=16046&clientId=550c37c9ba3d0621f829ae51&gclsrc=aw.ds&gad_source=1&gad_campaignid=21255784021&gbraid=0AAAADtqLJG-2eOCd56dvJY3CJJiGnCEN&gclid=CjwKCAiA8vXIBhAtEiwAf3B-g5PtHOqif01ATL5K1rL78oF7e778TbOJ9QzTHc8wOzb2MjPp86BBSRoCth8QAvD_BwE.

If you like, I can pull metadata like publisher dates or authors (if available) to make the citations more complete.

Appendix B: Cost equations and expanded calculations

To perform cost analysis the unit cost equation below was used in conjunction with the assumptions stated as the top of the economic analysis section.

$$C_U = \frac{(mC_m)}{1-f} + \frac{C_w}{n\prime} + \frac{C_t k}{n} + \left(\frac{1}{n'}\right)\left(\frac{C_e}{Lt_{wo}}\right)(q) + \frac{C_{OH}}{n\prime} \quad (1)$$

$$C_m: Material\ Cost$$

$$C_w: Labor\ Cost$$

$$C_t: Tooling\ Cost$$

$$C_e: Capital\ Cost$$

$$C_{OH}: Factory\ Overhead\ Cost$$

$$C_m: Material\ Cost$$

$$t_{wo}: Capital\ Write\ Off\ Time$$

$$L: Load\ Fraction$$

$$f: Fraction\ Of\ Scrap$$

$$n: Tooling\ Life$$

$$n': Production\ Rate$$

Expanded cost calculations for initial and final designs are attached below.

Calculated Cost Initial, Calculated Cost Final

CAD Drawings

Calculated Cost Initial Design

Part			Material Cost C_M				Labor Cost C_L			Tooling Cost C_T				Equipment Cost C_E						Overhead Cost C_OH			Total unit cost
			Weight of Material	Cost of Material	Fraction of scap material	Material Cost	Hourly Cost of wages	Production rate	Labor Cost	Cost of making tooling	# of Parts	Tooling wear factor (k=n/life of tool)	Tooling Cost	Cost of capital equipment	Production rate	Load factor, Fractional time equipment of productive	Capitol write off-time	Fraction of equipment sharing between products	Equipment Cost	Overhead Hourly	Production rate n'	Overhead Cost	Unit Cost
Part #	Part Number	Qty	m (lb)	c_m ($/lb)	(f)	C_M ($)	c_w ($/hr)	n' (units/hour)	C_L ($)	c_t ($)	n (units)	k	C_T ($)	c_e ($)	n' (units/hour)	L	t_wo (yrs)	q	C_E ($)	c_oh ($/hr)	(units/hour)	C_OH ($)	($/unit)
1	Upper Edge Holder	1	0.09073	1.47	0.1	$0.15	12	90	$0.13	$11,000.00	50000	2	$0.44	$50,000.00	90	0.3	3	0.35	$0.11	40	90	$0.44	$1.27
2	Lower Edge Holder	1	0.09933	1.47	0.1	$0.16	12	90	$0.13	$11,000.00	50000	2	$0.44	$50,000.00	90	0.3	3	0.35	$0.11	40	90	$0.44	$1.29
3	Wave Spring	1	0.039425	0.64	0.12	$0.03	12	48	$0.25	$38,000.00	50000	2	$1.52	$50,000.00	48	0.3	3	0.35	$0.20	40	48	$0.83	$2.83
4	Spring Dowels	5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$4.00
5	LED	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.96
6	Battery Pack	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$3.29
7	Battery	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$2.01
8	Wires	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.10
9	Double sided adhesive	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.10
10	Hinge Mount	1	0.00688	1.47	0.15	$0.01	12	130	$0.09	$7,000.00	50000	2	$0.28	$50,000.00	130	0.3	3	0.35	$0.07	40	130	$0.31	$0.77
11	Hinge Swivel	1	0.00301	1.47	0.15	$0.01	12	120	$0.10	$7,500.00	50000	2	$0.30	$50,000.00	120	0.3	3	0.35	$0.08	40	120	$0.33	$0.82
12	Hinge Fasteners	1	-	-		-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.04
13	Page Arm	1	0.00858	0.95	0.5	$0.02	14	4800	$0.00	$18,000.00	50000	2	$0.72	$50,000.00	4800	0.3	3	0.35	$0.00	40	4800	$0.01	$0.75
14	Rubber cover	1	0.0004	2.71	0.1	$0.00	12	250	$0.05	$14,000.00	50000	2	$0.56	$50,000.00	250	0.3	3	0.35	$0.04	40	250	$0.16	$0.81
15	Arm Fastner	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.07
16	Heat Set Inserts	4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	$0.68
																						Product Total Cost ($)	$19.79

Simplified Cost Calculated Sheet

Part #	Part Name	Qty	Material Cost C_M ($)	Labor Cost C_L ($)	Tooling Cost C_T ($)	Equipment Cost C_E ($)	Overhead Cost C_OH ($)	Unit Cost C_U ($)
1	Upper Edge Holder	1	$0.15	$0.13	$0.44	$0.11	$0.44	$1.27
2	Lower Edge Holder	1	$0.16	$0.13	$0.44	$0.11	$0.44	$1.29
3	Wave Spring	1	$0.03	$0.25	$1.52	$0.20	$0.83	$2.83
4	Spring Dowels	5	-	-	-	-	-	$4.00
5	LED	1	-	-	-	-	-	$0.96
6	Battery Pack	1	-	-	-	-	-	$3.29
7	Battery	1	-	-	-	-	-	$2.01
8	Wires	1	-	-	-	-	-	$0.10
9	Double sided adhesive	1	-	-	-	-	-	$0.10
10	Hinge Mount	1	$0.01	$0.09	$0.28	$0.07	$0.31	$0.77
11	Hinge Swivel	1	$0.01	$0.10	$0.30	$0.08	$0.33	$0.82
12	Hinge Fasteners	1	-	-	-	-	-	$0.04
13	Page Arm	1	$0.02	$0.00	$0.72	$0.00	$0.01	$0.75
14	Rubber cover	1	$0.00	$0.05	$0.56	$0.04	$0.16	$0.81
15	Arm Fastner	1	-	-	-	-	-	$0.07
16	Heat Set Inserts	4	-	-	-	-	-	$0.68
							Prodcut Unit C	$19.79

Order of Magnitude Cost Estimate

Part #	Part Number	Material	Manufacturing	Sale Price ($)
1	Upper Edge Holder	$0.15	$0.44	$1.32
2	Lower Edge Holder	$0.16	$0.48	$1.45
3	Wave Spring	$0.03	$0.08	$0.25
10	Hinge Mount	$0.01	$0.03	$0.10
11	Hinge Swivel	$0.01	$0.02	$0.05
14	Page Arm	$0.01	$0.04	$0.11
15	Rubber cover	$0.00	$0.00	$0.01

Break Even Point Q_B

Part #	Part Number	Total Varaible Cost, v ($)	Total Variable Fixed Costs ($)	Profit ($)	Break Even Price (50,000 units)
1	Upper Edge Holder	$1.27	$9,833.33	$1.59	$1.79
2	Lower Edge Holder	$1.29	$9,833.33	$1.61	$1.81
3	Wave Spring	$2.83	$9,833.33	$3.54	$3.74
4	Spring Dowels	$4.00	$9,833.33	$5.00	$5.20
5	LED	$0.96	$9,833.33	$1.20	$1.40
6	Battery Pack	$3.29	$9,833.33	$4.11	$4.31
7	Battery	$2.01	$9,833.33	$2.51	$2.71
8	Wires	$0.10	$9,833.33	$0.13	$0.32
9	Double sided adhesive	$0.10	$9,833.33	$0.13	$0.32
10	Hinge Mount	$0.77	$9,833.33	$0.96	$1.16
11	Hinge Swivel	$0.82	$9,833.33	$1.02	$1.22
12	Hinge Fasteners	$0.04	$9,833.33	$0.05	$0.25
13	Page Arm	$0.75	$9,833.33	$0.94	$1.13
14	Rubber cover	$0.81	$9,833.33	$1.01	$1.21
15	Arm Fastner	$0.07	$9,833.33	$0.09	$0.28
16	Heat Set Inserts	$0.68	$9,833.33	$0.85	$1.05
				Product Break Even Price	$27.88

Fixed Costs, f ($)				
G&A Exp	Depreciation (	Factory Expens	Sales on OH ($)	Total Variable Fixed Costs ($)
######	$833.33	$2,000.00	$4,000.00	$9,833.33

Calculated Cost Final Design

Part			Material Cost C_M				Labor Cost C_L			Tooling Cost C_T				Equipment Cost C_E						Overhead Cost C_OH			Unit Cost
Part Number	Part Number	Qty	Weight of Material m (lb)	Cost of Material c_m ($/lb)	Fraction of scap material (f)	Material Cost C_M ($)	Hourly Cost of wages c_w ($/hr)	Production rate n' (units/hour)	Labor Cost C_L ($)	Cost of making tooling c_t ($)	# of Parts n (units)	Tooling wear factor (k=n/life of tool) k	Tooling Cost C_T ($)	Cost of capital equipment c_e ($)	Production rate n' (units/hour)	Load factor, Fractional time equipment of productive L	Capitol write off-time t_wo (yrs)	Fraction of equipment sharing between products q	Equipment Cost C_E ($)	Overhead Hourly c_oh ($/hr)	Production rate n' (units/hour)	Overhead Cost C_OH ($)	Cost per unit off the shelf ($/unit)
1	Upper Edge Holder	1	0.09503	1.47	0.1	$0.16	12	90	$0.13	$14,500.00	50000	2	$0.58	$50,000.00	90	0.3	3	0.35	$0.11	40	90	$0.44	$1.42
2	Lower Edge Holder	1	0.09933	1.47	0.1	$0.16	12	90	$0.13	$12,500.00	50000	2	$0.50	$50,000.00	90	0.3	3	0.35	$0.11	40	90	$0.44	$1.35
3	Wave Spring	1	0.039425	0.64	0.12	$0.03	12	48	$0.25	$42,000.00	50000	2	$1.68	$50,000.00	48	0.3	3	0.35	$0.20	40	48	$0.83	$2.99
5	LED	1													0								$0.96
6	Battery Pack	1													0								$3.29
7	Battery	1													0								$2.01
8	Wires	1													0								$0.10
9	Double sided adhesive	1													0								$0.10
13	Page Arm	1	0.00858	0.95	0.5	$0.02	14	4800	$0.00	$18,000.00	50000	2	$0.72	$50,000.00	4800	0.3	3	0.35	$0.00	40	4800	$0.01	$0.75
14	Rubber cover	1	0.0004	2.71	0.1	$0.00	12	250	$0.05	$14,000.00	50000	2	$0.56	$50,000.00	250	0.3	3	0.35	$0.04	40	250	$0.16	$0.81
15	Arm Fastner	1																					$0.07
16	Heat Set Inserts	2																					$0.34
																						Product Total Cost ($)	$14.19

Simplified Cost Calculated Sheet

Part #	Part Name	Qty	Material Cost C_M ($)	Labor Cost C_L ($)	Tooling Cost C_T ($)	Equipment Cost C_E ($)	Overhead Cost C_OH ($)	Unit Cost C_U ($)
1	Upper Edge Holder	1	$0.16	$0.13	$0.58	$0.11	$0.44	$1.42
2	Lower Edge Holder	1	$0.16	$0.13	$0.50	$0.11	$0.44	$1.35
3	Wave Spring	1	$0.03	$0.25	$1.68	$0.20	$0.83	$2.99
5	LED	1	-	-	-	-	-	$0.96
6	Battery Pack	1	-	-	-	-	-	$3.29
7	Battery	1	-	-	-	-	-	$2.01
8	Wires	1	-	-	-	-	-	$0.10
9	Double sided adhesive	1	-	-	-	-	-	$0.10
13	Page Arm	1	$0.02	$0.00	$0.72	$0.00	$0.01	$0.75
14	Rubber cover	1	$0.00	$0.05	$0.56	$0.04	$0.16	$0.81
15	Arm Fastner	1	-	-	-	-	-	$0.07
16	Heat Set Inserts	2	-	-	-	-	-	$0.34

Order of Magnitude Cost Estimate

Part Number	Part Number	Material ($)	Manufacturing (	Sale Price ($)
1	Upper Edge Holder	$0.15	$0.46	$1.38
2	Lower Edge Holder	$0.16	$0.48	$1.45
3	Wave Spring	$0.03	$0.08	$0.25
14	Page Arm	$0.01	$0.04	$0.11
15	Rubber cover	$0.00	$0.00	$0.01

Break Even Point Q_B

Part Number	Part Number	Total Varaible Cost, v ($)	Total Variable Fixed Costs ($)	Profit ($)	Break Even Price (50,000 units)
1	Upper Edge Holder	$1.42	$9,833.33	$1.92	$2.12
2	Lower Edge Holder	$1.35	$9,833.33	$1.82	$2.02
3	Wave Spring	$2.99	$9,833.33	$4.04	$4.24
5	LED	$0.96	$9,833.33	$1.30	$1.49
6	Battery Pack	$3.29	$9,833.33	$4.44	$4.64
7	Battery	$2.01	$9,833.33	$2.71	$2.91
8	Wires	$0.10	$9,833.33	$0.14	$0.33
9	Double sided adhesive	$0.10	$9,833.33	$0.14	$0.33
13	Page Arm	$0.75	$9,833.33	$1.01	$1.21
14	Rubber cover	$0.81	$9,833.33	$1.09	$1.29
15	Arm Fastner	$0.07	$9,833.33	$0.09	$0.29
16	Heat Set Inserts	$0.34	$9,833.33	$0.46	$0.66
				Product Break Even Price	$21.52

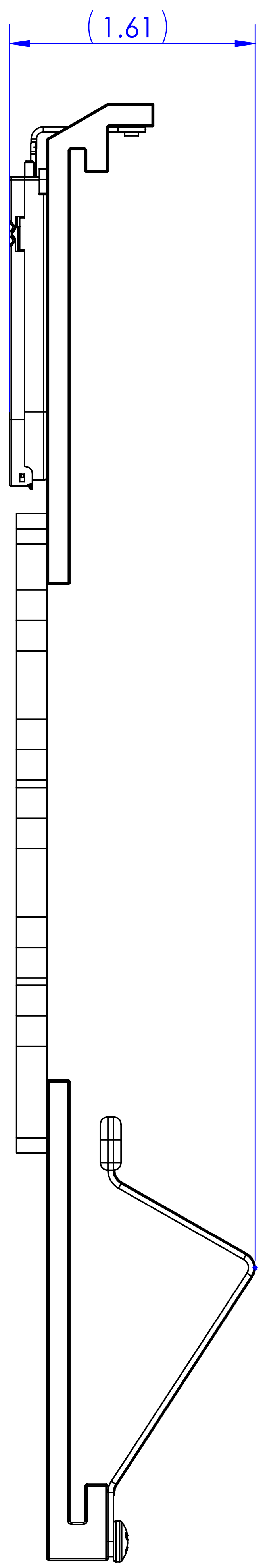
Fixed Costs, f ($)				
G&A Expense	Depreciation ($	Factory Exper	Sales on OH (	Total Variable Fixed Costs ($)
$3,000.00	$833.33	$2,000.00	$4,000.00	$9,833.33

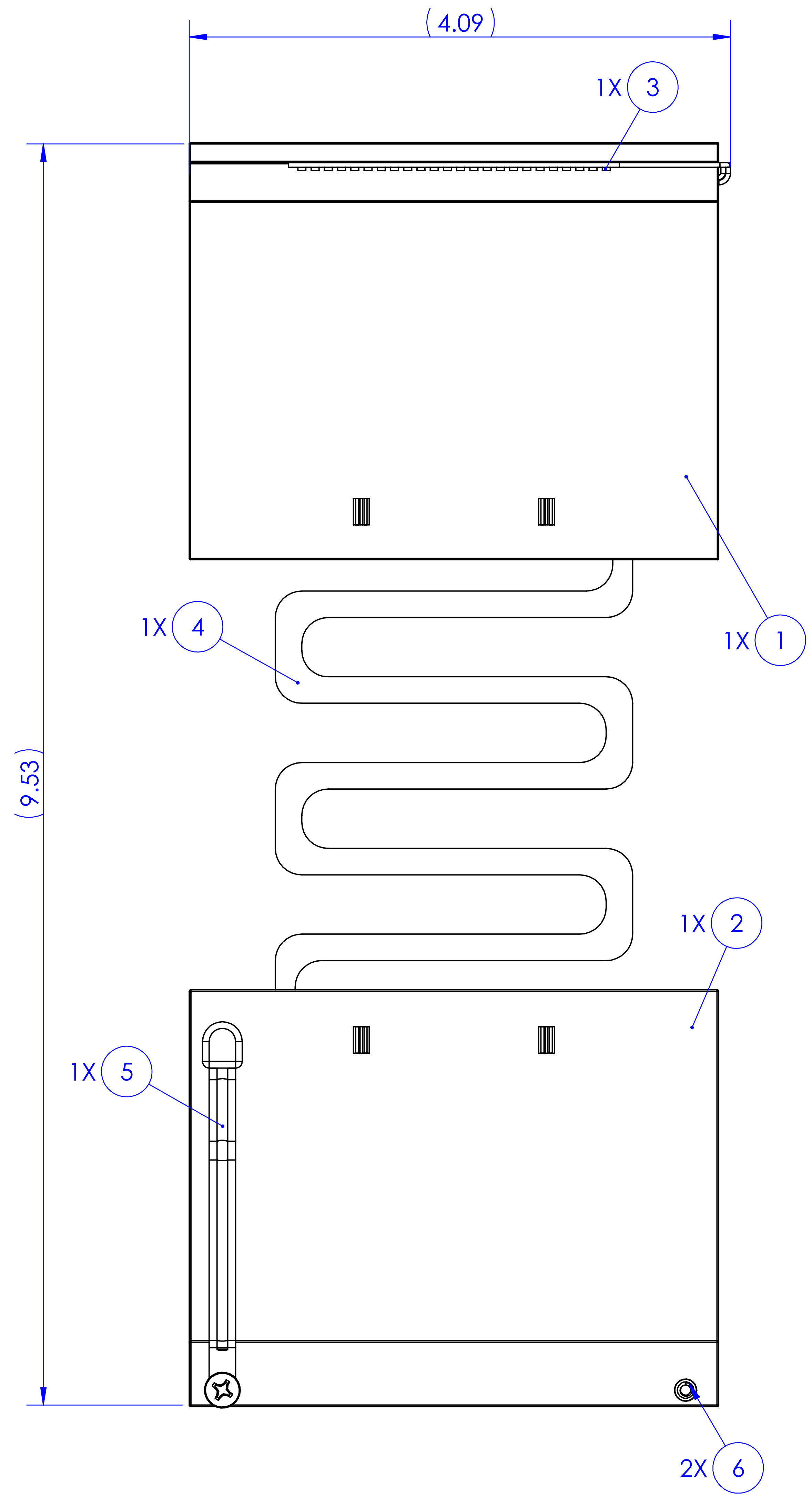
NOTES:

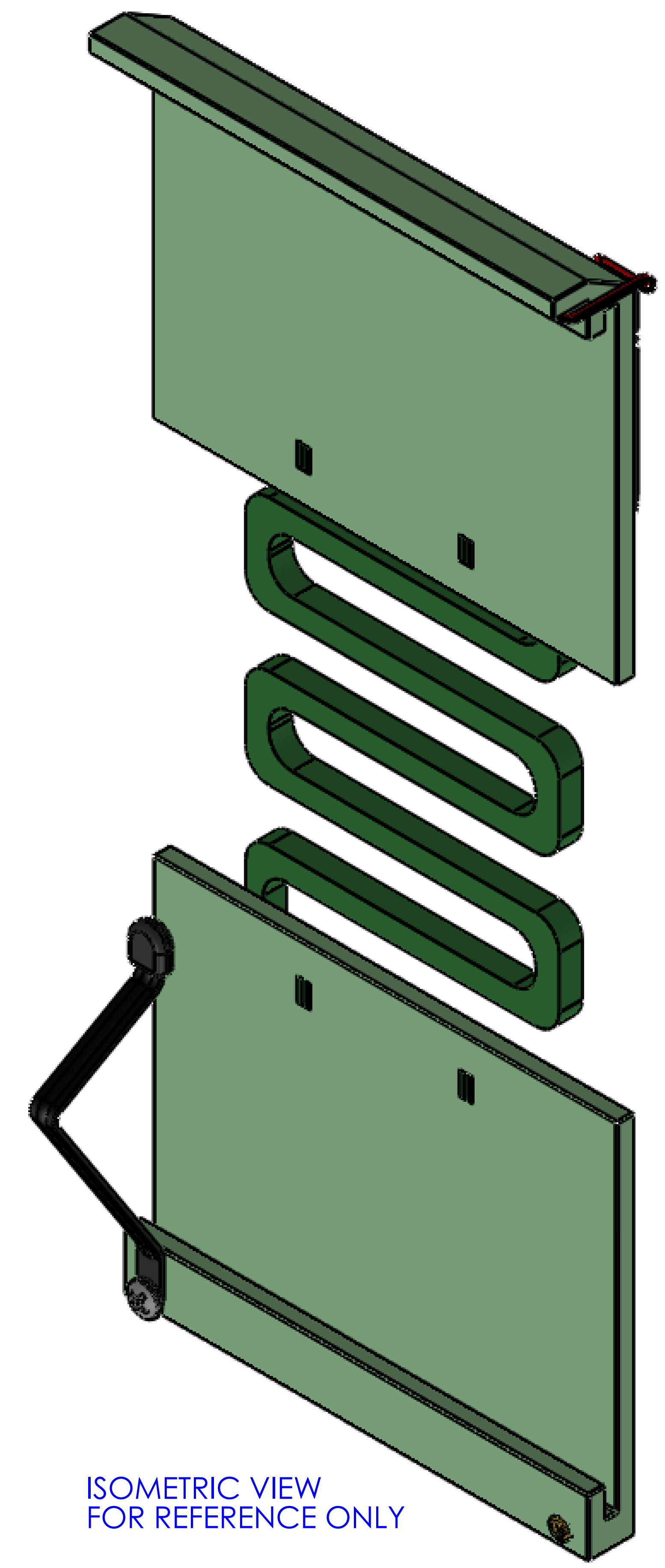
1. FULL ASSEMBLY WEIGHT: 0.3 LBS

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/17/2025

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	1 - UpperEdgeHolder		1
2	2 - LowerEdgeHolder		1
3	LightAssy		1
4	3 - WaveSpring		1
5	TensionArmAssembly		1
6	93365A120	Tapered Heat-Set Inserts for Plastic	2





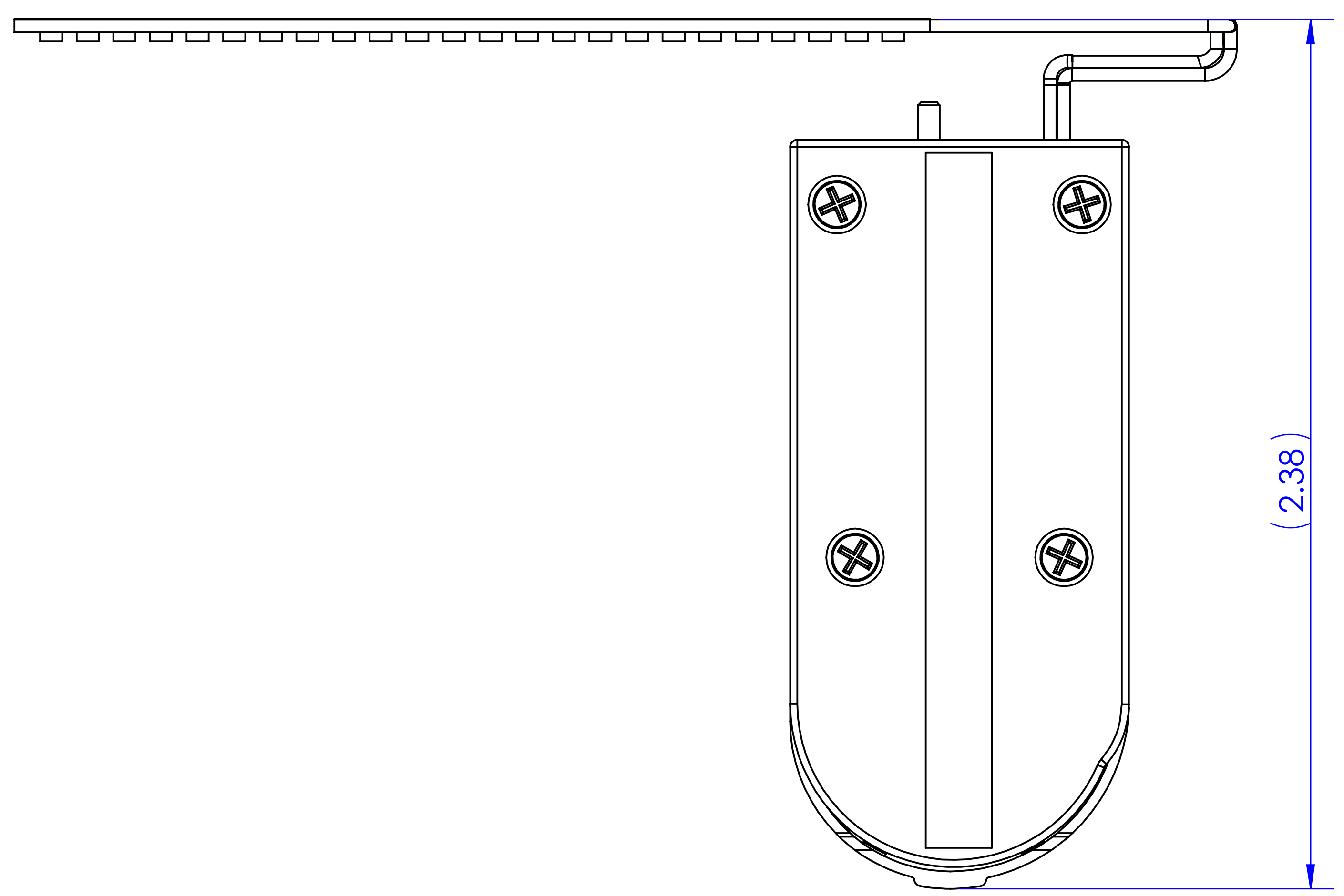


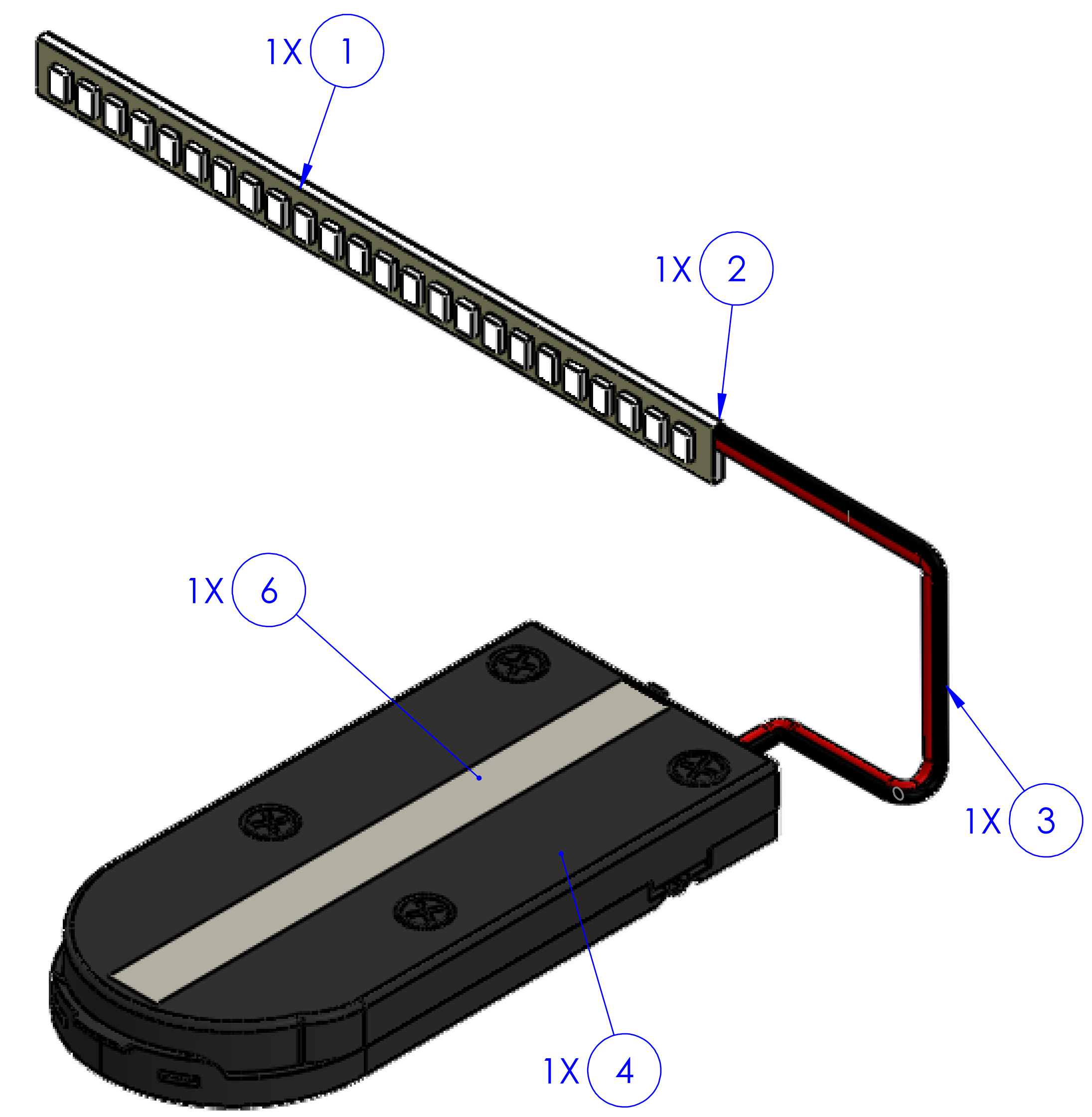
ISOMETRIC VIEW
FOR REFERENCE ONLY

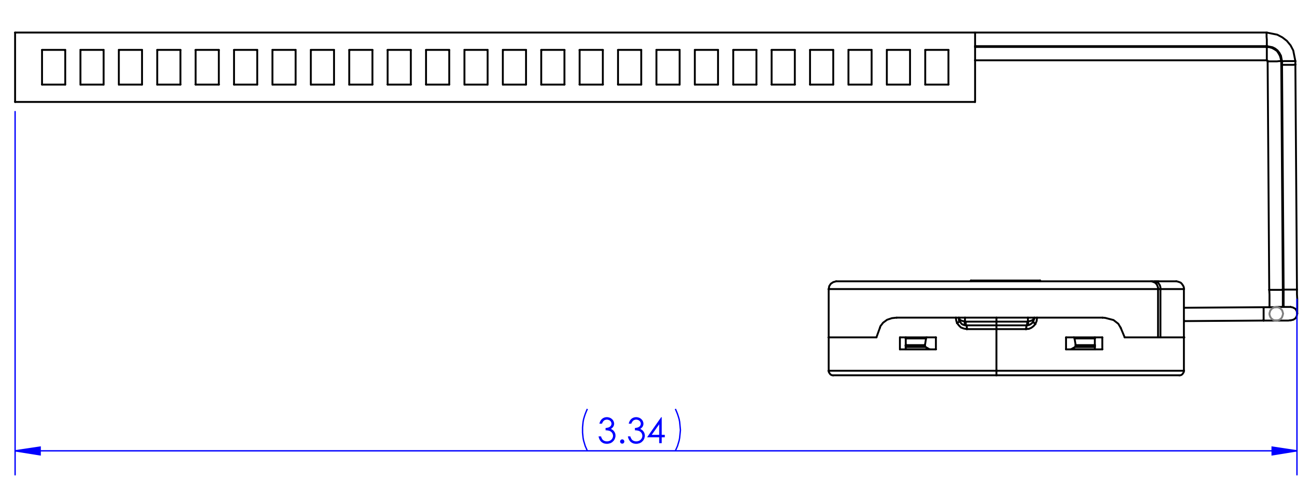
TOLERANCES UNLESS NOTED:	X.X ± .03 X.XX ± .01 X.XXX ± .005 X.X° ° 1	
UNITS: INCHES		
MATERIAL AS REQUIRED		DESCRIPTION BOOK NOOK FULL ASSEMBLY
FINISH N/A		AN 01 / REV A / SHEET 1 of 1

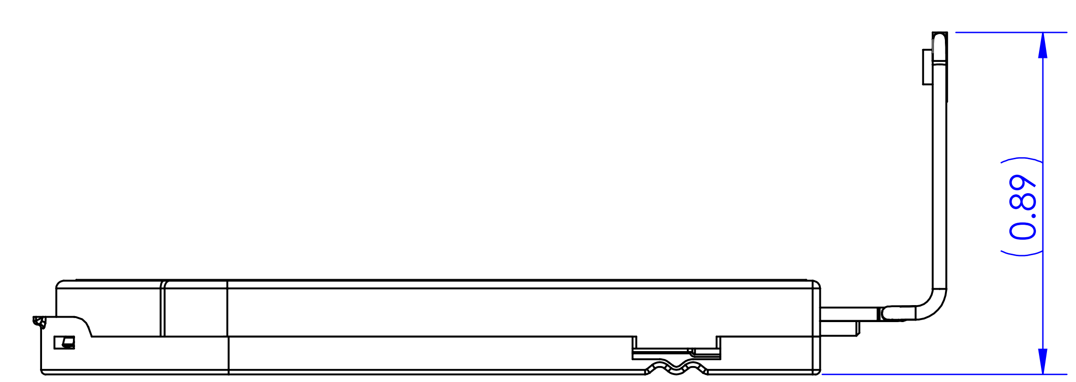
REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	5 - LED		1
2	9 - DoubleSidedAdhesive		1
3	8 - Wires		1
4	6 - BatteryPack	Battery Holder	1
5	7 - Battery		2
6	9 - DoubleSidedAdhesive		1









TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ± 1°; UNITS: INCHES			
MATERIAL: AS REQUIRED	DESCRIPTION: LIGHT ASSEMBLY		
FINISH: N/A	PN: 02	REV: A	SHEET: 1 of 1

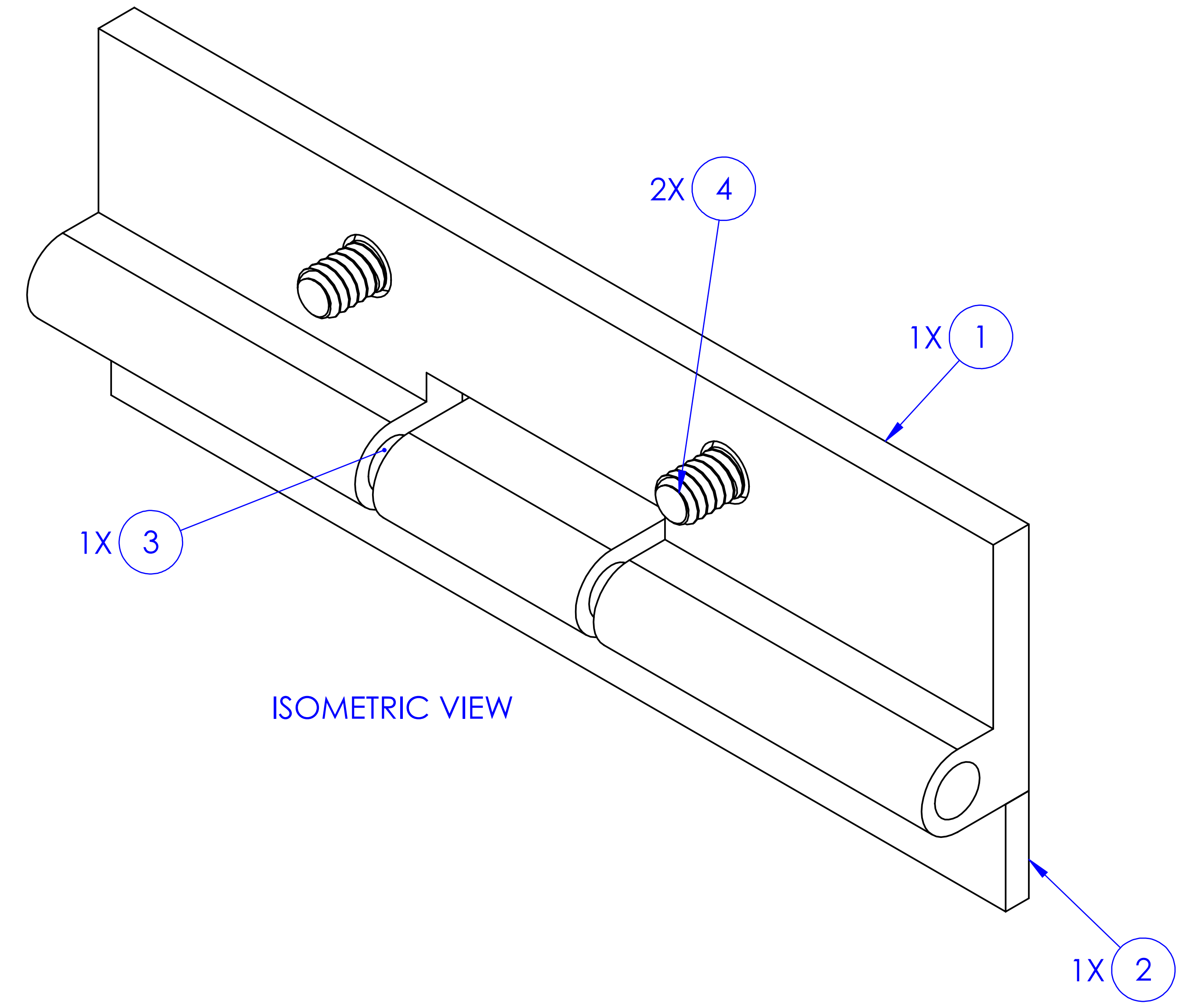
PROPRIETARY AND CONFIDENTIAL THE INFORMATION CONTAINED IN THIS DRAWING IS THE SOLE PROPERTY OF . ANY REPRODUCTION IN PART OR AS A WHOLE WITHOUT WRITTEN PERMISSION IS PROHIBITED.

NOTES:

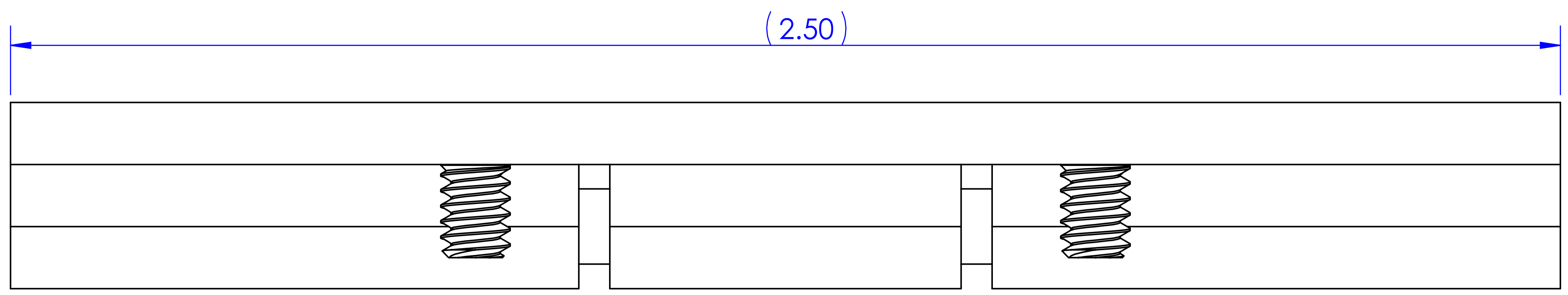
1. ITEM 3 IS PRESS FIT INTO ITEMS 1 AND 2

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

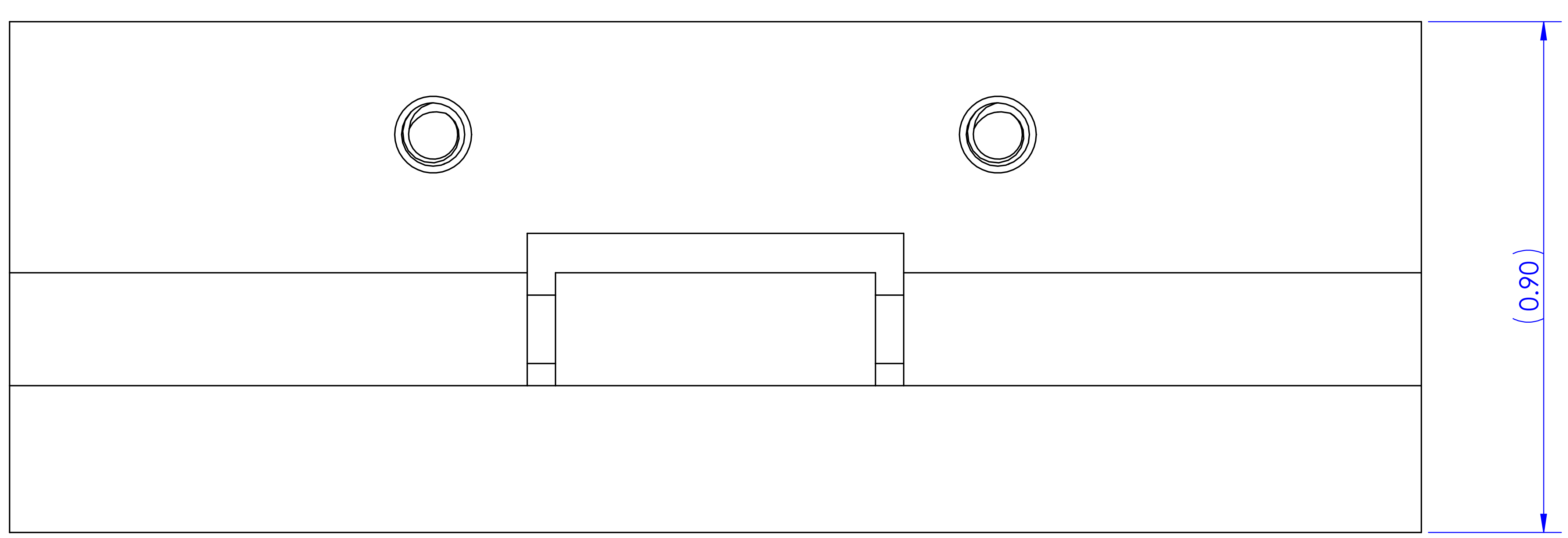
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	10 - HingeMount	HINGE MOUNT	1
2	11 - HingeSwivel	HINGE SWIVEL	1
3	12 - HingeRod	HINGE ROD	1
4	16 - 92210A105	18-8 Stainless Steel Hex Drive Flat Head Screw	2



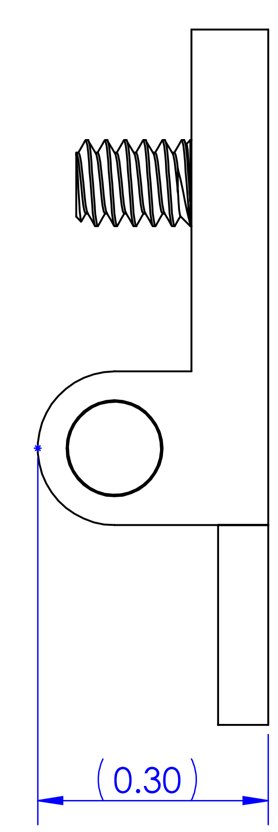
ISOMETRIC VIEW



TOP VIEW



FRONT VIEW

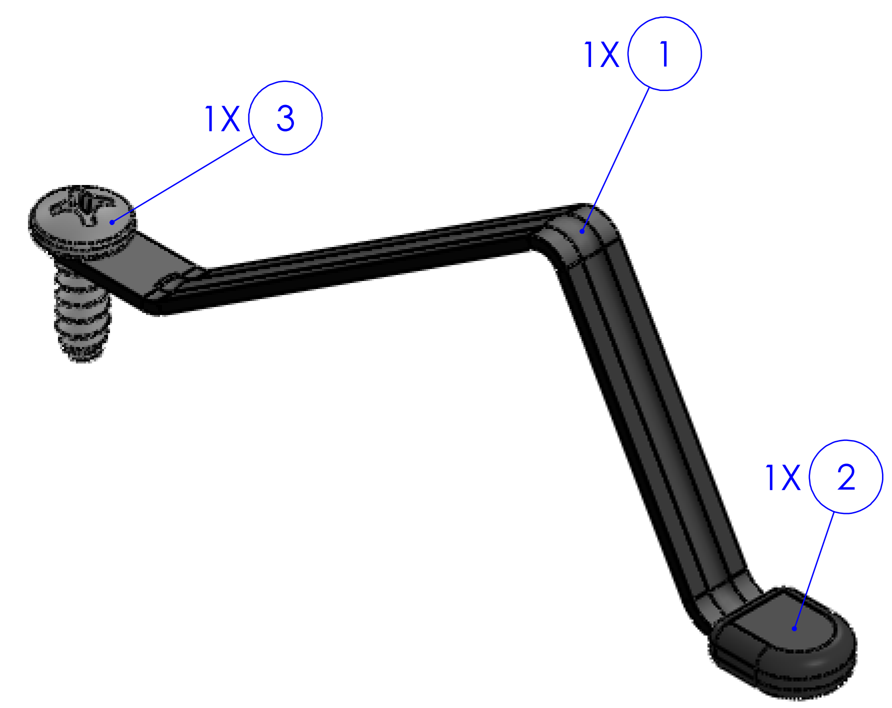


SIDE VIEW

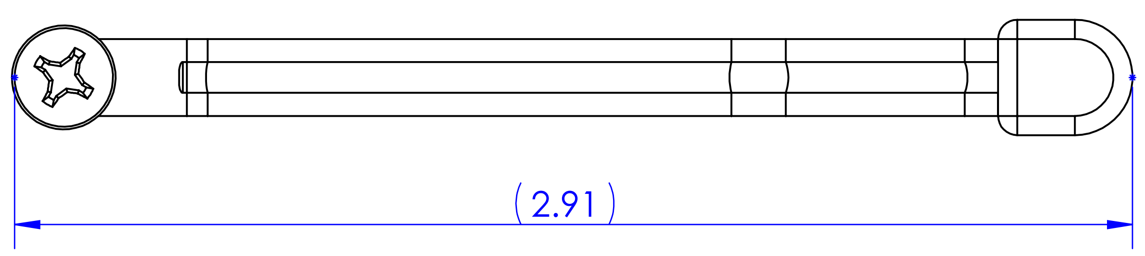
TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ± 1°; UNITS: INCHES		
MATERIAL: AS REQUIRED	DESCRIPTION: HINGE ASSEMBLY	
FINISH: N/A	AN: 01, REV: A	SHEET: 1 of 1

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

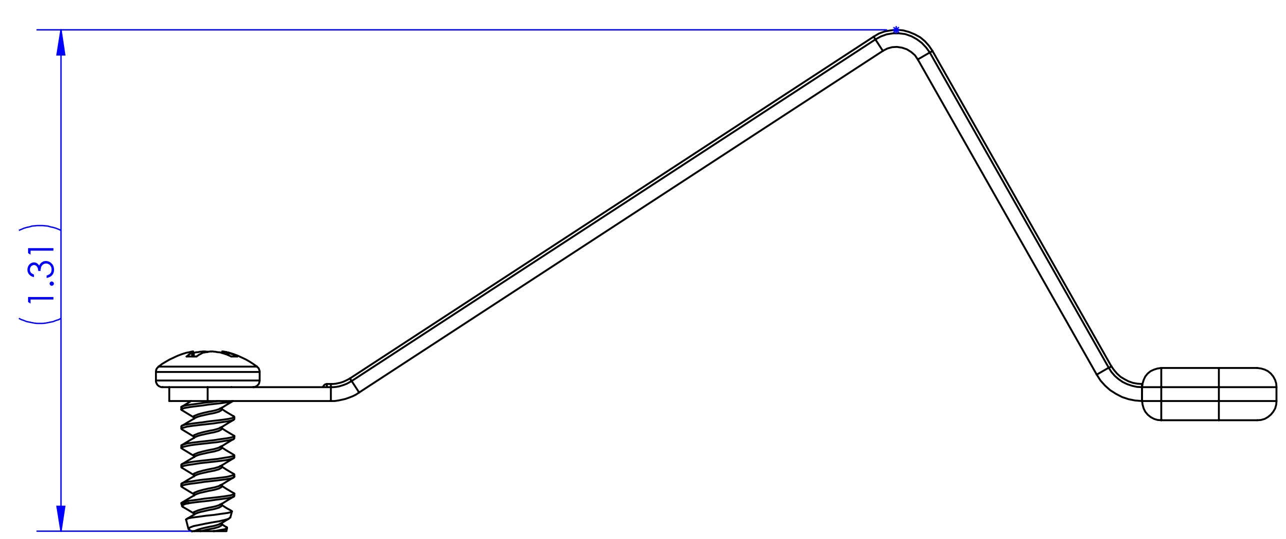
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	14 - Arm (20 Gauge)		1
2	15 - Rubber Cover		1
3	92525A315	Rounded Head Blunt Screws for Sheet Metal	1



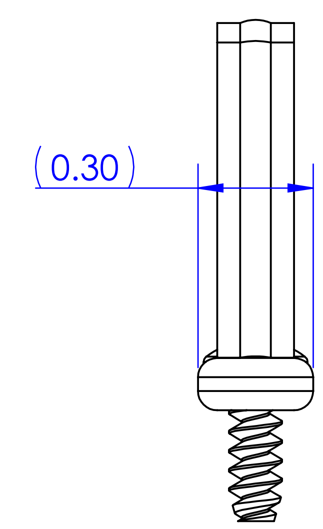
ISOMETRIC VIEW



TOP VIEW



FRONT VIEW



SIDE VIEW

TOLERANCES UNLESS NOTED: X.X ± .03; X.XX ± .01; X.XXX ± .005; X.X° ° 1; UNITS: INCHES	
MATERIAL AS REQUIRED	DESCRIPTION TENSION ARM ASSEMBLY
FINISH N/A	PN 04 — REV A — SHEET 1 of 1

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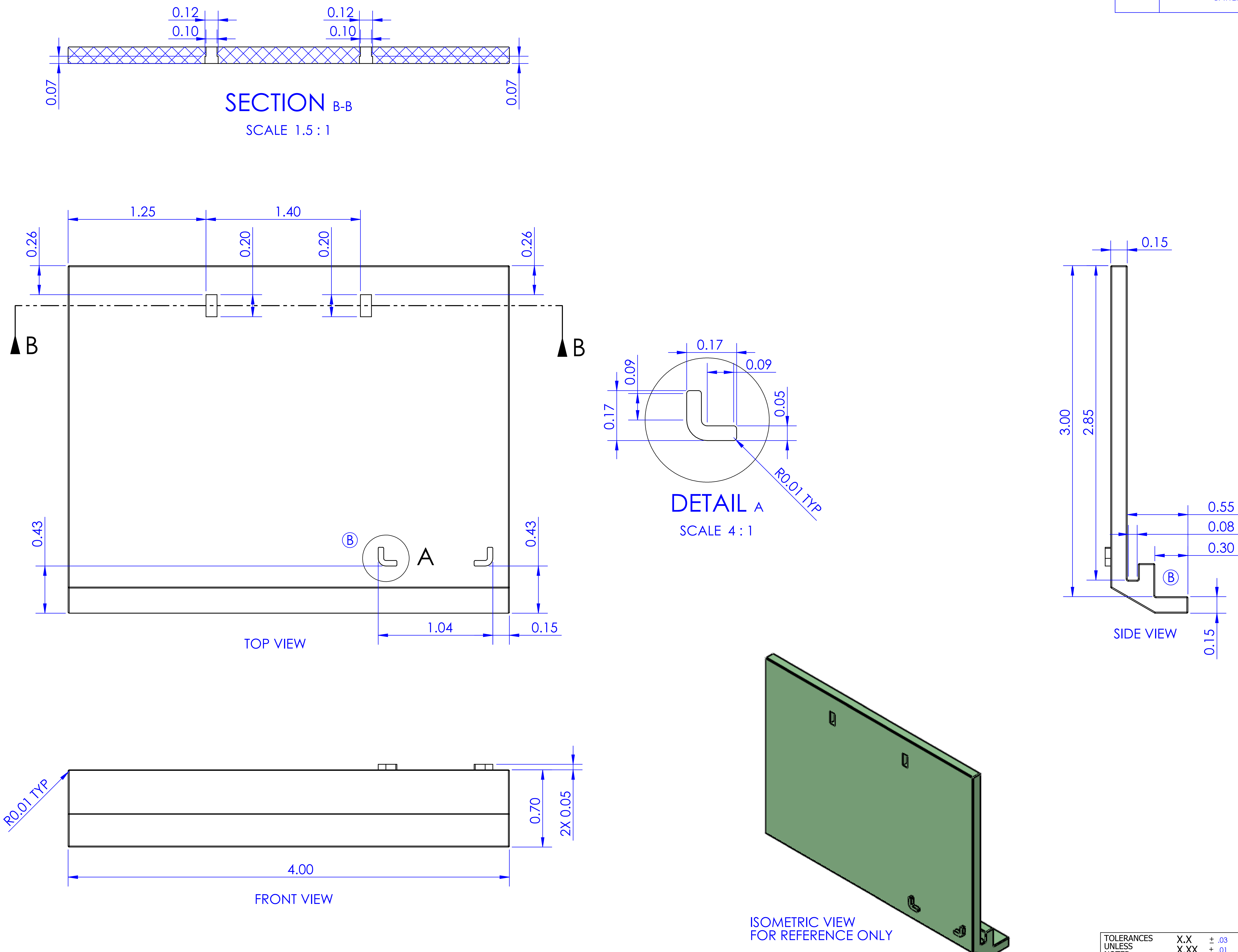
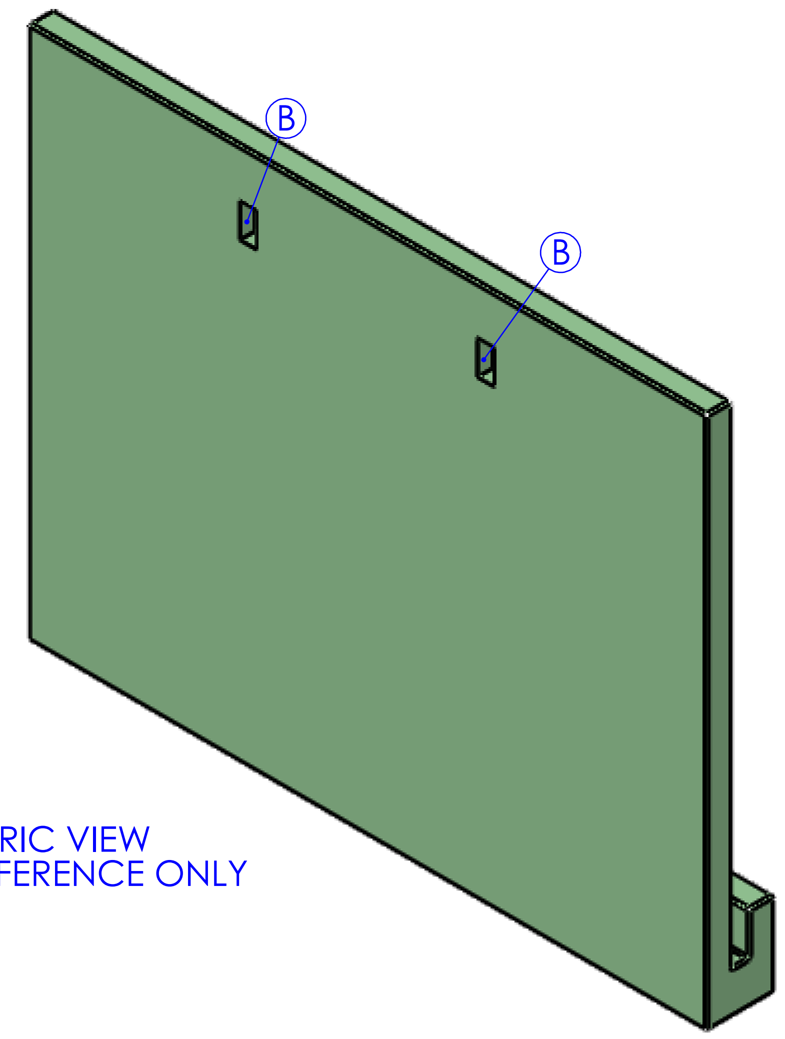
REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025
B	REMOVED SLOTS FOR WAVE SPRING AND ADDED HOLES FOR SNAP FIT, ADDED LIGHT OVERHANG - REMOVED TENSION HINGE TAPPED HOLES, ADDED BATTERY LOCATION MARKER	11/17/2025

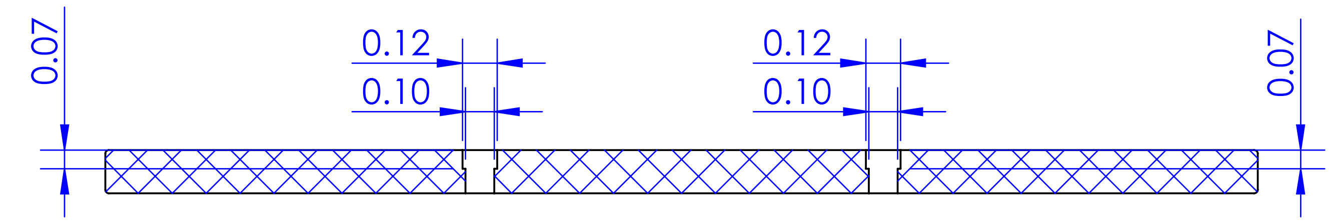


TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ° 1 UNITS: INCHES				
MATERIAL ABS	DESCRIPTION UPPER EDGE HOLDER			
FINISH NONE	PN 1	REV B		SHEET 1 of 1

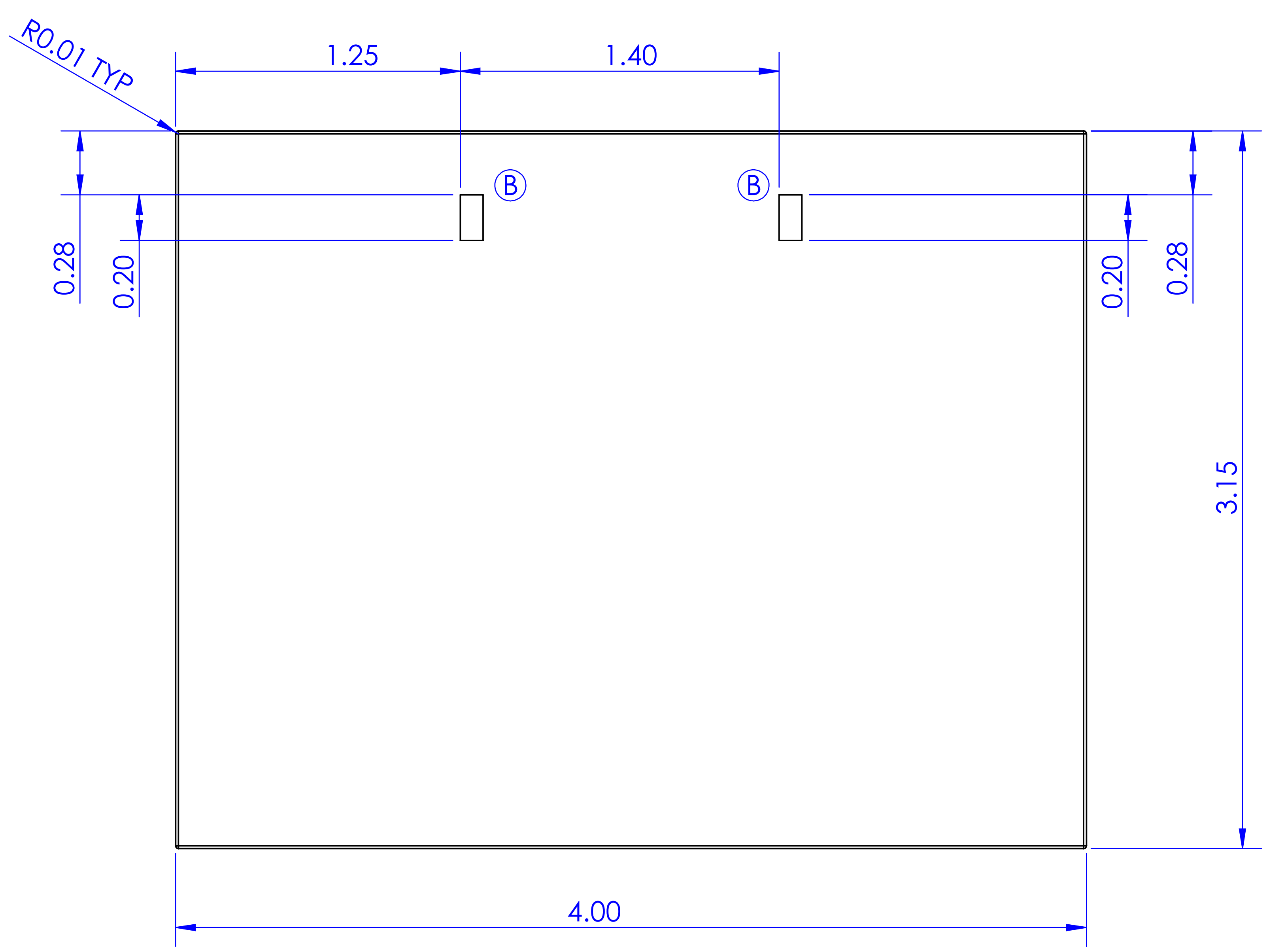
REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025
B	REMOVED SLOTS FOR WAVE SPRING AND ADDED HOLES FOR SNAP FIT	11/17/2025



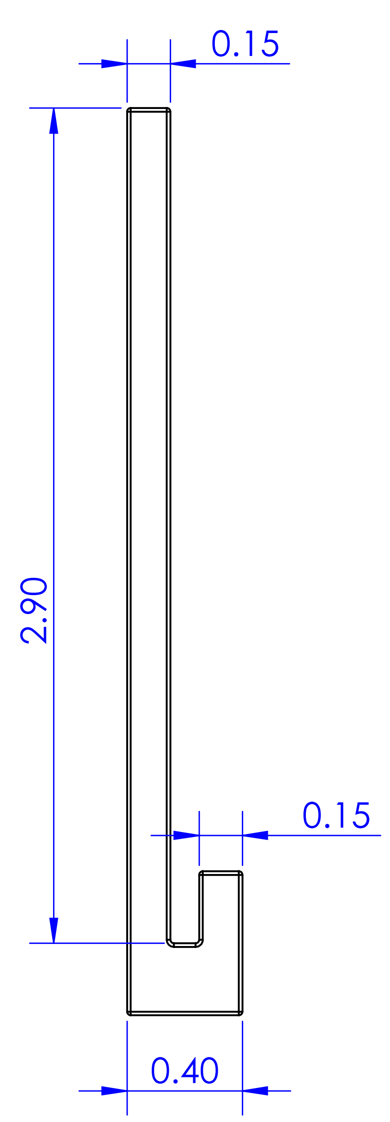
ISOMETRIC VIEW
FOR REFERENCE ONLY



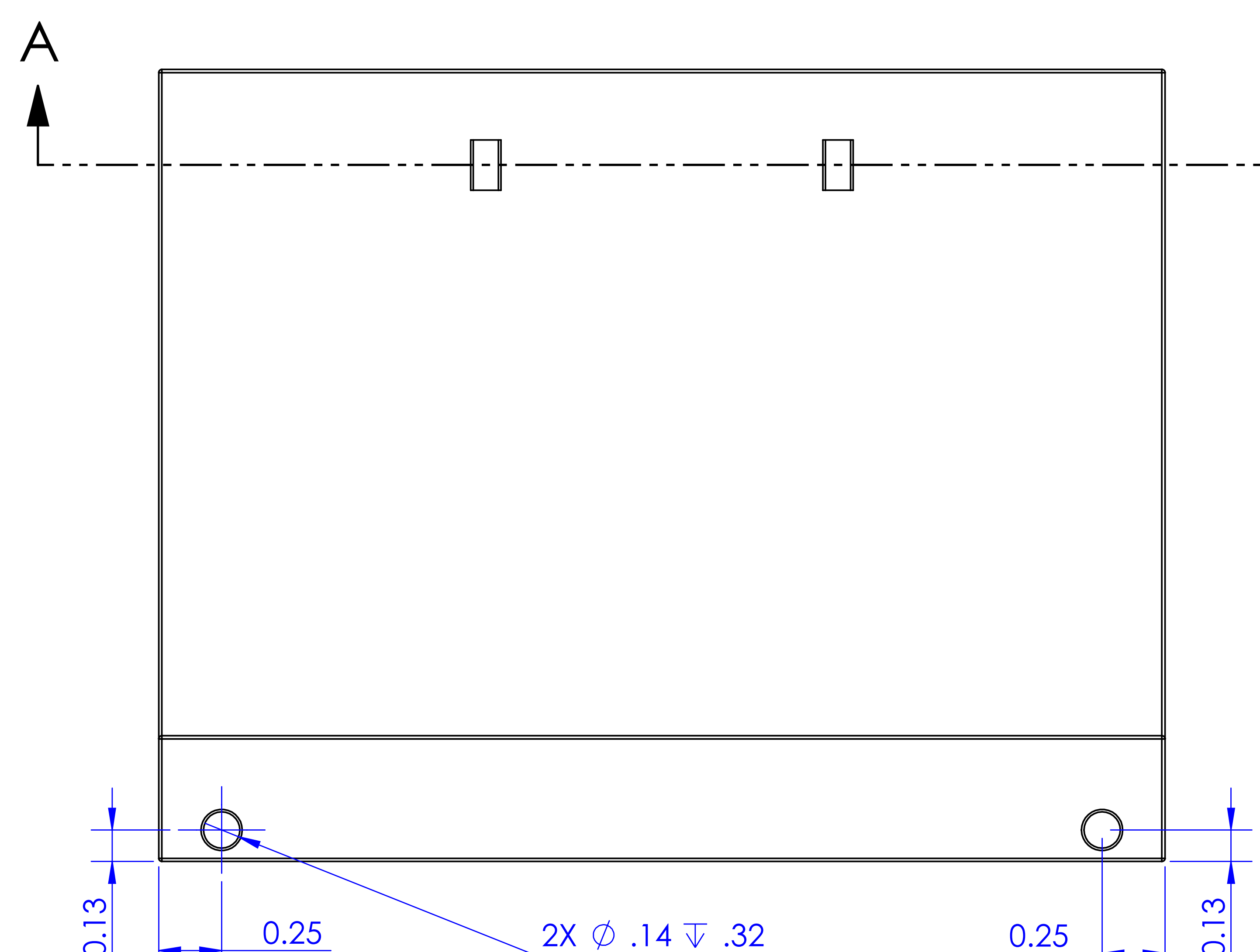
SECTION A-A
SCALE 1.5 : 1



FRONT VIEW



SIDE VIEW



BACK VIEW

TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ° 1; UNITS: INCHES				
MATERIAL ABS	DESCRIPTION LOWER EDGE HOLDER			
FINISH NONE	PN 2	REV B		SHEET 1 of 1

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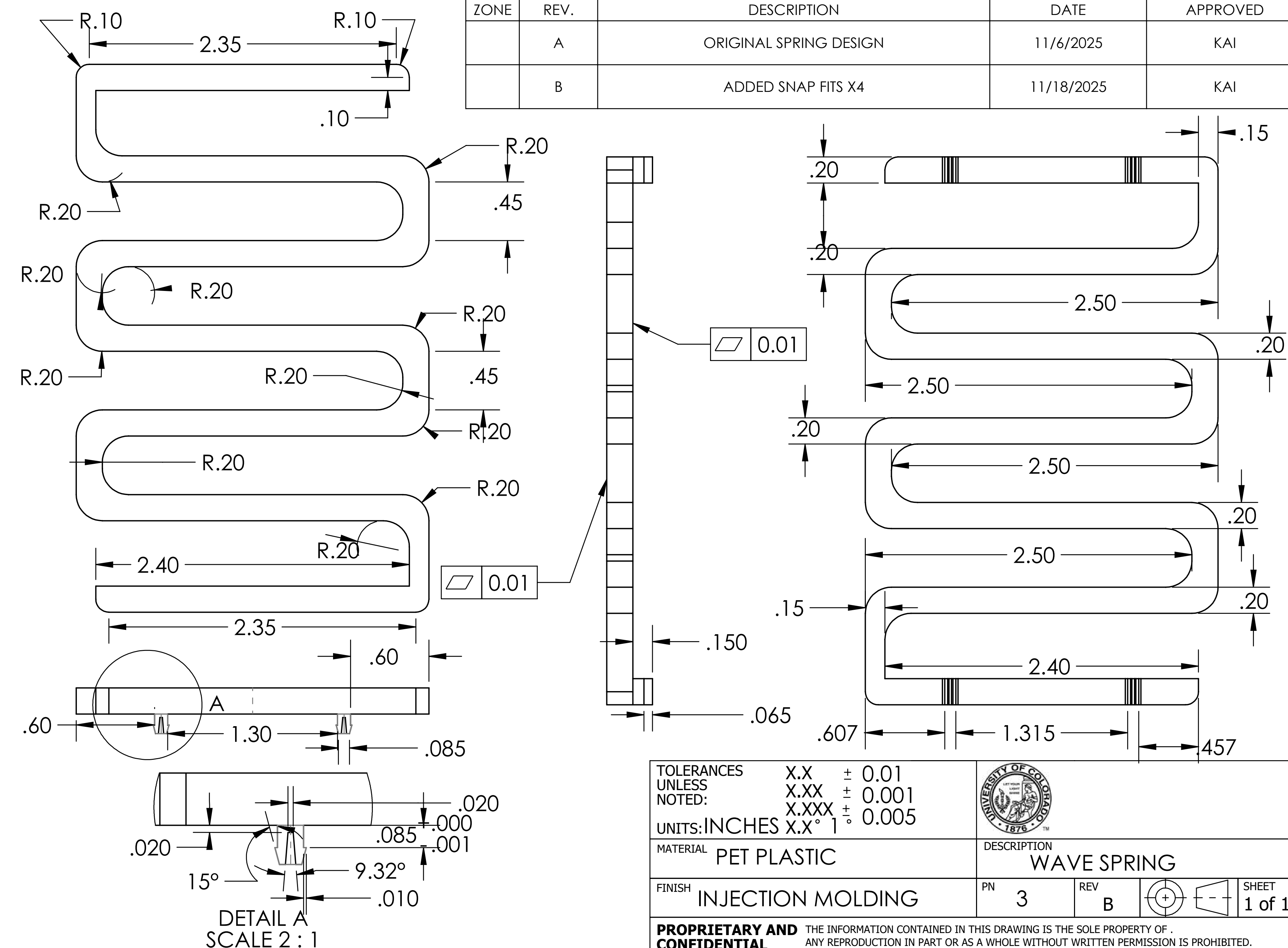
REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	ORIGINAL SPRING DESIGN	11/6/2025	KAI
	B	ADDED SNAP FITS X4	11/18/2025	KAI

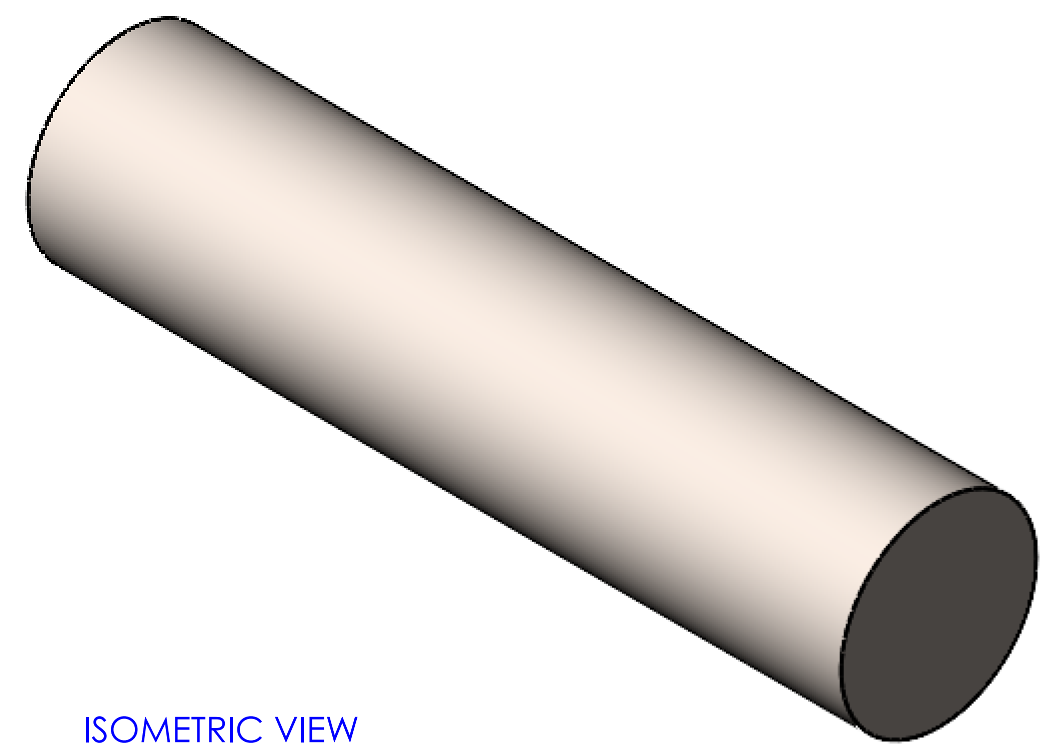


TOLERANCES UNLESS NOTED: X.X ± 0.01, X.XX ± 0.001, X.XXX ± 0.005, X.X° ± 1°			
UNITS: INCHES			
MATERIAL PET PLASTIC	DESCRIPTION WAVE SPRING		
FINISH INJECTION MOLDING	PN 3	REV B	SHEET 1 of 1

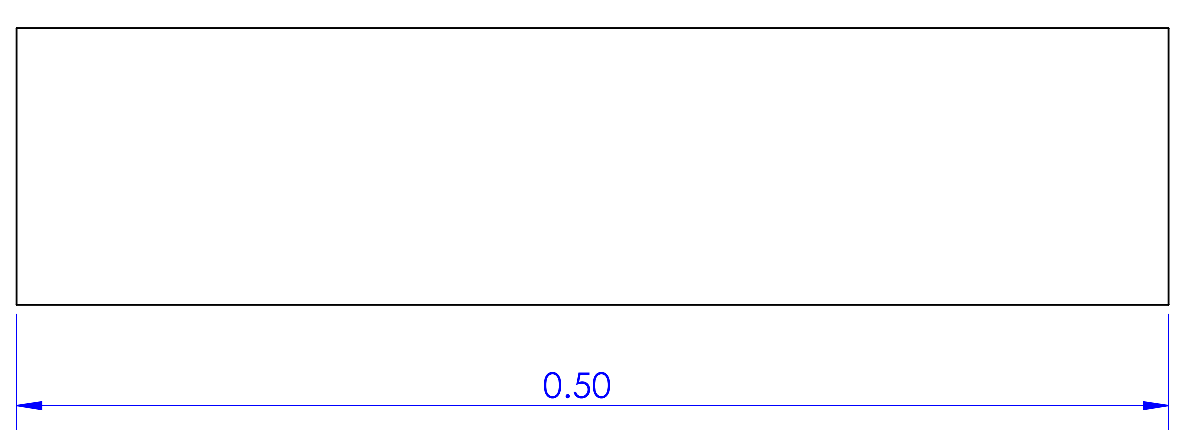
NOTES:

1. MAKE FROM MCMASTER PN 97395A441 OR EQUIVALNET

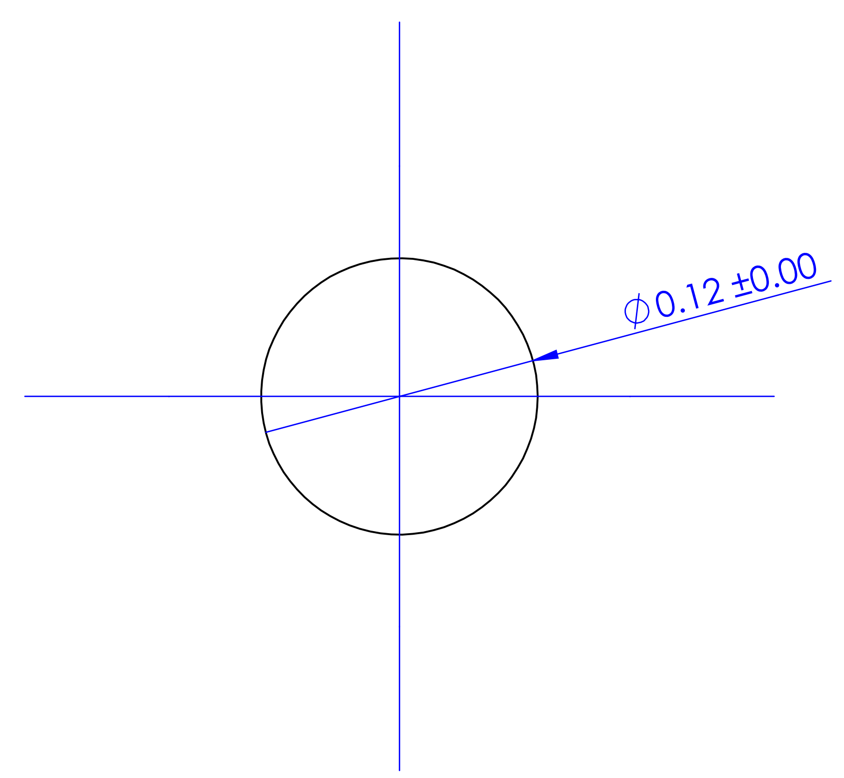
REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025



ISOMETRIC VIEW
FOR REFERENCE ONLY



FRONT VIEW



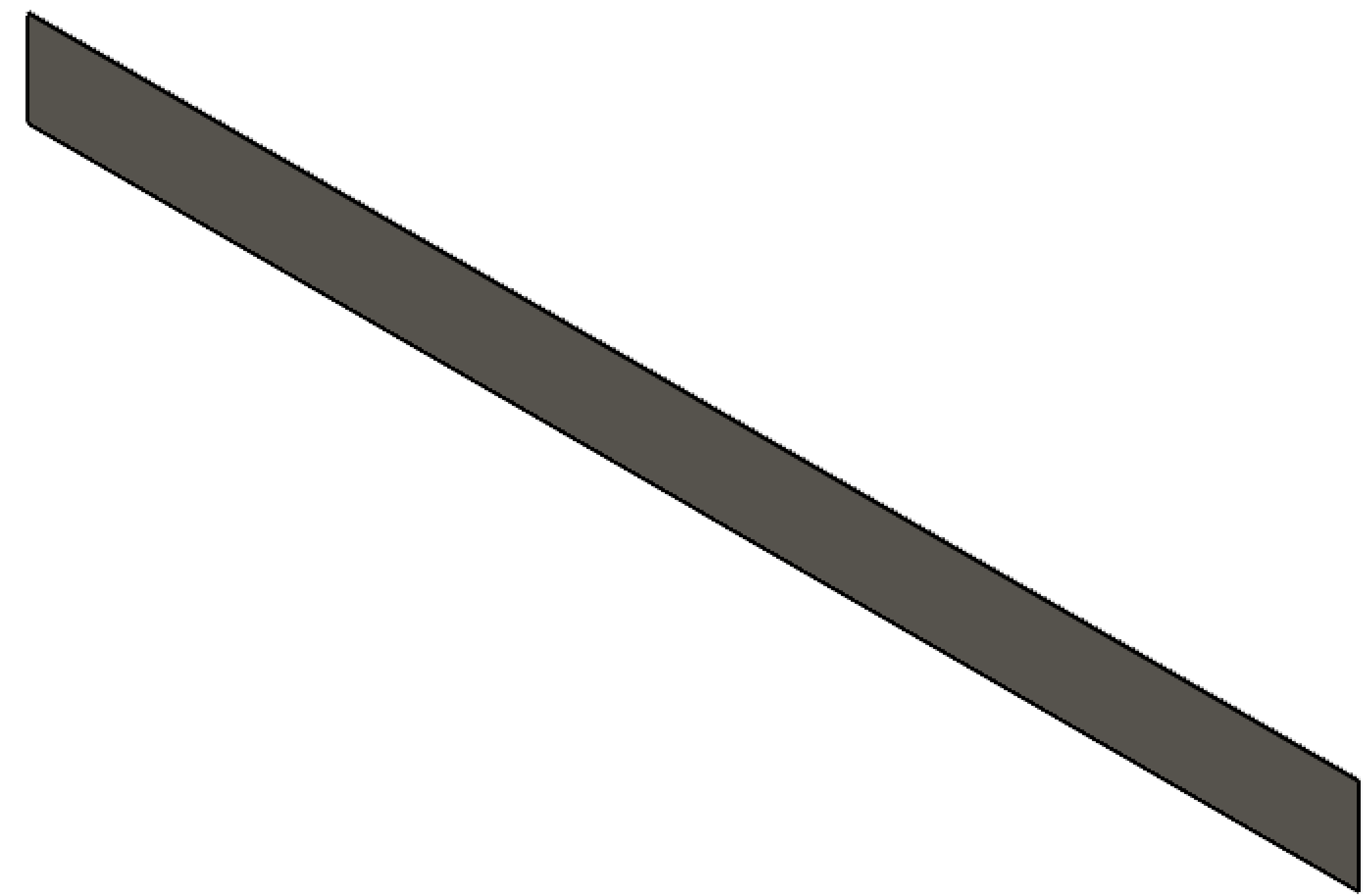
SIDE VIEW

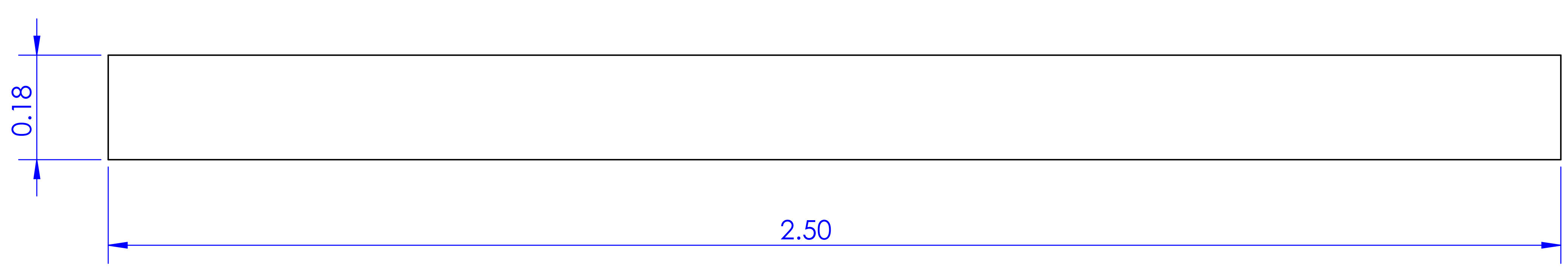
TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ° 1; UNITS: INCHES				
MATERIAL: 316 Stainless Steel	DESCRIPTION: SPRING DOWEL			
FINISH: NONE	PN: 4	REV: A		SHEET: 1 of 1

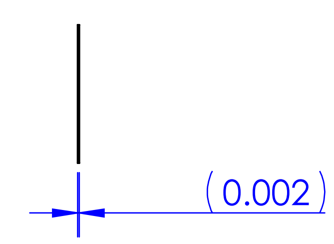
NOTES:

1. MAKE FROM MCMASTER PN 7628A49 OR EQUIVALENT

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

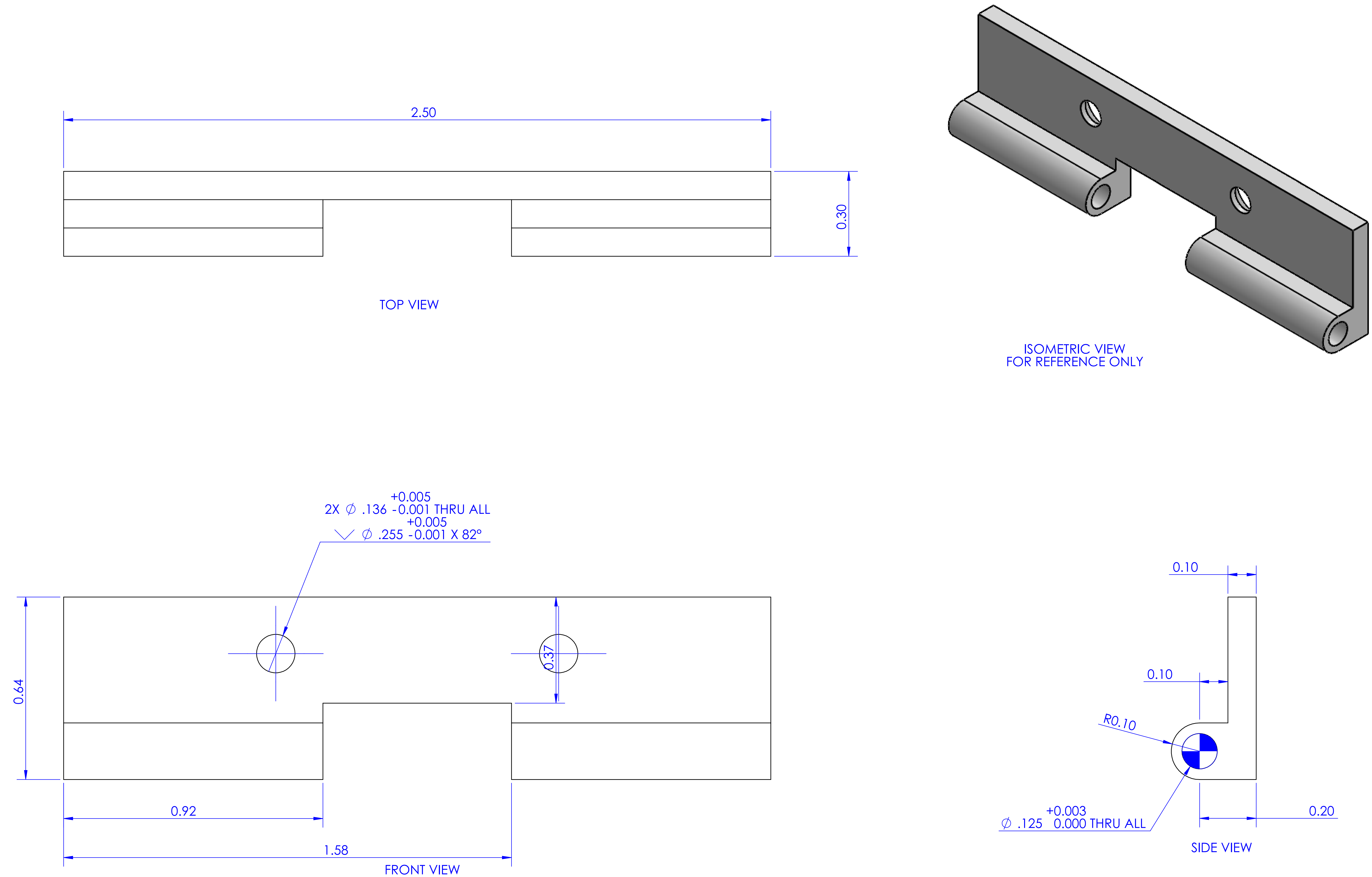






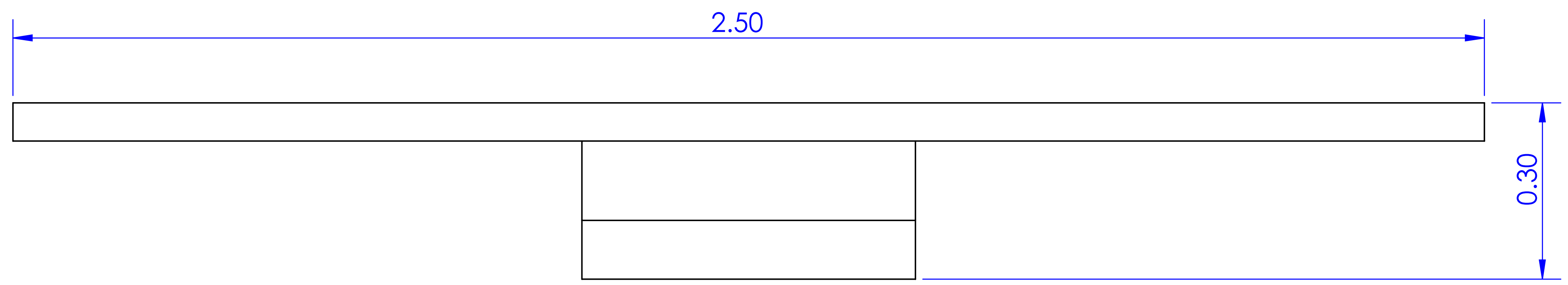
TOLERANCES UNLESS NOTED:	X.X ± .03 X.XX ± .01 X.XXX ± .005 X.X° ° 1		
UNITS: INCHES			
MATERIAL ADHESIVE	DESCRIPTION DOUBLE SIDED ADHESIVE		
FINISH NONE	PN 9	REV A	SHEET 1 of 1

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

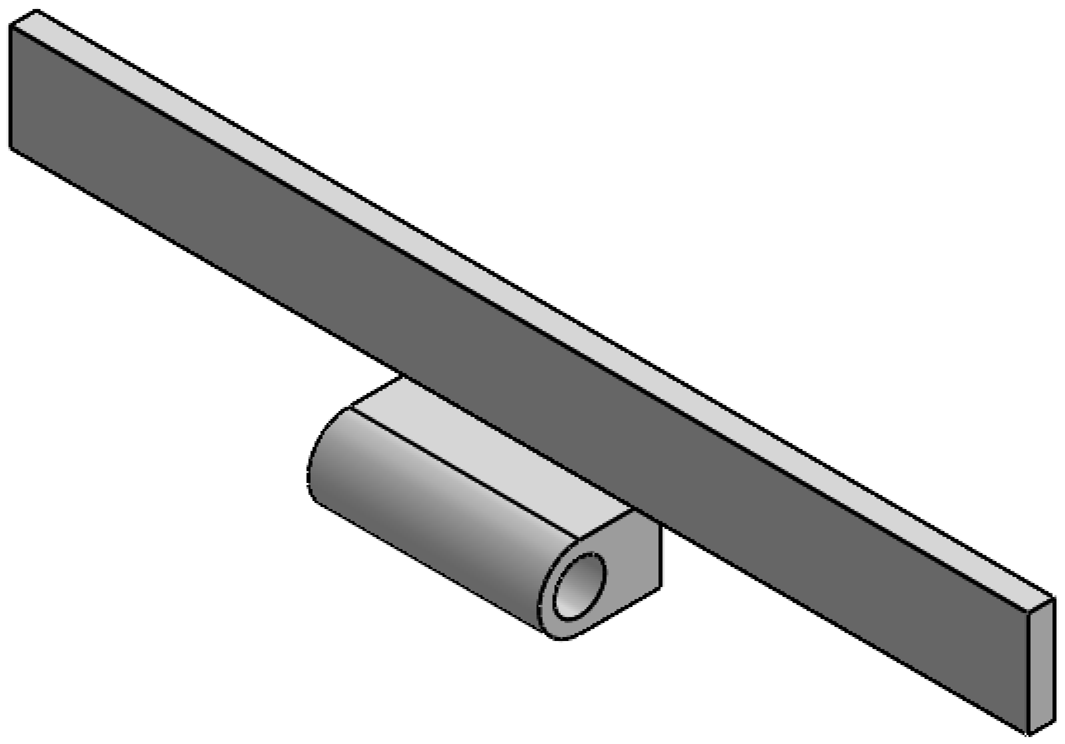


TOLERANCES UNLESS NOTED: X.X ± .03; X.XX ± .01; X.XXX ± .005; X.X° ° 1; UNITS: INCHES				
MATERIAL ABS	DESCRIPTION HINGE MOUNT			
FINISH NONE	PN 10	REV A		SHEET 1 of 1

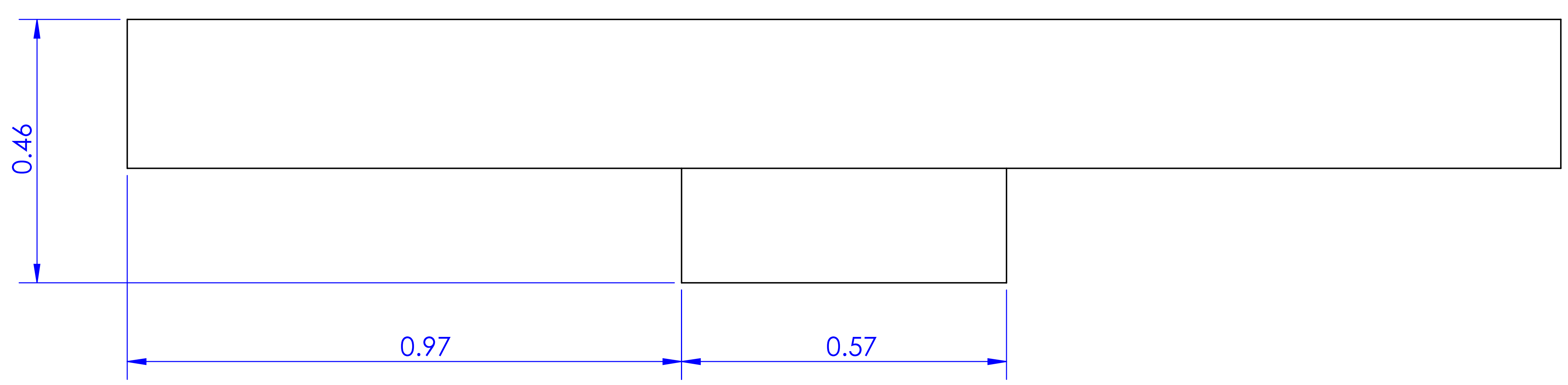
REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025



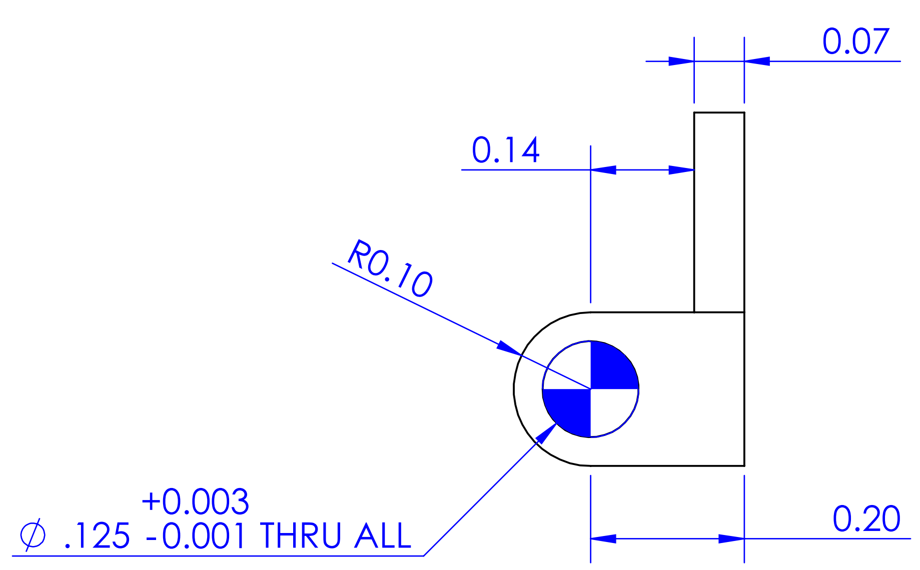
TOP VIEW



ISOMETRIC VIEW
FOR REFERENCE ONLY



FRONT VIEW



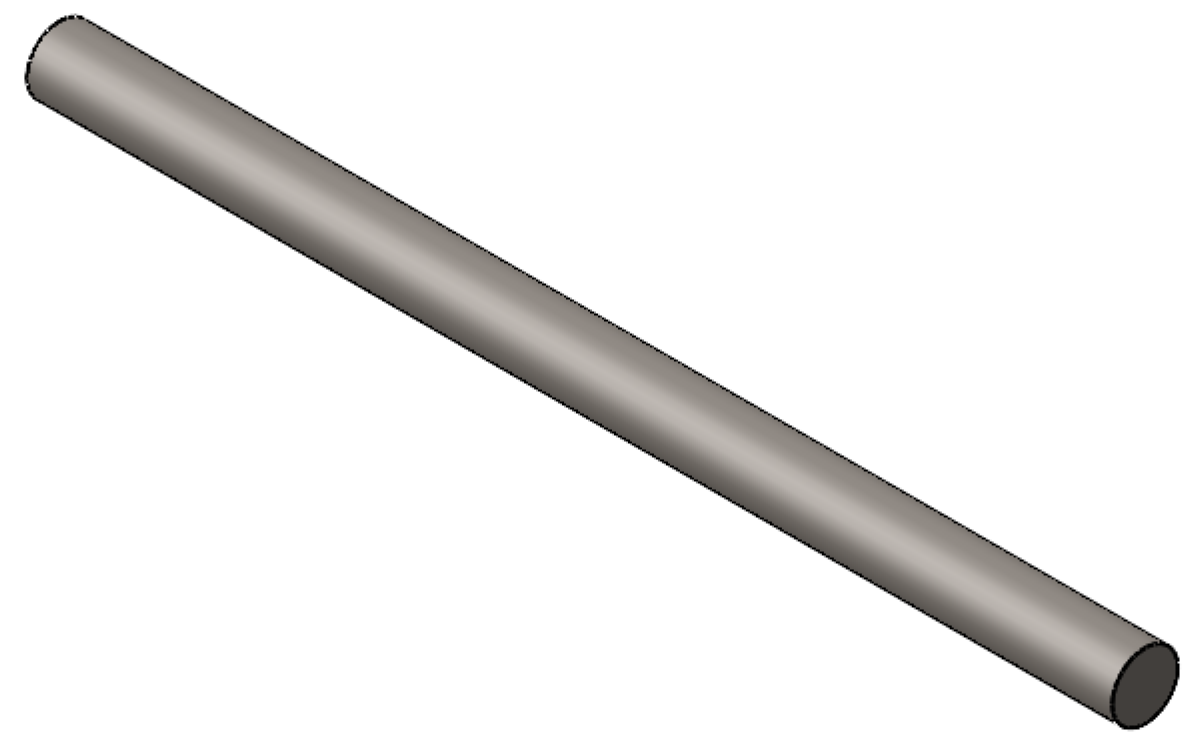
SIDE VIEW

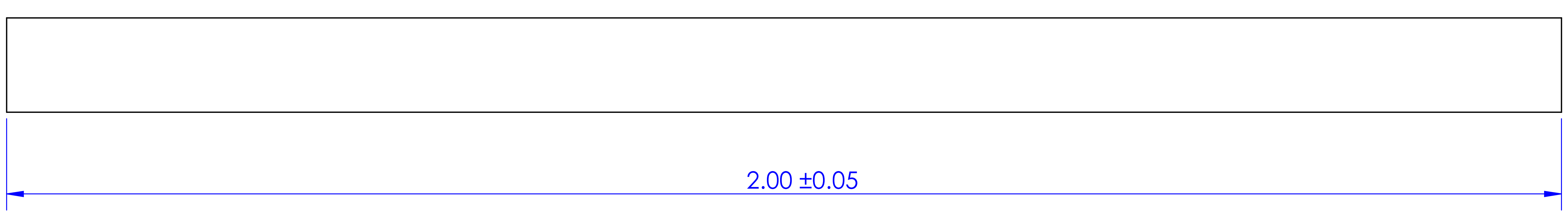
TOLERANCES UNLESS NOTED: X.X ± .03, X.XX ± .01, X.XXX ± .005, X.X° ° 1; UNITS: INCHES				
MATERIAL ABS	DESCRIPTION HINGE SWIVEL			
FINISH NONE	PN 11	REV A		SHEET 1 of 1

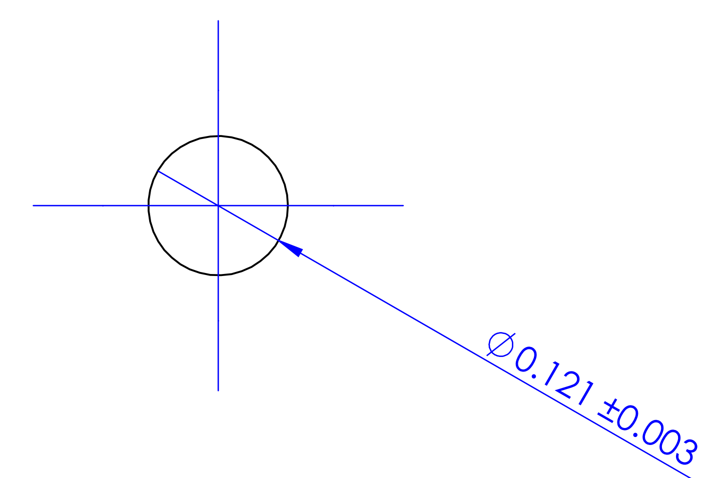
NOTES:

1. MAKE FROM MCMASTER PN 97395A441 OR EQUIVALENT

REVISIONS		
REV.	DESCRIPTION	DATE
A	INITIAL RELEASE	11/8/2025

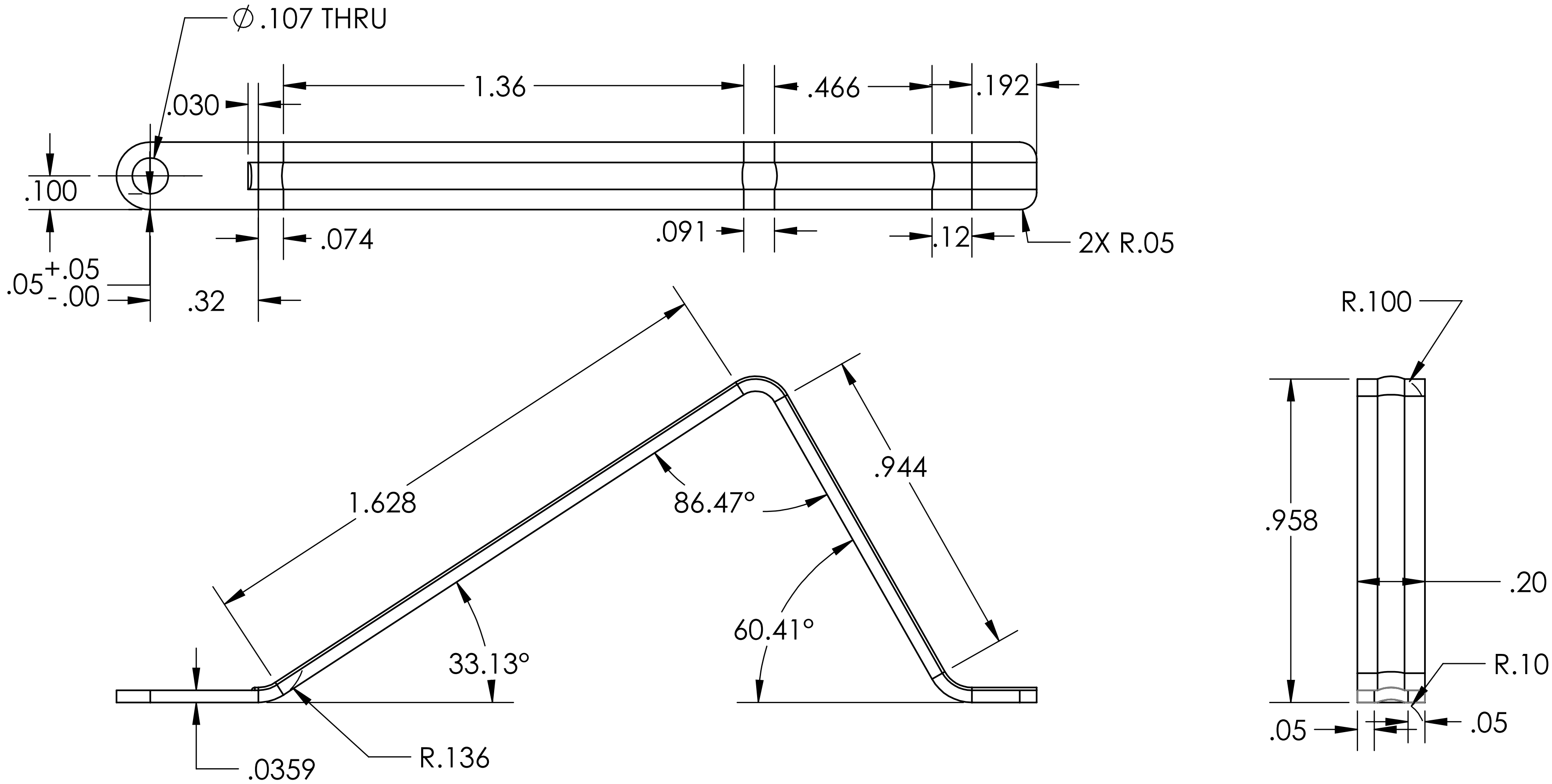






TOLERANCES UNLESS NOTED: X.X ± .03; X.XX ± .01; X.XXX ± .005; X.X° ° 1; UNITS: INCHES				
MATERIAL 316 STAINLESS STEEL	DESCRIPTION HINGE ROD			
FINISH NONE	PN 12	REV A		SHEET 1 of 1

REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL REVISION	11/6/2025	KAI
	B	ADDED BUMP FROM STAMPING FOR RIGIDITY AND RESISTANCE FOR CYCLIC LOADING	11/18/2025	KAI



TOLERANCES UNLESS NOTED: X.X ± X.XX ± X.XXX ± X.X° ± ° UNITS:				
MATERIAL 304 STAINLESS STEEL SHEET	DESCRIPTION ARM			
FINISH STAMPED	PN 14	REV B		SHEET 1 of 1

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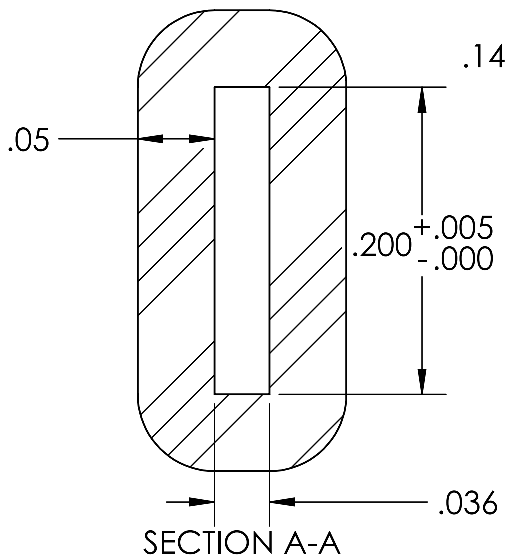
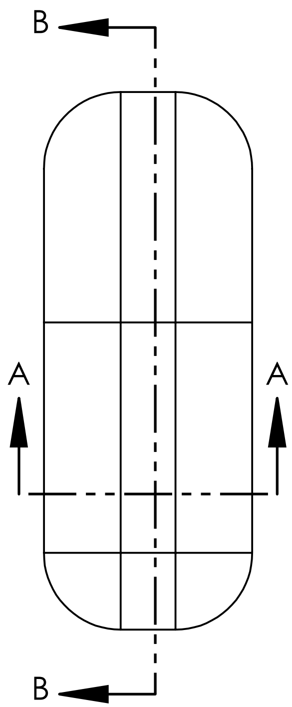
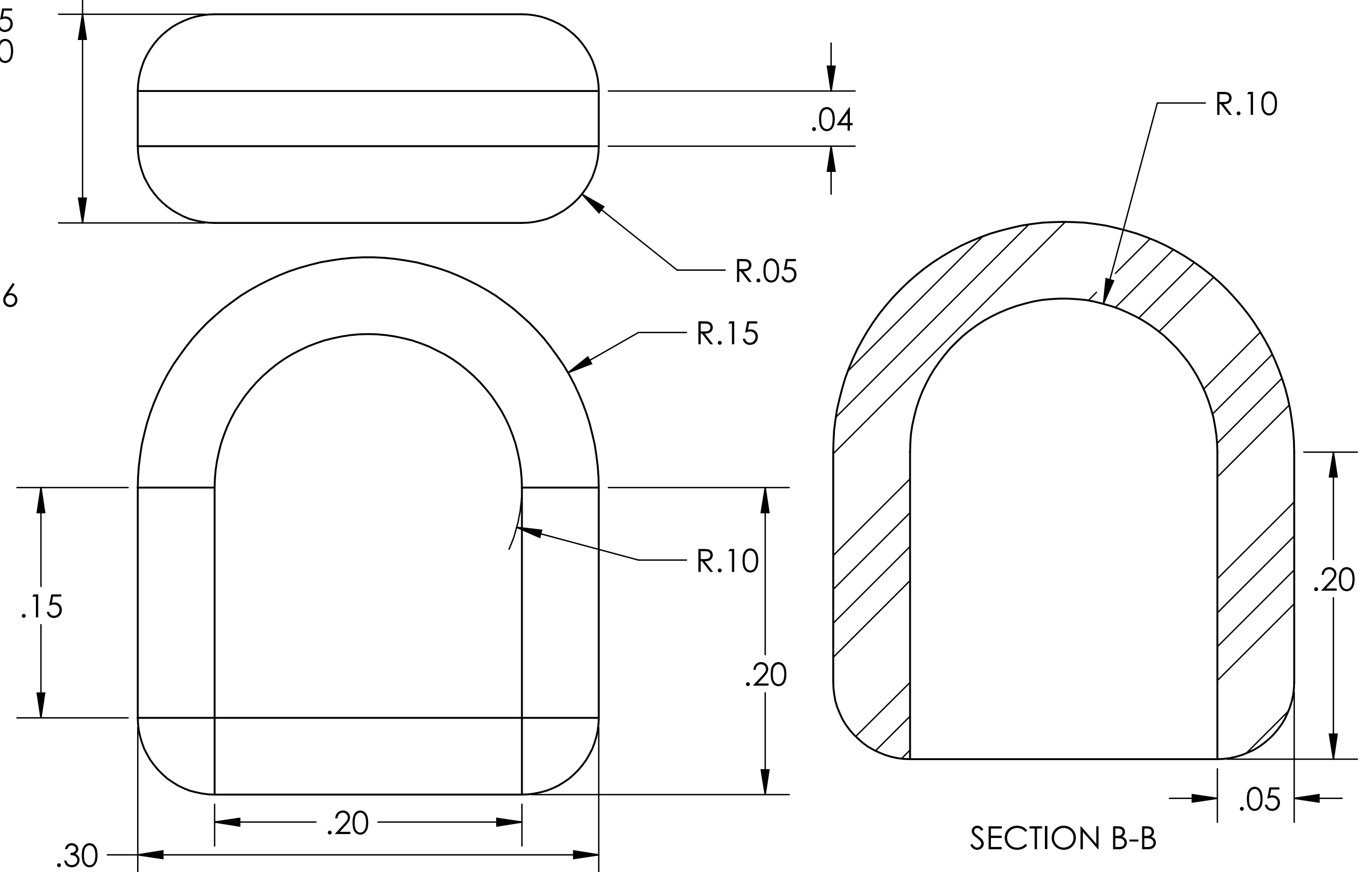
REVISIONS				
ZONE	REV.	DESCRIPTION	DATE	APPROVED
	A	INITIAL DESIGN	11/8/2025	KAI







TOLERANCES UNLESS NOTED: X.X ± X.XX ± X.XXX ± UNITS: X.X° ±°				
MATERIAL NATURAL RUBBER	DESCRIPTION RUBBER COVER			
FINISH MATTE	PN 15	REV A		SHEET 1 of 1

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